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Dimension Reduction for Derivative-Informed Operator Learning: An Analysis of Approximation Errors

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Abstract

We study the derivative-informed learning of nonlinear operators between infinite-dimensional Hilbert spaces. Such operators can arise as solution maps of partial differential equations, and their approximation by accurate *surrogate models* can accelerate simulation-intensive tasks of scientific and engineering interest, including inference, control, and uncertainty quantification. Since efficiently performing such tasks often requires an accurate representation of the operator's derivatives, we analyze the approximation capabilities of neural operators measured by Sobolev norms over infinite-dimensional Gaussian input measures. We focus on the reduced basis neural operator, which employs linear encoders/decoders defined on dominant input/output subspaces. We study two methods for generating the subspaces: principal component analysis (PCA) and derivative-informed subspaces (DIS). These use the dominant eigenvectors of the covariance of the data or the derivatives as bases for the subspaces, respectively. We derive bounds for errors arising from both dimension reduction and latent neural network approximations, including sampling errors associated with the empirical estimation of the PCA/DIS subspaces. Our analysis is validated through numerical experiments, which demonstrate that subspaces informed by the underlying operator (DIS or output PCA) yield smaller generalization errors in the Sobolev norm, while input PCA may underperform unless ranks and training sample sizes are sufficiently large.

Keywords: operator learning, derivative-informed neural operators, Sobolev training, dimension reduction, universal approximation

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1. Introduction

In this work, we consider the task of learning neural network representations of nonlinear operators between infinite-dimensional spaces in terms of both their outputs and their derivatives. Such operators arise, for example, as the solution operators of partial differential equations (PDEs), which model many physical systems and play important roles in scientific discovery and engineering design. For complex PDEs, their solution operators can only be accessed using numerical methods with significant computational costs, which often renders their use in simulation-intensive outer-loop tasks—such as model-based design, control, inference, and uncertainty quantification—extremely computationally expensive, if not intractable. To this end, neural network approximations of the expensive-to-solve PDEs can be effective *surrogate models* that have the potential to significantly accelerate the aforementioned outer-loop tasks.

In recent years, a vast selection of neural network architectures for operator learning has been proposed. Examples include architectures using reduced-basis representations such as PCANet, POD-DL-ROM, and DIPNet (Hesthaven and Ubbiali, 2018; Bhattacharya et al., 2021; Fresca and Manzoni, 2022; O'Leary-Roseberry, Villa, et al., 2022), deep operator networks (DeepONets) (Lu et al., 2021, 2022; Jin et al., 2022), Fourier neural operators (FNOs) (Li et al., 2021; Bonev et al., 2023), and many more recent developments (see Kovachki et al., 2023, 2024 for more comprehensive discussions). Typically, these neural networks are trained in a conventional, supervised manner, where the PDE's input parameters and solutions are used as input-output pairs to define a loss function based on the misfit between the neural network predictions and the true outputs.

In addition, it has been demonstrated that, when available, derivatives of the underlying input-output maps (or approximations thereof) can be used to improve the accuracy of the surrogates (Liu and Batill, 2000; Czarnecki et al., 2017; Bouhlel et al., 2020; Tsay, 2021; Yu et al., 2022; O'Leary-Roseberry et al., 2024). In particular, derivatives of the input-output maps can be used as training data in addition to the input-output pairs used in conventional supervised learning, and incorporated into the training loss with a Sobolev-type norm, thereby encouraging the resulting function approximation to be accurate not only in its predicted output values, but also in its derivatives with respect to the input. This has been considered for constructing finite-dimensional neural network approximations in works such as Liu and Batill (2000); Czarnecki et al. (2017); Bouhlel et al. (2020); Tsay (2021); Yu et al. (2022), where the approach has been termed *gradient-enhanced neural networks* or *Sobolev training*.

In O'Leary-Roseberry et al. (2024), Sobolev training is considered for the approximation of formally infinite-dimensional mappings that arise from the solution operators of PDEs, where it is termed *derivative-informed neural operators* (DINOs). Additional works have shown that Sobolev training of neural operators, which we will also refer to as *derivative-informed operator learning*, can lead to notable benefits when used to construct surrogates for outer-loop tasks involving PDEs (Bouhlel et al., 2020; Luo et al., 2025; Go and Chen, 2025a; Cao et al., 2025; Go and Chen, 2025b; Cao et al., 2026). For example, derivative-informed surrogates have been applied to PDE-constrained optimization, both deterministic and under uncertainty (Bouhlel et al., 2020; Luo et al., 2025), acceleration of geometric Markov Chain Monte Carlo and measure transport algorithms for Bayesian inverse problems (Cao et al., 2025, 2026), and Bayesian optimal experimental design (Go and Chen, 2025a,b). Due to the important roles played by the derivatives in these outer-loop tasks, the improved derivative accuracy in the surrogates provided by Sobolev training can be crucial to obtaining the observed performance gains.

Since the input and output spaces in operator learning can be arbitrarily high dimensional upon discretization, the discretized derivative (i.e., the Jacobian matrix) can be extremely expensive to compute. To address this, O'Leary-Roseberry et al. (2024), along with subsequent works (Luo

et al., 2025; Go and Chen, 2025a; Cao et al., 2025; Qiu et al., 2024; Go and Chen, 2025b; Cao et al., 2026), have leveraged the reduced basis neural operator (RBNO) architecture to enable scalable and efficient Sobolev training. That is, the operator surrogate uses linear encoding and decoding of the input and output spaces given in terms of reduced bases. These are constructed before training the neural network, which defines only the mapping between the reduced (latent) representations of the inputs and/or outputs. This choice of architecture allows the data generation and training costs associated with the derivatives to scale only with the intrinsic dimensionality of the mapping (i.e., rank of the reduced bases), as only the actions of the derivative within the reduced subspaces are needed in training.

In this approach, the selection of the reduced bases, or dimension reduction technique, plays an important role. A common approach is principal component analysis (PCA) on function space, which uses the dominant eigenvectors of the input and output covariance operators as reduced bases (Hesthaven and Ubbiali, 2018; Bhattacharya et al., 2021; Fresca and Manzoni, 2022). The active subspace (AS) method (Constantine et al., 2014; Zahm et al., 2020) has also been considered for input dimension reduction (O'Leary-Roseberry, Villa, et al., 2022; O'Leary-Roseberry, Du, et al., 2022). This approach uses the dominant eigenvectors of the covariance of the derivative operator's adjoint to construct a subspace of the input space to which the outputs are most sensitive.

In this work, our goal is to analyze the approximation error of reduced-basis architectures in the context of derivative-informed operator learning. Specifically, we focus on operators between (real-valued) infinite-dimensional separable Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, where the input space is equipped with a Gaussian measure γ. We are interested in studying not only the approximation error of the operator in the usual L^2_γ Bochner sense, but also the approximation error of the derivative. That is, we consider the approximations of operators in the H^1_γ Sobolev norm over infinite-dimensional Gaussian measures as defined in Malliavin calculus (Bogachev, 1998, 2010; Nualart, 2006), which is closely related to the classical Sobolev spaces for finite-dimensional inputs.

Since the choice of dimension reduction plays an important role in reduced-basis architectures, we investigate how different choices of reduced bases impact the overall approximation error. In addition to the PCA, we also consider a generalization of the AS method to both input and output dimension reduction, where the output dimension reduction uses the covariance of the derivative operator instead of its adjoint. We refer to this approach as the input/output derivative-informed subspaces (DIS). Our goal is then to obtain combined approximation error bounds in terms of both the dimension reduction and the neural network approximation, where we are additionally interested in the statistical properties of the dimension reduction errors due to the use of sample-based covariance estimators in computing the PCA and DIS.

1.1 Related Work

1.1.1 Approximation Theory for Operator Learning

Many of the existing works for analyzing the approximation errors of operator surrogates focus on the approximation of the operator's output values, typically providing error bounds in terms of L^p_γ Bochner norms or supremum norms over compact sets. In this context, the works of Bhattacharya et al. (2021); Lanthaler (2023) are most closely related to ours. These works establish approximation error bounds for PCANet—a reduced-basis architecture that uses PCA for dimension reduction—in which the error decomposes into dimension reduction errors and neural network approximation errors. Similar approximation error results have also been developed for the POD-DL-ROM architecture (Brivio et al., 2024), which uses the proper orthogonal decomposition for output dimension reduction (i.e., PCA), but considers the setting of low-dimensional parametric inputs. Additionally, we mention that the approximation error of linear encoder-decoder architectures for operator

learning has also been studied in Herrmann et al. (2024); Adcock et al. (2024a,b); Reinhardt et al. (2026); Adcock et al. (2025); Galimberti et al. (2026); Cheng et al. (2026), which additionally establish approximation rates with respect to the number of neural network parameters and/or training sample size under assumptions such as Lipschitz continuity, Fréchet differentiability, or holomorphy of the underlying operator.

In addition to the aforementioned works, approximation theoretic results also exist for a wide range of neural operator architectures, such as DeepONets (Lu et al., 2021; Lanthaler et al., 2022), Fourier neural operators (Kovachki et al., 2021), attention-based architectures (Alvarez et al., 2025; Calvello et al., 2025), and more general nonlinear architectures (Lanthaler et al., 2025). Again, universality and approximation rates (under suitable regularity assumptions) have been established in terms of L^p_γ or supremum norms over compact sets.

1.1.2 Sobolev Norm Approximation by Neural Networks

On the other hand, approximation results for operator surrogates involving the derivative accuracy in a Sobolev-type norm are much more limited, typically given in the finite-dimensional setting. The approximation accuracy of neural networks in finite dimensions is considered in works as early as that of Hornik (1991), which proves universal approximation results for functions in C^k (the space of k times continuously differentiable functions) by single-layer neural networks using smooth, sigmoid-like activation functions, with errors measured in either weighted Sobolev norms or C^k norms over compact sets. A more recent extension of this result is given by Neufeld and Schmocker (2025), which extends to single-layer neural networks with a broad class of possibly non-saturating activation functions.

We note that modern results for approximation in classical Sobolev norms are available (e.g., Gühring et al., 2019; De Ryck et al., 2021; Opschoor et al., 2021; Feng and Zeng, 2022; Kratsios et al., 2026), often with approximation rates for the size of the neural network under regularity assumptions. However, these results are typically presented in the context of Sobolev norms over compact input domains as opposed to weighted Sobolev norms over unbounded domains.

1.1.3 Gradient-based Dimension Reduction

Our analysis of dimension reduction via the DIS also draws on the works of Cui et al. (2014); Zahm et al. (2020); Lam et al. (2020); Cui and Tong (2022); Baptista et al. (2022); Chen et al. (2025). For Gaussian input measures, our input DIS coincides with the vector-valued active subspace (AS) of Zahm et al. (2020), which has an established bound on the L^2_γ error when used for input dimension reduction. The results of Zahm et al. (2020) have subsequently been applied to finite-dimensional neural network approximations in O'Leary-Roseberry, Villa, et al. (2022); O'Leary-Roseberry, Du, et al. (2022). Moreover, the DIS also coincides with the likelihood-informed subspace (LIS) in the context of input dimension reduction for Bayesian inverse problems (Cui et al., 2014; Zahm et al., 2022; Cui and Tong, 2022) as well as subsequent extensions to joint input and output dimension reduction (Baptista et al., 2022; Chen et al., 2025). Again, the aforementioned works typically considered finite-dimensional input/output spaces. Additionally, the input DIS (LIS) is considered for an infinite-dimensional input space but a finite-dimensional output space in Cao et al. (2025, 2026).

Sampling errors associated with PCA in Hilbert spaces are well studied (e.g., Blanchard et al., 2006; Mas and Ruymgaart, 2014; Reiß and Wahl, 2020). Analogous results are also used in the analyses of PCANet in Bhattacharya et al. (2021); Lanthaler (2023). Additionally, Lam et al. (2020); Cui and Tong (2022) consider the sampling errors associated with constructing the AS/LIS matrices, though in the finite-dimensional setting.

1.2 Contributions

In this work, we analyze the errors associated with approximating both an operator and its derivatives by an RBNO architecture. Our main contribution is a universal approximation result in the H^1_γ norm by RBNOs that decomposes the error into neural network errors, dimension reduction errors, and sampling errors, which we summarize in Theorem 13. The error contributions are then numerically studied in the context of learning the solution operator of elliptic PDEs with Gaussian random field inputs.

A key objective of our analysis is to understand how dimension reduction strategies such as PCA and DIS, which were originally developed based on their L^2_γ reconstruction properties, perform in terms of the H^1_γ approximation error, i.e., for both the operator and its derivatives. To this end, we make the following contributions:

- We formulate the input/output DIS (AS/LIS) in the infinite-dimensional Hilbert space setting based on the theory of Sobolev classes over infinite-dimensional Gaussian measures.
- For both PCA and DIS, we develop analogous results for the reconstruction errors and ridge function errors in the H^1_γ sense. We show that under suitable assumptions (Assumptions 8 to 11), both PCA and DIS can yield output and derivative errors that are bounded by the truncated eigenvalues of their respective covariance operators.
- We follow the approaches of Reiß and Wahl (2020); Bhattacharya et al. (2021); Lanthaler (2023) to provide bounds for the sampling errors associated with the DIS operators in the Hilbert space setting, taking the form of Monte Carlo rates with respect to the number of samples used in the covariance estimation.

This allows us to unify several earlier results that have been developed for various choices of input and output bases in either the finite- or infinite-dimensional settings. A comparison of the problem settings in several existing works is presented in Table 1.

We then combine our dimension reduction error bounds with finite-dimensional universal approximation results to obtain the approximation result for the RBNO architecture in the H^1_γ norm. To this end, we adapt the universal approximation results of Hornik (1991); Neufeld and Schmocker (2025) in Sobolev norms for single-layer neural networks, which we extend to the case of deep neural networks. We also provide regularity and latent-dimension-dependent approximation rates when using ReLU^k activation functions for the Gaussian-weighted Sobolev space, which are analogous to standard rates for ReLU networks in classical Sobolev norms over compact domains, e.g., by Gühring et al. (2019). While these extensions are conceptually straightforward, they allow us to obtain a full characterization of the RBNO error in practically relevant settings.

The remainder of this paper is organized as follows. In Section 2, we introduce the mathematical setup of derivative-informed operator learning with reduced-basis architectures over an infinite-dimensional Gaussian input measure, including the definition of Sobolev classes over Gaussian measures, the formulations for PCA and DIS, and the relevant universal approximation results for neural networks in finite-dimensional Sobolev norms. In Section 3, we present our approximation error results, providing bounds for the different sources of error—dimension reduction, neural network approximation, and sampling. In Section 4, we provide two PDE-based examples for which the assumptions used in the analysis can be verified and therefore the results of our main theorems can be applied. We then present a suite of numerical experiments in Section 5, where we validate our theoretical bounds while numerically comparing the effectiveness of PCA and DIS as dimension reduction strategies in derivative learning with RBNOs. Finally, we conclude in Section 6 with a summary of our findings and discussion of key limitations and potential future directions.

	Input dim.	Output dim.	Input red.	Output red.
Bhattacharya et al. (2021)	Infinite	Infinite	PCA	PCA
Lanthaler (2023)	Infinite	Infinite	PCA	PCA
Cui et al. (2014)	Finite	Finite	DIS	-
Zahm et al. (2022)	Finite	Finite	DIS	-
Cui and Tong (2022)	Finite	Finite	DIS	-
Zahm et al. (2020)	Finite	Finite	DIS, PCA	-
Lam et al. (2020)	Finite	Finite	DIS, PCA	-
O'Leary-Roseberry et al. (2024)	Finite	Finite	DIS, PCA	PCA
O'Leary-Roseberry, Du, et al. (2022)	Finite	Finite	DIS, PCA	PCA
Baptista et al. (2022)	Finite	Finite	DIS, PCA	DIS, PCA
Chen et al. (2025)	Finite	Finite	DIS, PCA	DIS, PCA
Cao et al. (2025)	Infinite	Finite	DIS, PCA	-
Cao et al. (2026)	Infinite	Finite	DIS, PCA	-
This work	Infinite	Infinite	DIS, PCA	DIS, PCA

Table 1: A summary of the function space settings (finite or infinite dimensions) and dimension reduction strategies (PCA or DIS) considered in a selection of relevant prior work. Our work considers both PCA and DIS in the infinite-dimensional setting and includes results for the approximation error in H^1_γ. Note that our use of the term DIS additionally includes the AS and LIS strategies.

2. Learning an Operator and its Derivatives

2.1 Mathematical Setup and Notation

Let $\mathcal{X}$ and $\mathcal{Y}$ be real separable Hilbert spaces with inner products $\langle\cdot,\cdot\rangle_{\mathcal{X}}$ and $\langle\cdot,\cdot\rangle_{\mathcal{Y}}$. We will consider the case where $\mathcal{X}$ is equipped with a Gaussian measure $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ defined over the Borel sigma algebra $\mathcal{B}(\mathcal{X})$, assumed to be centered. In this setting, the covariance operator $C_{\mathcal{X}} = \mathbb{E}_{x\sim\gamma}[x \otimes x]$ is a positive, self-adjoint, trace-class linear operator on $\mathcal{X}$, where we use $\otimes$ to denote the outer product, such that $(x \otimes x)v = \langle x, v\rangle_{\mathcal{X}}\, x$ for $x, v \in \mathcal{X}$. Additionally, let $\mathcal{E} = \text{Range}(C_{\mathcal{X}}^{1/2})$ be the Cameron–Martin space associated with $\mathcal{X}$, which itself is a separable Hilbert space with the inner product $\langle w_1, w_2\rangle_{\mathcal{E}} = \left\langle C_{\mathcal{X}}^{-1/2} w_1, C_{\mathcal{X}}^{-1/2} w_2 \right\rangle_{\mathcal{X}}$.

For a pair of separable Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, we will use $\mathcal{L}_k(\mathcal{H}_1, \mathcal{H}_2)$ to define the space of continuous k-linear operators, with $\mathcal{L}(\mathcal{H}_1, \mathcal{H}_2) = \mathcal{L}_1(\mathcal{H}_1, \mathcal{H}_2)$. For $k > 1$, we will write $A(h_1, \ldots, h_k) \in \mathcal{H}_2$ for the action of $A \in \mathcal{L}_k(\mathcal{H}_1, \mathcal{H}_2)$ on input directions $h_1, \ldots, h_k \in \mathcal{H}_1$. Moreover, we write $A(h_1, \ldots, h_{k-1}, \cdot) \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ as the linear operator defined by leaving one of the input directions open (in this case, the kth input direction). The operator norm for $A \in \mathcal{L}_k(\mathcal{H}_1, \mathcal{H}_2)$ is given by

$$\|A\|_{\mathcal{L}_k(\mathcal{H}_1,\mathcal{H}_2)} = \sup_{h_1,\ldots,h_k\neq 0} \frac{\|A(h_1, \ldots, h_k)\|_{\mathcal{H}_2}}{\|h_1\|_{\mathcal{H}_1} \cdots \|h_k\|_{\mathcal{H}_1}}.$$

Similarly, we will consider the space of k-linear Hilbert–Schmidt operators $\text{HS}_k(\mathcal{H}_1, \mathcal{H}_2)$, where the inner product generating the Hilbert–Schmidt norm $\|\cdot\|_{\text{HS}_k(\mathcal{H}_1,\mathcal{H}_2)}$ is given by

$$\langle A, B\rangle_{\text{HS}_k(\mathcal{H}_1,\mathcal{H}_2)} = \sum_{i_1,\ldots,i_k=1}^{\infty} \langle A(e_{i_1}, \ldots, e_{i_k}), B(e_{i_1}, \ldots, e_{i_k})\rangle_{\mathcal{H}_2},$$

with any orthonormal basis $(e_i)_{i=1}^{\infty}$ of the input space $\mathcal{H}_1$. Specifically, we write $\mathrm{HS}(\mathcal{H}_1, \mathcal{H}_2) = \mathrm{HS}_1(\mathcal{H}_1, \mathcal{H}_2)$ with inner product

$$\langle A, B\rangle_{\mathrm{HS}(\mathcal{H}_1,\mathcal{H}_2)} = \sum_{i=1}^{\infty} \langle Ae_i, Be_i\rangle_{\mathcal{H}_2} = \sum_{i=1}^{\infty} \langle e_i, A^*Be_i\rangle_{\mathcal{H}_1} = \mathrm{Tr}(A^*B)$$

for $A, B \in \mathrm{HS}(\mathcal{H}_1, \mathcal{H}_2)$, where $\mathrm{Tr}(\cdot)$ denotes the trace and A^* denotes the adjoint of A. Note that the Hilbert–Schmidt norm of A is simply $\sqrt{\mathrm{Tr}(A^*A)}$.

For any (real) separable Hilbert space $\mathcal{H}$, let $L^p(\gamma, \mathcal{H})$ denote the space of functions from $\mathcal{X}$ to $\mathcal{H}$ that are integrable in the Bochner sense with finite pth moments, $p \in [1, \infty)$, such that $\|F\|^p_{L^p(\gamma,\mathcal{H})} = \mathbb{E}_\gamma[\|F\|^p_{\mathcal{H}}] < \infty$ for $F \in L^p(\gamma, \mathcal{H})$. For $m \in \mathbb{N}_0$, we can then introduce the Sobolev class of functions $W^{m,p}(\gamma, \mathcal{H})$, which are functions with Sobolev derivatives up to order m that belong to L^p. Note that we are using the convention that $\mathbb{N}$ is the set of positive integers and $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ is the set of non-negative integers. The space $W^{m,p}(\gamma, \mathcal{H})$ has a number of equivalent definitions. Here, we describe the definition using the completion of smooth cylindrical functions, and refer to Bogachev (1998); Nualart (2006) for further details.

Let $\mathcal{FC}^\infty$ denote the collection of functions $F : \mathcal{X} \to \mathcal{H}$ that have the representation

$$F(x) = \sum_{i=1}^{l} \phi_i(\langle v_1, x\rangle_{\mathcal{X}}, \ldots, \langle v_n, x\rangle_{\mathcal{X}})u_i,$$

for some $l, n \in \mathbb{N}$, where $v_1, \ldots, v_n \in \mathcal{X}$, $u_1, \ldots, u_l \in \mathcal{H}$, and $\phi_1, \ldots, \phi_l \in C_b^\infty(\mathbb{R}^n, \mathbb{R})$, where $C_b^\infty(\mathbb{R}^n, \mathbb{R})$ refers to the space of smooth functions from $\mathbb{R}^n$ to $\mathbb{R}$ that are bounded and have bounded derivatives of all orders. The class $\mathcal{FC}^\infty$ is referred to as the class of smooth cylindrical functions. For $F \in \mathcal{FC}^\infty$, the restriction of the derivatives D^kF to directions in the Cameron–Martin space, denoted by $D^k_{\mathcal{E}}F(x) = D^kF(x)|_{\mathcal{E}}$, is Hilbert–Schmidt (i.e., $D^k_{\mathcal{E}}F(x) \in \mathrm{HS}_k(\mathcal{E}, \mathcal{H})$), and has bounded moments. We define the norm $\|\cdot\|_{W^{m,p}(\gamma,\mathcal{H})}$ as

$$\|F\|_{W^{m,p}(\gamma,\mathcal{H})} = \left(\sum_{k=0}^{m} \int \|D^k_{\mathcal{E}}F(x)\|^p_{\mathrm{HS}_k(\mathcal{E},\mathcal{H})}\, \gamma(dx)\right)^{\frac{1}{p}} = \left(\sum_{k=0}^{m} \|D^k_{\mathcal{E}}F\|^p_{L^p(\gamma,\mathrm{HS}_k(\mathcal{E},\mathcal{H}))}\right)^{\frac{1}{p}}. \tag{1}$$

The space $W^{m,p}(\gamma, \mathcal{H})$ is then defined as the completion of $\mathcal{FC}^\infty$ under $\|\cdot\|_{W^{m,p}}$. For functions $F \in W^{m,p}(\gamma, \mathcal{H})$, we define their Sobolev derivatives $D_{\mathcal{E}}F$ as the unique limit of sequences $(D_{\mathcal{E}}F_i)_{i=1}^{\infty}$ for $F_i \in \mathcal{FC}^\infty$, $i \in \mathbb{N}$ that converges to F in $W^{m,p}(\gamma, \mathcal{H})$. Notably, the Sobolev derivatives behave as the stochastic Gâteaux derivatives (Bogachev, 1998, Chapter 5). That is, for $F \in W^{1,p}(\gamma, \mathcal{H})$, the residual

$$\frac{F(x + th) - F(x)}{t} - D_{\mathcal{E}}F(x)h$$

converges to zero in measure γ as $t \to 0$ for any $h \in \mathcal{E}$. Analogous results hold for higher-order derivatives.

In this work, we focus on the case $p = 2$, for which we introduce the notation $H^m(\gamma, \mathcal{H}) := W^{m,2}(\gamma, \mathcal{H})$, along with the shorthand $L^2_\gamma := L^2(\gamma, \mathcal{Y})$, $H^m_\gamma := H^m(\gamma, \mathcal{Y})$, and $W^{m,p}_\gamma := W^{m,p}(\gamma, \mathcal{Y})$ specifically when the output space is $\mathcal{Y}$. Moreover, we will often consider mappings F that are additionally twice continuously Fréchet differentiable, $F \in C^2(\mathcal{X}, \mathcal{Y})$, where the first and second derivatives have bounded second moments, i.e., $DF \in L^2(\gamma, \mathcal{L}(\mathcal{X}, \mathcal{Y}))$ and $D^2F \in L^2(\gamma, \mathcal{L}_2(\mathcal{X}, \mathcal{Y}))$. Note that Fréchet differentiability is a stronger notion of differentiability than the Sobolev differentiability that defines $W^{m,p}(\gamma, \mathcal{Y})$. Since the inclusion map $\iota : \mathcal{E} \to \mathcal{X}$ is in $\mathrm{HS}(\mathcal{E}, \mathcal{X})$, the restriction of an operator $A \in \mathcal{L}(\mathcal{X}, \mathcal{Y})$, $A|_{\mathcal{E}} = A \circ \iota$, is also in $\mathrm{HS}(\mathcal{E}, \mathcal{Y})$. In particular, this implies $DF|_{\mathcal{E}} = D_{\mathcal{E}}F$

and its adjoint with respect to $(\mathcal{E}, \mathcal{Y})$ is given by $D_{\mathcal{E}}F(x)^* = C_{\mathcal{X}}DF(x)^*$, where $DF(x)^*$ is the adjoint of $DF(x)$ defined on $\mathcal{L}(\mathcal{X}, \mathcal{Y})$. Thus, these assumptions on Fréchet differentiability would imply that $F \in H^2(\gamma, \mathcal{Y})$.

For such F, we aim to analyze the approximation accuracy of operator surrogates $\widetilde{F} \approx F$, quantified in terms of both the function value and its derivatives. We will measure the function value in the $L^2_\gamma = L^2(\gamma, \mathcal{Y})$ norm, and the derivative with the seminorm $|\cdot|_{H^1_\gamma}$ defined by

$$|F|^2_{H^1_\gamma} := \|D_{\mathcal{E}}F\|^2_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}]. \tag{2}$$

That is, we consider overall approximation error estimates in the H^1_γ norm,

$$\|F\|_{H^1_\gamma} = \left(\|F\|^2_{L^2_\gamma} + |F|^2_{H^1_\gamma}\right)^{1/2}. \tag{3}$$

This H^1_γ norm, despite restricting differentiation directions to the Cameron–Martin space $\mathcal{E}$, is a natural choice for measuring the derivative accuracy of operator surrogates when the input measure is Gaussian. For example, it plays a direct role in the accuracy and efficiency of several surrogate-based methods for Bayesian inference under Gaussian priors (Cao et al., 2025, 2026).

2.2 Operator Surrogates using Deep Neural Networks

In this work, we consider operator surrogates based on feedforward neural networks. For an activation function $\psi : \mathbb{R} \to \mathbb{R}$, we say $\varphi_{\psi,\theta} : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ is a neural network with activation function ψ and parameters θ if it takes the form

$$\begin{aligned} z_0 &= x, \\ z_i &= \psi(\mathbf{W}_i z_{i-1} + \mathbf{b}_i), \quad i = 1, \dots, d_L, \\ \varphi_{\psi,\theta}(x) &= \mathbf{W}_{d_L+1} z_{d_L} + \mathbf{b}_{d_L+1}, \end{aligned}$$

where $\theta = (\mathbf{W}_i, \mathbf{b}_i)_{i=1}^{d_L+1}$ denotes the parameters of the neural network, consisting of weights $\mathbf{W}_1 \in \mathbb{R}^{d_{W,1} \times d_{\text{in}}}$, $\mathbf{W}_i \in \mathbb{R}^{d_{W,i} \times d_{W,i-1}}$ for $i = 2, \dots, d_L$, $\mathbf{W}_{d_L+1} \in \mathbb{R}^{d_{\text{out}} \times d_{W,d_L}}$, and biases $\mathbf{b}_i \in \mathbb{R}^{d_{W,i}}$, for $i = 1, \dots, d_L$ and $\mathbf{b}_{d_L+1} \in \mathbb{R}^{d_{\text{out}}}$. We define the depth of the neural network as the number of hidden layers, and the width of the neural network as the maximum width of its hidden layers, i.e.,

$$\mathrm{depth}(\varphi_{\psi,\theta}) = d_L, \qquad \mathrm{width}(\varphi_{\psi,\theta}) = \max_{i=1,\dots,d_L} d_{W,i}.$$

We say that $\varphi_{\psi,\theta}$ has constant width if $d_{W,i} = d_W$ for all $i = 1, \dots, d_L$, and note that any neural network can be transformed into a constant-width neural network by appropriately padding its weights and biases with zeros. Additionally, we define the size of the neural network as the total number of non-zero parameters, i.e.,

$$\mathrm{size}(\varphi_{\psi,\theta}) = \sum_{i=1}^{d_L+1} \left(\|\mathbf{W}_i\|_0 + \|\mathbf{b}_i\|_0\right),$$

where $\|\cdot\|_0$ denotes the number of non-zero entries in a matrix or vector.

Our focus will be on activation functions that are smooth and non-saturating with at most polynomial growth. We then define the class of activation functions $\mathbb{A}^m$ for $m \in \mathbb{N}_0$ as follows.

Definition 1 (Non-polynomial functions with bounded derivative growth)**.** *Given $m \in \mathbb{N}_0$, let $C^{m,1}(\mathbb{R})$ denote the space of functions $\psi : \mathbb{R} \to \mathbb{R}$ such that $\psi \in C^m(\mathbb{R})$ and its mth derivative $\psi^{(m)}$ is globally Lipschitz continuous. We define the class of activation functions as*

$$\mathbb{A}^m := \{\psi : \psi \in C^{k,1}(\mathbb{R}) \text{ for some } k \in \mathbb{N}_0,\ k \geq m \text{ and } \psi \text{ is non-polynomial}\}.$$

This includes commonly used smooth activation functions in machine learning such as softplus, the Gaussian-error linear unit (GELU), and the sigmoid linear unit (SiLU). It also includes ReLU^k activation functions, i.e., $\mathrm{ReLU}^k(t) = \max(0,t)^k$ for $k \in \mathbb{N}, k \geq m+1$. In this work, we will consider ReLU^k activation functions with integer powers k only.

With activation functions $\psi \in \mathbb{A}^m$, we can show that deep neural networks with $d_L \geq 1$ layers are universal approximators in Sobolev norms. Specifically, we provide an extension to the result of Hornik (1991) for functions on $\mathbb{R}^d$ over finite measures. Given a finite measure ν over $\mathbb{R}^{d_{\text{in}}}$, $m \in \mathbb{N}_0$, and $p \in [1,\infty)$, we follow Hornik (1991) to define the class of functions

$$C_\nu^{m,p}(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}}) := \{f \in C^m(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}}) : \|f\|_{W^{m,p}(\nu,\mathbb{R}^{d_{\text{out}}})} < \infty\},$$

where the $W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})$ norm is defined as

$$\|f\|_{W^{m,p}(\nu,\mathbb{R}^{d_{\text{out}}})} = \left(\int_{\mathbb{R}^{d_{\text{in}}}} \|f(x)\|_2^p d\nu(x) + \sum_{1 \leq |\alpha| \leq m} \int_{\mathbb{R}^{d_{\text{in}}}} \|\partial^\alpha f(x)\|_2^p d\nu(x) \right)^{\frac{1}{p}}.$$

Here, $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_{d_{\text{in}}}) \in \mathbb{N}_0^{d_{\text{in}}}$ is a multi-index, $\partial^\alpha = \partial_{x_1}^{\alpha_1} \ldots \partial_{x_{d_{\text{in}}}}^{\alpha_{d_{\text{in}}}}$ denotes the partial derivatives with order α_i in coordinate x_i, $|\alpha| := \sum_{i=1}^{d_{\text{in}}} \alpha_i$ denotes the total order of differentiation, and $\|\cdot\|_2$ refers to the standard Euclidean norm on $\mathbb{R}^{d_{\text{out}}}$. Note that this definition of the $W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})$ norm is equivalent to our earlier definition given in (1) when $\nu = \gamma$ is a Gaussian over the finite-dimensional space $\mathbb{R}^{d_{\text{in}}}$ and $\mathcal{Y} = \mathbb{R}^{d_{\text{out}}}$ is also finite-dimensional. The original result of Hornik (1991) then states that single-layer neural networks with smooth and bounded activation functions $\psi \in C_b^m(\mathbb{R})$ are universal approximators of the class of functions $C_\nu^{m,p}(\mathbb{R}^d, \mathbb{R})$ in the $W^{m,p}(\nu, \mathbb{R})$ norm (we restate this in Theorem 55).

In practice, deep neural networks with non-saturating activation functions are often preferred. To this end, Neufeld and Schmocker (2025) provide a universal approximation result in Sobolev spaces over unbounded domains by single-layer neural networks with very general non-saturating activation functions that include the class $\mathbb{A}^m$. For deep neural networks (i.e., multiple hidden layers), one can verify that this universality is maintained when using $\psi \in \mathbb{A}^m$. While this extension is straightforward, it is not typically stated in the literature, and thus we present it here for completeness. The proof of the following result is presented in Appendix B.3 and is directly based on the original result of Hornik (1991) or equivalently that of Neufeld and Schmocker (2025).

Theorem 2 (Deep universal approximation with $\mathbb{A}^m$ activation functions)**.** *Let $m \in \mathbb{N}_0$, $p \in [1,\infty)$, $\psi \in \mathbb{A}^m$, and suppose $g \in C_\nu^{m,p}(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}})$ where ν is a finite measure on $\mathbb{R}^{d_{\text{in}}}$. Then, for any $\epsilon > 0$ and $d_L \in \mathbb{N}$, there exists a d_L-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ such that*

$$\|g - \varphi_{\psi,\theta}\|_{W^{m,p}(\nu,\mathbb{R}^{d_{\text{out}}})} \leq \epsilon.$$

Note that since $\mathbb{A}^m$ only imposes Lipschitz continuity on the highest derivative, neural networks with activation functions in $\mathbb{A}^m$ are not guaranteed to have bounded $W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})$ norms in general. Instead, the proof is based on constructing a neural network with $\psi \in \mathbb{A}^m$ that belongs to $C_b^m(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}})$, which does have bounded $W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})$ norm. Nonetheless, for measures such as Gaussians, which have finite moments of arbitrary order, neural networks with activation functions $\psi \in \mathbb{A}^m$ can be shown to belong to $W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})$ since Lipschitz continuity of $\psi^{(k)}$ for some $k \geq m$ implies that ψ and its derivatives up to order m have at most polynomial growth.

For Gaussian measures γ on finite-dimensional spaces $\mathbb{R}^{d_{\text{in}}}$, we can also obtain approximation rates in the Gaussian-weighted Sobolev norm H_γ^m with ReLU^k neural networks (i.e., $\psi = \mathrm{ReLU}^k$) when the underlying function belongs to a smoother Sobolev space, say H_γ^l with $l > m$.

Theorem 3 (Sobolev approximation rates for ReLU^k networks)**.** *Let $d_{\text{in}}, d_{\text{out}} \in \mathbb{N}$, $k \in \mathbb{N}$ with $k \geq 2$, $l, m \in \mathbb{N}_0$ with $l > m$, and let $\gamma = \gamma_{d_{\text{in}}} = \mathcal{N}(0, I_{d_{\text{in}}})$ be the standard Gaussian measure on $\mathbb{R}^{d_{\text{in}}}$. Consider a function $g \in H^l_\gamma = H^l(\gamma_{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}})$. Then, for any $\epsilon > 0$, there exists a* ReLU^k *neural network $\varphi_{\psi,\theta}$ such that*

$$\|g - \varphi_{\psi,\theta}\|_{H^m_\gamma} \leq \epsilon,$$

where the neural network $\varphi_{\psi,\theta}$ can be chosen to have depth and size satisfying

$$\text{depth}(\varphi_{\psi,\theta}) \leq \kappa_1 \left(\log(\epsilon^{-1}\|g\|_{H^l_\gamma} + 1) + 1 \right), \qquad \text{size}(\varphi_{\psi,\theta}) \leq \kappa_2 \left(\left(\epsilon^{-1}\|g\|_{H^l_\gamma} \right)^{\frac{2d_{\text{in}}}{l-m}} + 1 \right).$$

Here, $\kappa_1 = \kappa_1(d_{\text{in}}, k, l, m)$ and $\kappa_2 = \kappa_2(d_{\text{in}}, d_{\text{out}}, k, l, m)$ are constants.

The proof of this result uses standard techniques (Canuto and Quarteroni, 1982; DeVore, 1998; DeVore et al., 2021; Opschoor et al., 2021), but we present it in Appendix B.4 since it is not typically stated in the literature for Gaussian Sobolev spaces. Similar to Theorem 2, although general ReLU^k networks are not guaranteed to belong to H^m_γ, the proof of Theorem 3 is based on a specific construction of ReLU^k networks that exactly represent polynomial approximants and hence do belong to H^m_γ. Note that we have explicitly included $\|g\|_{H^l_\gamma}$ in the above expressions to emphasize the dependence on the regularity of g.

2.3 Reduced Basis Neural Operators

Due to the high-dimensionality of the spaces $\mathcal{X}$ and $\mathcal{Y}$, a naive feedforward neural network may consist of a large number of weights and biases, and can be prone to overfitting (O'Leary-Roseberry, Villa, et al., 2022). Instead, more appropriate architectures have been considered for the approximation of PDE solution maps, such as DeepONets, FNOs, and reduced-basis architectures such as PCANet, POD-DL-ROM, and DIPNet. We will specifically focus on reduced-basis architectures, which are encoder-decoder architectures that use a neural network to approximate the mapping F within suitably-chosen $r_{\mathcal{X}}$- and $r_{\mathcal{Y}}$-dimensional subspaces of $\mathcal{X}$ and $\mathcal{Y}$, respectively. This allows the neural network to be constructed as a mapping between the low-dimensional latent spaces, reducing the required dimensionality of its parameters (and thereby the risk of overfitting) as well as the training cost.

To this end, let us first introduce linear encoder and decoder operators $\mathfrak{E}_{\mathcal{X}} \in \mathcal{L}(\mathcal{X}, \mathbb{R}^r)$ and $\mathfrak{D}_{\mathcal{X}} \in \mathcal{L}(\mathbb{R}^r, \mathcal{X})$ on the input space and $\mathfrak{E}_{\mathcal{Y}} \in \mathcal{L}(\mathcal{Y}, \mathbb{R}^r)$ and $\mathfrak{D}_{\mathcal{Y}} \in \mathcal{L}(\mathbb{R}^r, \mathcal{Y})$ on the output space. Note that for clarity of presentation, we will assume $r_{\mathcal{X}} = r_{\mathcal{Y}} = r$ in our analysis. Cases with $r_{\mathcal{X}} \neq r_{\mathcal{Y}}$ follow as corollaries. We will then consider neural operator surrogates of the form

$$\widetilde{F}_\theta = \mathfrak{D}_{\mathcal{Y}} \circ \varphi_{\psi,\theta} \circ \mathfrak{E}_{\mathcal{X}} + b$$

where $b \in \mathcal{Y}$ is a bias term, and $\varphi_{\psi,\theta} : \mathbb{R}^r \to \mathbb{R}^r$ is a neural network that maps between the latent spaces.

In the reduced-basis architecture, the encoder and decoder operators are defined using orthonormal bases. For the output space, we consider r orthonormal basis vectors in $\mathcal{Y}$, $U_r = (u_i)_{i=1}^r$. The corresponding decoder for this reduced basis is given by $\mathfrak{D}_{\mathcal{Y}}(\xi) = U_r\xi = \sum_{i=1}^r \xi_i u_i$ for $\xi = (\xi_1, \ldots, \xi_r) \in \mathbb{R}^r$, where we have overloaded the notation U_r to additionally denote the decoding operator $U_r = \mathfrak{D}_{\mathcal{Y}} \in \mathcal{L}(\mathbb{R}^r, \mathcal{Y})$. Similarly, for the input space, we consider r orthonormal basis vectors in the Cameron–Martin space $\mathcal{E} \subset \mathcal{X}$, $V_r = (v_i)_{i=1}^r$, and analogously write the decoder as $\mathfrak{D}_{\mathcal{X}}(\xi) = V_r\xi = \sum_{i=1}^r \xi_i v_i$. The corresponding encoders $\mathfrak{E}_{\mathcal{X}} = V_r^\dagger$ and $\mathfrak{E}_{\mathcal{Y}} = U_r^\dagger$ can then be defined as left inverses of the decoders. Note that we will also refer to the size of the reduced basis r as its rank. Evidently, the approximation properties of $\widetilde{F}_\theta$ depend on the choices of the bases U_r and V_r.

Remark 4 (Comment on notation)**.** *In the remainder of the paper, unless otherwise specified, we will use λ, u, U, P to denote eigenvalues, eigenvectors, bases/decoders, and projections on the output space $\mathcal{Y}$, and μ, v, V, and Q for that on the input space $\mathcal{X}$ or the Cameron–Martin space $\mathcal{E}$.*

2.4 Sobolev Training of Reduced Basis Neural Operators

To construct operator surrogates that are accurate in the H^1_γ norm, one can pose the supervised learning problem directly in the H^1_γ norm. That is, for a neural operator architecture parametrized by θ, one solves the following minimization problem,

$$\min_\theta \|F - \widetilde{F}_\theta\|^2_{H^1_\gamma} = \mathbb{E}_\gamma[\|F - \widetilde{F}_\theta\|^2_{\mathcal{Y}}] + \mathbb{E}_\gamma[\|D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_\theta\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}]. \tag{4}$$

The incorporation of derivatives in the training loss is termed Sobolev training, and its application to operator learning leads to the so-called derivative-informed neural operators. When using a reduced basis architecture of the form $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + b$, where U_r and V_r are fixed orthonormal bases of $\mathcal{Y}$ and $\mathcal{E}$, the H^1_γ training problem can be equivalently written in the latent/reduced spaces as

$$\min_\theta \mathbb{E}_{x\sim\gamma}[\|U_r^\dagger(F(x) - b) - \varphi_{\psi,\theta}(V_r^\dagger x)\|_2^2] + \mathbb{E}_{x\sim\gamma}[\|U_r^\dagger D_{\mathcal{E}}F(x)V_r - D\varphi_{\psi,\theta}(V_r^\dagger x)\|_F^2], \tag{5}$$

where $\|\cdot\|_2$ refers to the ℓ^2 norm on $\mathbb{R}^r$ and $\|\cdot\|_F$ refers to the matrix Frobenius norm on $\mathbb{R}^{r\times r}$.

In practice, the training problem is solved using a finite number of training samples, which are tuples of the form $(x_k, F(x_k), U_r^\dagger D_{\mathcal{E}}F(x_k)V_r)$, $k = 1, \dots, N$, where x_k are sampled from the input measure γ. Moreover, the training loss is often normalized to balance the contributions of the output and derivative terms. This leads to the following practical form of the empirical risk minimization problem,

$$\min_\theta \frac{1}{N}\left(\sum_{k=1}^N \frac{1}{a_{0,k}}\|U_r^\dagger(F(x_k) - b) - \varphi_{\psi,\theta}(V_r^\dagger x_k)\|_2^2 + \frac{1}{a_{1,k}}\|U_r^\dagger D_{\mathcal{E}}F(x_k)V_r - D\varphi_{\psi,\theta}(V_r^\dagger x_k)\|_F^2\right), \tag{6}$$

where $(a_{0,k})_{k=1}^N$ and $(a_{1,k})_{k=1}^N$ are positive weights scaling the output and derivative terms, respectively. Common choices of normalization include uniformly normalizing by the second moments of the training data, $a_{0,k} = \sum_{i=1}^N \|U_r^\dagger(F(x_i) - b)\|_2^2$ and $a_{1,k} = \sum_{i=1}^N \|U_r^\dagger D_{\mathcal{E}}F(x_i)V_r\|_F^2$, or normalizing each data point by its norm, $a_{0,k} = \|U_r^\dagger(F(x_k) - b)\|_2^2 + \tau$ and $a_{1,k} = \|U_r^\dagger D_{\mathcal{E}}F(x_k)V_r\|_F^2 + \tau$, where, if required, τ can be chosen to be a small constant to avoid division by zero.

Notably, the derivative-informed training problem requires evaluations of both the operator's output values and its derivatives at the sampled points. However, since the reduced-basis architecture only maps between the subspaces spanned by U_r and V_r, only the projection of the derivative $U_r^\dagger D_{\mathcal{E}}F(x)V_r$ is needed. For PDE-based mappings, this can be efficiently computed using forward/adjoint sensitivity methods (see, for example, O'Leary-Roseberry et al., 2024), making the Sobolev training of RBNOs computationally feasible.

In contrast, conventional supervised learning of operator surrogates typically minimizes the error in the L^2_γ norm. That is,

$$\min_\theta \|F - \widetilde{F}_\theta\|^2_{L^2_\gamma} = \mathbb{E}_\gamma[\|F - \widetilde{F}_\theta\|^2_{\mathcal{Y}}],$$

or in the practical setting with the reduced-basis architecture, normalization, and finite training samples,

$$\min_\theta \frac{1}{N}\sum_{k=1}^N \frac{1}{a_{0,k}}\|U_r^\dagger(F(x_k) - b) - \varphi_{\psi,\theta}(V_r^\dagger x_k)\|_2^2, \tag{7}$$

where the output data can be normalized in the same way as described above for the Sobolev training. While this can be computationally cheaper than H^1_γ Sobolev training, L^2_γ training does not explicitly control the derivative error, and thus may lead to surrogates that are inaccurate in the H^1_γ norm.

2.5 Choices of Dimension Reduction

2.5.1 Principal Component Analysis

We now consider choices of reduced bases V_r and U_r for the input and output spaces, respectively. One approach is to use subspaces spanned by the PCA bases. For the input space, the PCA coincides with the Karhunen Loève expansion (KLE) of the Gaussian measure γ. Since we have assumed γ is centered, this simply corresponds to the eigenvectors of the covariance operator,

$$C_{\mathcal{X}} v_i^{\text{PCA}} = \mu_i^{\text{PCA}} v_i^{\text{PCA}},$$

where the eigenvalues are written in decreasing order $\mu_1^{\text{PCA}} \geq \cdots \geq \mu_r^{\text{PCA}} \geq \cdots > 0$. Since $C_{\mathcal{X}}$ is self-adjoint and of trace class, its full collection of eigenvectors, $(v_i)_{i=1}^{\infty}$, yields an orthonormal basis in $\mathcal{X}$. Alternatively, they can be normalized in the Cameron–Martin norm such that they are also an orthonormal basis in $\mathcal{E}$. For consistency, we will take this latter approach.

To reduce to an r-dimensional subspace, we take the first r eigenvectors as a reduced basis, $V_r^{\text{PCA}} := (v_i^{\text{PCA}})_{i=1}^r$. As previously described, we overload V_r^{PCA} to denote the decoder $V_r^{\text{PCA}} \in \mathcal{L}(\mathbb{R}^r, \mathcal{X})$ along with the corresponding encoder (its left inverse) $(V_r^{\text{PCA}})^\dagger \in \mathcal{L}(\mathcal{X}, \mathbb{R}^r)$ such that for $\xi = (\xi_1, \ldots, \xi_r) \in \mathbb{R}^r$ and $x \in \mathcal{X}$, we have

$$V_r^{\text{PCA}} \xi = \sum_{i=1}^r \xi_i v_i^{\text{PCA}}, \qquad (V_r^{\text{PCA}})^\dagger x = \left(\left\langle C_{\mathcal{X}}^{-1} v_1^{\text{PCA}}, x \right\rangle_{\mathcal{X}}, \ldots, \left\langle C_{\mathcal{X}}^{-1} v_r^{\text{PCA}}, x \right\rangle_{\mathcal{X}} \right).$$

In particular, when restricted to elements in the Cameron–Martin space $h \in \mathcal{E}$, we have

$$(V_r^{\text{PCA}})^\dagger h = \left(\left\langle v_1^{\text{PCA}}, h \right\rangle_{\mathcal{E}}, \ldots, \left\langle v_r^{\text{PCA}}, h \right\rangle_{\mathcal{E}} \right).$$

Thus,

$$Q_r^{\text{PCA}} := V_r^{\text{PCA}} (V_r^{\text{PCA}})^\dagger = \sum_{i=1}^r v_i^{\text{PCA}} \otimes C_{\mathcal{X}}^{-1} v_i^{\text{PCA}} = \sum_{i=1}^r v_i^{\text{PCA}} \otimes v_i^{\text{PCA}} / \mu_i^{\text{PCA}}$$

defines an orthogonal projection on $\mathcal{X}$, and its restriction to $\mathcal{E}$, $Q_r^{\text{PCA}}|_{\mathcal{E}} = \sum_{i=1}^r v_i^{\text{PCA}} \otimes_{\mathcal{E}} v_i^{\text{PCA}}$, is also an orthogonal projection on $\mathcal{E}$. Here, we have used $\otimes$ to denote the outer product implied by the space of its arguments and use the subscript $\otimes_{\mathcal{E}}$ to specifically denote the outer product given by the Cameron–Martin inner product, i.e., $v_1 \otimes v_2 = v_1 \langle v_2, \cdot \rangle_{\mathcal{X}}$ and $v_1 \otimes_{\mathcal{E}} v_2 = v_1 \langle v_2, \cdot \rangle_{\mathcal{E}}$.

In this work, we will assume that the covariance operator $C_{\mathcal{X}}$ is known and can be accessed. This is an appropriate assumption in the setting of operator learning for surrogate modeling, where the training distribution $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ is often known a priori, and the covariance operator $C_{\mathcal{X}}$ is used to draw samples $x \sim \gamma$ to generate the training data.

Analogously, one can consider a PCA of the output space $\mathcal{Y}$. In this work, we will focus on the PCA obtained with the centered covariance

$$C_{\mathcal{Y}} = \mathbb{E}_\gamma[(F - \bar{F}) \otimes (F - \bar{F})],$$

which leads to favorable properties for surrogate modeling compared to the uncentered covariance. We discuss this fact in Section 2.5.3. The output PCA basis is then obtained by solving the eigenvalue problem

$$C_{\mathcal{Y}} u_i^{\text{PCA}} = \lambda_i^{\text{PCA}} u_i^{\text{PCA}},$$

with $\lambda_1^{\mathrm{PCA}} \geq \cdots \geq \lambda_r^{\mathrm{PCA}} \geq 0$. We can then take the dominant eigenvectors, $U_r^{\mathrm{PCA}} := (u_i^{\mathrm{PCA}})_{i=1}^r$, as the output basis, such that the decoding and encoding operators are

$$U_r^{\mathrm{PCA}}\xi = \sum_{i=1}^r \xi_i u_i^{\mathrm{PCA}}, \qquad (U_r^{\mathrm{PCA}})^\dagger y = \left(\left\langle u_1^{\mathrm{PCA}}, y\right\rangle_{\mathcal{Y}}, \ldots, \left\langle u_r^{\mathrm{PCA}}, y\right\rangle_{\mathcal{Y}}\right).$$

Unlike the covariance of the inputs (which is assumed to be known), this covariance is generally not known, and in practice, is typically approximated empirically from samples. That is,

$$\widehat{C}_{\mathcal{Y}} = \frac{1}{N}\sum_{k=1}^N (F(x_k) - \widehat{F}) \otimes (F(x_k) - \widehat{F}) = \frac{1}{N}\sum_{k=1}^N F(x_k) \otimes F(x_k) - \widehat{F} \otimes \widehat{F},$$

where $\widehat{F} = \frac{1}{N}\sum_{k=1}^N F(x_k)$ is an estimator of the mean using N i.i.d. samples from γ, denoted as $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$. We note that this covariance estimator is biased, where the bias decreases with sample size. We will write the corresponding eigenvalue problem using the covariance estimator as

$$\widehat{C}_{\mathcal{Y}}\widehat{u}_i^{\mathrm{PCA}} = \widehat{\lambda}_i^{\mathrm{PCA}}\widehat{u}_i^{\mathrm{PCA}},$$

again with $\widehat{\lambda}_1^{\mathrm{PCA}} \geq \cdots \geq \widehat{\lambda}_r^{\mathrm{PCA}} \geq 0$. Thus, we have $\widehat{U}_r^{\mathrm{PCA}} := (\widehat{u}_i^{\mathrm{PCA}})_{i=1}^r$ and

$$\widehat{U}_r^{\mathrm{PCA}}\xi = \sum_{i=1}^r \xi_i \widehat{u}_i^{\mathrm{PCA}}, \qquad (\widehat{U}_r^{\mathrm{PCA}})^\dagger y = \left(\left\langle \widehat{u}_1^{\mathrm{PCA}}, y\right\rangle_{\mathcal{Y}}, \ldots, \left\langle \widehat{u}_r^{\mathrm{PCA}}, y\right\rangle_{\mathcal{Y}}\right),$$

as the decoder and encoder corresponding to the empirical output PCA basis, defined in the same manner as before. The projection operators

$$P_r^{\mathrm{PCA}} := U_r^{\mathrm{PCA}}(U_r^{\mathrm{PCA}})^\dagger \quad \text{and} \quad \widehat{P}_r^{\mathrm{PCA}} := \widehat{U}_r^{\mathrm{PCA}}(\widehat{U}_r^{\mathrm{PCA}})^\dagger$$

are both orthogonal projections on $\mathcal{Y}$.

The input and output PCA bases minimize the mean squared reconstruction errors of the inputs x and outputs $F(x)$ measured in $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{Y}}$, respectively. We will state this result in Section 3. However, neither the input nor the output PCA bases explicitly capture the sensitivity of the map F. In particular, the input PCA basis is independent of F, i.e., it does not account for any dependence of the outputs on the inputs. On the other hand, the output PCA basis implicitly captures the dependence of the output with respect to the input, as it is constructed using the dominant features of $F(x)$. As we will see, this has implications for the approximation errors of the derivatives.

2.5.2 Derivative-Informed Subspaces

We also consider an alternative set of reduced bases that are constructed directly using the derivatives of the map F. To this end, we define operators $H_{\mathcal{X}} : \mathcal{E} \to \mathcal{E}$ and $H_{\mathcal{Y}} : \mathcal{Y} \to \mathcal{Y}$, where

$$H_{\mathcal{X}} := \mathbb{E}_{x\sim\gamma}[D_{\mathcal{E}}F(x)^* D_{\mathcal{E}}F(x)] = C_{\mathcal{X}}\mathbb{E}_{x\sim\gamma}[DF(x)^* DF(x)], \tag{8}$$
$$H_{\mathcal{Y}} := \mathbb{E}_{x\sim\gamma}[D_{\mathcal{E}}F(x) D_{\mathcal{E}}F(x)^*] = \mathbb{E}_{x\sim\gamma}[DF(x) C_{\mathcal{X}} DF(x)^*], \tag{9}$$

where the latter equalities in (8) and (9) hold if we assume $F \in C^1(\mathcal{X}, \mathcal{Y})$ with bounded second moments on the derivatives, i.e., $DF \in L^2(\gamma, \mathcal{L}(\mathcal{X}, \mathcal{Y}))$, such that $DF^*DF \in L^1(\gamma, \mathcal{L}(\mathcal{X}, \mathcal{X}))$. More generally, when $F \in H^1_\gamma$, the sensitivity operators $H_{\mathcal{X}} : \mathcal{E} \to \mathcal{E}$ and $H_{\mathcal{Y}} : \mathcal{Y} \to \mathcal{Y}$ are both self-adjoint, positive semi-definite, trace-class operators. Thus, their eigenvectors form orthonormal

bases of $\mathcal{E}$ and $\mathcal{Y}$. The reduced bases are then given by the dominant eigenvectors of $H_{\mathcal{X}}$ and $H_{\mathcal{Y}}$. In this work, we will refer to $H_{\mathcal{X}}$ as the input sensitivity operator and $H_{\mathcal{Y}}$ as the output sensitivity operator, and refer to the subspaces generated by this approach as the input/output derivative-informed subspaces.

For the input, we consider the eigenvalue problem

$$H_{\mathcal{X}} v_i^{\text{DIS}} = \mu_i^{\text{DIS}} v_i^{\text{DIS}}, \tag{10}$$

with $\mu_1^{\text{DIS}} \geq \mu_2^{\text{DIS}} \geq \cdots \geq 0$, and take the first r eigenvectors $V_r^{\text{DIS}} := (v_i^{\text{DIS}})_{i=1}^r$ as the input basis/decoder, i.e., we recognize that v_i^{DIS} are orthonormal in $\mathcal{E}$, and so for any $x \in \mathcal{X}$, we write the decoder and encoder as

$$V_r^{\text{DIS}} \xi = \sum_{i=1}^r \xi_i v_i^{\text{DIS}}, \qquad (V_r^{\text{DIS}})^\dagger x := \left(\left\langle C_{\mathcal{X}}^{-1} v_1^{\text{DIS}}, x \right\rangle_{\mathcal{X}}, \ldots, \left\langle C_{\mathcal{X}}^{-1} v_r^{\text{DIS}}, x \right\rangle_{\mathcal{X}} \right),$$

where the encoder is well-defined in $\mathcal{L}(\mathcal{X}, \mathbb{R}^r)$ whenever $C_{\mathcal{X}}^{-1} v_i^{\text{DIS}}$ is well-defined in $\mathcal{X}$ (i.e., $C_{\mathcal{X}}^{-1} v_i^{\text{DIS}}$ are elements of $\mathcal{X}$). We also note that the projection

$$Q_r^{\text{DIS}} := V_r^{\text{DIS}} (V_r^{\text{DIS}})^\dagger$$

is not necessarily orthogonal in $\mathcal{X}$. However, when restricted to $\mathcal{E}$, we do have

$$(V_r^{\text{DIS}})^\dagger h = \left(\left\langle v_1^{\text{DIS}}, h \right\rangle_{\mathcal{E}}, \ldots, \left\langle v_r^{\text{DIS}}, h \right\rangle_{\mathcal{E}} \right),$$

such that $Q_r^{\text{DIS}}|_{\mathcal{E}} = \sum_{i=1}^r v_i^{\text{DIS}} \otimes_{\mathcal{E}} v_i^{\text{DIS}}$ is orthogonal in $\mathcal{E}$.

Remark 5. *For convenience, in the remainder of this work, we will assume that the input DIS bases (and their empirical counterparts) do satisfy $C_{\mathcal{X}}^{-1} v_i^{\text{DIS}} \in \mathcal{X}$ for $i = 1, \ldots, r$. From (8) and (10), we see that this is true when $DF \in L^2(\gamma, \mathcal{L}(\mathcal{X}, \mathcal{Y}))$ and $\mu_i^{\text{DIS}} > 0$. If $H_{\mathcal{X}}$ has finite rank $r_{H_{\mathcal{X}}}$, then $\mu_i^{\text{DIS}} = 0$ for $i > r_{H_{\mathcal{X}}}$. In this case, we will adopt the convention that for $i > r_{H_{\mathcal{X}}}$, v_i^{DIS} are constructed by Gram–Schmidt orthogonalization of the PCA basis starting with the basis set $(v_i^{\text{DIS}})_{i=1}^{r_{H_{\mathcal{X}}}}$. This ensures that for any finite r, $(C_{\mathcal{X}}^{-1} v_i^{\text{DIS}})_{i=1}^r$ are well-defined in $\mathcal{X}$.*

The output derivative-informed subspace is found by the eigenvalue problem

$$H_{\mathcal{Y}} u_i^{\text{DIS}} = \lambda_i^{\text{DIS}} u_i^{\text{DIS}}, \tag{11}$$

with $\lambda_1^{\text{DIS}} \geq \lambda_2^{\text{DIS}} \geq \cdots \geq 0$, and take the first r eigenvectors $U_r^{\text{DIS}} := (u_i^{\text{DIS}})_{i=1}^r$ as the output basis such that the decoder and encoder are given by

$$U_r^{\text{DIS}} \xi = \sum_{i=1}^r \xi_i u_i^{\text{DIS}}, \qquad (U_r^{\text{DIS}})^\dagger y = \left(\left\langle u_1^{\text{DIS}}, y \right\rangle_{\mathcal{Y}}, \ldots, \left\langle u_r^{\text{DIS}}, y \right\rangle_{\mathcal{Y}} \right).$$

This again yields an orthogonal projection on $\mathcal{Y}$,

$$P_r^{\text{DIS}} := U_r^{\text{DIS}} (U_r^{\text{DIS}})^\dagger.$$

As we will show, by nature of their construction, the subspaces spanned by V_r^{DIS} and U_r^{DIS} have optimal reconstruction properties for the derivative operator.

Similar to the output PCA, the operators $H_{\mathcal{X}}$ and $H_{\mathcal{Y}}$ need to be approximated using i.i.d. samples $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$. This leads to the estimators

$$\widehat{H}_{\mathcal{X}} := \frac{1}{N}\sum_{k=1}^{N} D_{\mathcal{E}}F(x_k)^* D_{\mathcal{E}}F(x_k),$$
$$\widehat{H}_{\mathcal{Y}} := \frac{1}{N}\sum_{k=1}^{N} D_{\mathcal{E}}F(x_k) D_{\mathcal{E}}F(x_k)^*.$$

Their corresponding eigenvalue problems are then given by

$$\widehat{H}_{\mathcal{X}}\widehat{v}_i^{\mathrm{DIS}} = \widehat{\mu}_i^{\mathrm{DIS}}\widehat{v}_i^{\mathrm{DIS}},$$
$$\widehat{H}_{\mathcal{Y}}\widehat{u}_i^{\mathrm{DIS}} = \widehat{\lambda}_i^{\mathrm{DIS}}\widehat{u}_i^{\mathrm{DIS}},$$

again with $\widehat{\mu}_1^{\mathrm{DIS}} \geq \widehat{\mu}_2^{\mathrm{DIS}} \geq \cdots \geq 0$ and $\widehat{\lambda}_1^{\mathrm{DIS}} \geq \widehat{\lambda}_2^{\mathrm{DIS}} \geq \cdots \geq 0$. The resulting input and output bases are given by $\widehat{V}_r^{\mathrm{DIS}} := (\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ and $\widehat{U}_r^{\mathrm{DIS}} := (\widehat{u}_i^{\mathrm{DIS}})_{i=1}^r$, such that corresponding decoders and encoders are given by

$$\widehat{V}_r^{\mathrm{DIS}}\xi := \sum_{i=1}^{r}\xi_i\widehat{v}_i^{\mathrm{DIS}}, \qquad (\widehat{V}_r^{\mathrm{DIS}})^\dagger x := \left(\left\langle C_{\mathcal{X}}^{-1}\widehat{v}_1^{\mathrm{DIS}}, x\right\rangle_{\mathcal{X}}, \ldots \left\langle C_{\mathcal{X}}^{-1}\widehat{v}_r^{\mathrm{DIS}}, x\right\rangle_{\mathcal{X}}\right),$$
$$\widehat{U}_r^{\mathrm{DIS}}\xi := \sum_{i=1}^{r}\xi_i\widehat{u}_i^{\mathrm{DIS}}, \qquad (\widehat{U}_r^{\mathrm{DIS}})^\dagger y := \left(\left\langle \widehat{u}_1^{\mathrm{DIS}}, y\right\rangle_{\mathcal{Y}}, \ldots \left\langle \widehat{u}_r^{\mathrm{DIS}}, y\right\rangle_{\mathcal{Y}}\right),$$

which yield $\mathcal{E}$ and $\mathcal{Y}$ orthogonal projections

$$\widehat{Q}_r^{\mathrm{DIS}} := \widehat{V}_r^{\mathrm{DIS}}(\widehat{V}_r^{\mathrm{DIS}})^\dagger, \qquad \widehat{P}_r^{\mathrm{DIS}} := \widehat{U}_r^{\mathrm{DIS}}(\widehat{U}_r^{\mathrm{DIS}})^\dagger.$$

The operator $H_{\mathcal{X}}$ coincides with that of the AS method for finite-dimensional vector-valued functions studied in Zahm et al. (2020). In the context of Bayesian inverse problems with Gaussian prior and noise, $H_{\mathcal{X}}$ and $H_{\mathcal{Y}}$ are analogous to the diagnostic matrices used in the dimension reduction strategies of Cui et al. (2014); Zahm et al. (2022); Baptista et al. (2022); Chen et al. (2025). In this work, we will extend the previous analysis by formulating these subspaces for Gaussian measures on infinite-dimensional separable Hilbert spaces.

2.5.3 Satisfaction of Nonparametric Linear Constraints

Before moving on to analyzing the approximation errors, we point out an additional property of interest for the output bases. In mappings that arise from PDE solution operators, it is common for the output $F(x)$, i.e., the solution of the PDE, to satisfy linear constraints, $BF(x) = h$, for some continuous linear operator $B \in \mathcal{L}(\mathcal{Y}, \mathcal{H})$ mapping to another separable Hilbert space $\mathcal{H}$, independent of the input $x \in \mathcal{X}$. For example, these can arise from Dirichlet boundary conditions or divergence-free conditions on the solution. When using the centered PCA and DIS bases for the output space, we can ensure that the surrogates of the form $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + \bar{F}$ also satisfy the constraint, i.e., $B\widetilde{F}_\theta(x) = h$.

Proposition 6. *Suppose $F \in H^1_\gamma$. Let $\mathcal{H}$ be another separable Hilbert space and $B \in \mathcal{L}(\mathcal{Y}, \mathcal{H})$ be a continuous linear operator such that*

$$BF(x) = h \qquad \forall x \in \mathcal{X},$$

for some fixed $h \in \mathcal{H}$. Then, given $r \in \mathbb{N}$, the affine space $\bar{F} + \text{span}(u_1, \dots, u_r)$ corresponding to the output PCA and DIS bases with nonzero eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$ satisfies

$$By = h \qquad \forall y \in \bar{F} + \text{span}((u_i)_{i=1}^r).$$

The same holds almost surely with respect to the sampling of $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$ when $\bar{F}$ is replaced by the estimator $\widehat{F}$ and when the PCA or DIS bases are replaced by their estimators $(\widehat{u}_i)_{i=1}^r$ with nonzero eigenvalues $\widehat{\lambda}_1 \geq \cdots \geq \widehat{\lambda}_r > 0$.

Proof Since $BF(x) = h$ for all x, we have $B\bar{F} = B\mathbb{E}_\gamma[F] = \mathbb{E}_\gamma[BF] = h$. Thus, it remains to verify that $Bu_i = 0$ for the reduced bases $u_i = u_i^{\text{DIS}}$ or $u_i = u_i^{\text{PCA}}$. For the centered PCA, we have $u_i^{\text{PCA}} = C_{\mathcal{Y}} u_i^{\text{PCA}} / \lambda_i^{\text{PCA}}$ when $\lambda_i^{\text{PCA}} > 0$. This implies

$$Bu_i^{\text{PCA}} = \frac{1}{\lambda_i^{\text{PCA}}} B\mathbb{E}_\gamma[(F - \bar{F}) \otimes (F - \bar{F})] u_i^{\text{PCA}} = \frac{1}{\lambda_i^{\text{PCA}}} \mathbb{E}_\gamma[B(F - \bar{F}) \otimes (F - \bar{F})] u_i^{\text{PCA}} = 0.$$

For the DIS, we begin by noting that $BF(x) = h$ implies $D_{\mathcal{E}}(BF(x)) = BD_{\mathcal{E}}F(x) = 0$. We then have,

$$Bu_i^{\text{DIS}} = \frac{1}{\lambda_i^{\text{DIS}}} BH_{\mathcal{Y}} u_i^{\text{DIS}} = \frac{1}{\lambda_i^{\text{DIS}}} \mathbb{E}_\gamma[BD_{\mathcal{E}}FD_{\mathcal{E}}F^*] u_i^{\text{DIS}} = 0,$$

which again holds for all $i \leq r$ with $\lambda_i^{\text{DIS}} > 0$. Results for the empirical mean $\widehat{F}$ and bases $\widehat{u}_i^{\text{PCA}}$ and $\widehat{u}_i^{\text{DIS}}$ can be shown by replacing $\mathbb{E}_\gamma$ with the sum over the samples, where the result holds almost surely with respect to the sampling. ■

3. Error Analysis

3.1 Overview and Assumptions

In this section, we analyze the approximation errors of reduced basis neural operators (RBNOs) in the Sobolev norm H^1_γ, where γ is a Gaussian measure. We will focus on the case where the neural operator is constructed as

$$\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + \widehat{F},$$

with reduced bases U_r and V_r that are orthonormal in $\mathcal{Y}$ and $\mathcal{E}$, a latent space neural network φ, and a sample-based mean estimate $\widehat{F} = \frac{1}{N}\sum_{k=1}^N F(x_k)$. This represents a practical setup, where the neural operator is constructed to fit the (empirical) mean-shifted training data $F - \widehat{F}$.

To begin, we introduce some additional assumptions that are used to bound approximation errors for specific choices of dimension reduction. As previously mentioned, we will always assume centered Gaussian measures γ on the input space $\mathcal{X}$ with non-degenerate covariance operators. Nonetheless, our results can translate to the uncentered case with mean $\bar{x}$ provided that we can shift the input data by the mean when constructing our RBNO. That is, we can instead aim to approximate the shifted operator $x' \mapsto F(x' + \bar{x})$, where $x' \sim \mathcal{N}(0, C_{\mathcal{X}})$ is now centered.

Assumption 7 (Gaussian input measure)**.** *The input measure γ is a centered Gaussian measure on $\mathcal{X}$ with a non-degenerate covariance operator $C_{\mathcal{X}}$.*

When using the input PCA, we will require the additional assumptions that F is twice continuously Fréchet differentiable, and derivatives of F up to second order also have bounded second moments. These assumptions also ensure that the DIS projection operators are bounded linear operators on $\mathcal{X}$ (see Remark 5).

Assumption 8 (Bounded second moments of Fréchet derivatives)**.** *The operator F belongs to $C^2(\mathcal{X},\mathcal{Y})$ with derivatives $DF \in L^2(\gamma,\mathcal{L}(\mathcal{X},\mathcal{Y}))$ and $D^2F \in L^2(\gamma,\mathcal{L}_2(\mathcal{X},\mathcal{Y}))$.*

To ensure the sampling errors associated with the empirical PCA/DIS are bounded, we additionally require that F has bounded fourth moments in its output and its Sobolev derivatives, i.e., $F \in W^{1,4}_\gamma$. Similar assumptions are typical in the analysis of sampling errors in PCA (e.g., Reiß and Wahl, 2020; Bhattacharya et al., 2021).

Assumption 9 (Bounded fourth moments of Sobolev derivatives)**.** *The operator F belongs to $W^{1,4}_\gamma$.*

We will also make use of Assumption 10 and Assumption 11 in the analysis of output PCA and input DIS, respectively. Intuitively, these state that lower-order derivatives of F have control over its higher-order derivatives. This ensures that subspaces computed from lower-order derivatives have controlled error when approximating higher derivatives of F.

Assumption 10 (Derivative inverse inequality)**.** *For $F \in H^1_\gamma$, there exists a constant $K_D = K_D(F) \geq 0$ such that*

$$\langle u, H_{\mathcal{Y}} u\rangle_{\mathcal{Y}} \leq K_D \langle u, C_{\mathcal{Y}} u\rangle_{\mathcal{Y}} \quad \forall u \in \mathcal{Y}. \tag{12}$$

Assumption 11 (Hessian inverse inequality)**.** *For $F \in H^2_\gamma$, there exists a constant $K_H = K_H(F) \geq 0$ such that*

$$\mathbb{E}_{x\sim\gamma}[\|D^2_{\mathcal{E}}F(x)(\cdot,v)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \leq K_H \mathbb{E}_\gamma\left[\|D_{\mathcal{E}}Fv\|^2_{\mathcal{Y}}\right] = K_H \langle v, H_{\mathcal{X}} v\rangle_{\mathcal{E}}, \quad \forall v \in \mathcal{E}. \tag{13}$$

Here, $D^2_{\mathcal{E}}F(x)(\cdot,v)$ refers to the linear operator $\mathcal{E} \ni h \mapsto D^2_{\mathcal{E}}F(x)(h,v) \in \mathcal{Y}$, such that

$$\|D^2_{\mathcal{E}}F(x)(\cdot,v)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} = \sum_{j=1}^{\infty} \|D^2_{\mathcal{E}}F(x)(w_j,v)\|^2_{\mathcal{Y}},$$

for any orthonormal basis $(w_j)_{j=1}^\infty$ of $\mathcal{E}$.

We note that measurable polynomials of degree n satisfy these assumptions with $K_D = n$ and $K_H = n-1$. This is made precise in Proposition 54, Appendix A.5. We also analytically consider two examples of nonlinear PDE-based operators in Section 4. In particular, the examples highlight that the inverse inequalities are not necessarily tied to additional regularity of F.

3.2 Main Results

We begin by presenting our two universal approximation results, Theorem 12 and Theorem 13. Their proofs are presented in Appendices C.3 and C.7 using results developed throughout the rest of this section. Our first result is that the RBNO architecture is indeed universal in the Sobolev norm H^m_γ given orthonormal reduced bases U_r and V_r of $\mathcal{Y}$ and $\mathcal{E}$, respectively. While this result is not difficult to obtain, we believe it may be of independent interest and therefore state it before proceeding to the more detailed analysis.

Theorem 12 (Generic universal approximation)**.** *Suppose $F \in H^m_\gamma$ for some $m \in \mathbb{N}_0$. Let $(u_i)_{i=1}^\infty$ be an orthonormal basis of $\mathcal{Y}$ and $(v_i)_{i=1}^\infty$ be an orthonormal basis of $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1}v_i)_{i=1}^\infty$ are well-defined in $\mathcal{X}$. Additionally, let $\psi \in \mathbb{A}^m$ and $d_L \in \mathbb{N}$. Then, for any $\epsilon > 0$, there exists a rank $r \in \mathbb{N}$ and a d_L-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ such that the neural operator approximation $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger$ with the reduced bases $U_r = (u_i)_{i=1}^r$ and $V_r = (v_i)_{i=1}^r$ achieves the error bound*

$$\|F - \widetilde{F}_\theta\|_{H^m_\gamma} \leq \epsilon.$$

When $\mathcal{X}$ and $\mathcal{Y}$ are finite-dimensional, Theorem 12 trivially reduces to the universal approximation theorem of neural networks in finite dimensions. We also note that it does not quantify the effectiveness of the chosen dimension reduction strategies and how they impact the approximation errors.

To more explicitly determine the contribution of errors, we proceed by performing a more careful accounting of the errors arising from the dimension reduction. Here, the approximation error can essentially be decomposed into (i) reconstruction errors arising from representing the outputs in reduced bases, (ii) ridge function errors arising from truncating the input by the input reduced basis, and (iii) neural network approximation errors. Moreover, when using empirical estimators to construct the PCA/DIS, the reconstruction and ridge function errors additionally have a statistical error component.

Similar to the analyses provided by Zahm et al. (2020); Bhattacharya et al. (2021); Lanthaler (2023), the reconstruction and ridge function errors can be bounded in terms of summations over trailing eigenvalues ($i > r$) of the covariance or sensitivity operators that are used to define the dimension reduction, while statistical errors have a Monte Carlo-type bound related to the sample size N. On the other hand, for almost every realization of the samples (and hence the input and output bases), the neural network approximation error can be made arbitrarily small via the universal approximation result for weighted Sobolev norms (Theorems 2 and 3) using activation functions $\psi \in \mathbb{A}^1$. Additionally, a bound on the size of the neural network can be given when using the ReLU^k activation function with $k \geq 2$.

We summarize these results in the form of a theorem, which is the main result of this work. Its proof draws on the results developed throughout the rest of this section and is provided in Appendix C.7.

Theorem 13 (Detailed universal approximation)**.** *Suppose $F \in H^m_\gamma$ for some $m \in \mathbb{N}, m \geq 2$ and F satisfies Assumptions 8 to 11. Given $r \in \mathbb{N}$, let $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + \widehat{F}$ be an RBNO with a neural network $\varphi_{\psi,\theta}$ using activation function $\psi \in \mathbb{A}^1$ along with the bias $\widehat{F} = \frac{1}{N}\sum_{k=1}^N F(x_k)$ and the reduced bases $U_r = \widehat{U}_r^{\text{PCA}}$ or $\widehat{U}_r^{\text{DIS}}$ and $V_r = V_r^{\text{PCA}}$ or $\widehat{V}_r^{\text{DIS}}$, where $\widehat{F}$, $\widehat{U}_r^{\text{PCA}}$, $\widehat{U}_r^{\text{DIS}}$, and $\widehat{V}_r^{\text{DIS}}$ are estimated from $N \in \mathbb{N}$ i.i.d. samples $\mathcal{S}_N := (x_k)_{k=1}^N \overset{iid}{\sim} \gamma$. Then, $\widetilde{F}_\theta$ simultaneously satisfies the following error decomposition bounds:*

$$\|F - \widetilde{F}_\theta\|^2_{L^2_\gamma} \leq \mathscr{E}_\varphi^{(1)} + K_{\mathcal{Y}}^{(1)} \left(\sum_{i=r+1}^{\infty} \lambda_i + \mathscr{E}_{\mathcal{Y}}(\mathcal{S}_N, r) \right) + K_{\mathcal{X}}^{(1)} \left(\sum_{i=r+1}^{\infty} \mu_i + \mathscr{E}_{\mathcal{X}}(\mathcal{S}_N, r) \right) + \mathscr{E}_F(\mathcal{S}_N), \tag{14}$$

$$|F - \widetilde{F}_\theta|^2_{H^1_\gamma} \leq \mathscr{E}_\varphi^{(2)} + K_{\mathcal{Y}}^{(2)} \left(\sum_{i=r+1}^{\infty} \lambda_i + \mathscr{E}_{\mathcal{Y}}(\mathcal{S}_N, r) \right) + K_{\mathcal{X}}^{(2)} \left(\sum_{i=r+1}^{\infty} \mu_i + \mathscr{E}_{\mathcal{X}}(\mathcal{S}_N, r) \right), \tag{15}$$

where $\mathscr{E}_\varphi^{(1)}$ and $\mathscr{E}_\varphi^{(2)}$ are the neural network approximation errors, λ_i and μ_i are the eigenvalues (given in descending order) of the covariance or sensitivity operators used to define the dimension reduction, i.e.,

$$\lambda_i = \begin{cases} \lambda_i^{\text{PCA}} & \text{if } U_r = \widehat{U}_r^{\text{PCA}} \\ \lambda_i^{\text{DIS}} & \text{if } U_r = \widehat{U}_r^{\text{DIS}} \end{cases}, \quad \mu_i = \begin{cases} \mu_i^{\text{PCA}} & \text{if } V_r = V_r^{\text{PCA}} \\ \mu_i^{\text{DIS}} & \text{if } V_r = \widehat{V}_r^{\text{DIS}} \end{cases},$$

$K_{\mathcal{X}}^{(i)}$, $K_{\mathcal{Y}}^{(i)}$, $i = 1, 2$ are non-negative constants depending on the choice of dimension reduction and the function F, and $\mathscr{E}_F(\mathcal{S}_N)$, $\mathscr{E}_{\mathcal{X}}(\mathcal{S}_N, r)$, and $\mathscr{E}_{\mathcal{Y}}(\mathcal{S}_N, r)$ are bounds on the sampling errors

associated with the mean estimation, input dimension reduction, and output dimension reduction, respectively. In particular, we have the following results on the components in (14) *and* (15)*:*

1. *For any $\epsilon > 0$, $d_L \in \mathbb{N}$, and almost every realization of the samples $\mathcal{S}_N$, there exists a d_L-layer neural network $\varphi_{\psi,\theta}$ such that simultaneously $\mathscr{E}_\varphi^{(1)} < \epsilon$ and $\mathscr{E}_\varphi^{(2)} < \epsilon$. If specifically $\psi = \mathrm{ReLU}^k$ with $k \in \mathbb{N}, k \geq 2$, then there exist constants $\kappa_1 = \kappa_1(r,k,m)$ and $\kappa_2 = \kappa_2(r,k,m)$ such that $\varphi_{\psi,\theta}$ can be chosen to have depth and size satisfying*

$$\mathrm{depth}(\varphi_{\psi,\theta}) \leq \kappa_1 \left(\log(\epsilon^{-\frac{1}{2}}\|F\|_{H^m_\gamma} + 1) + 1\right), \quad \mathrm{size}(\varphi_{\psi,\theta}) \leq \kappa_2 \left(\left(\epsilon^{-1}\|F\|^2_{H^m_\gamma}\right)^{\frac{r}{m-1}} + 1\right).$$

   Note that the exponent of ϵ differs from that in Theorem 3 by a factor of two since (14) *and* (15) *are stated in terms of squared errors.*

2. *The sampling error bounds satisfy*

$$\begin{aligned}
\mathbb{E}_N[\mathscr{E}_F(\mathcal{S}_N)] &\leq \frac{\|F - \bar{F}\|^2_{L^2_\gamma}}{N}, \\
\mathbb{E}_N[\mathscr{E}_\mathcal{X}(\mathcal{S}_N, r)] &\leq \min\left\{\sqrt{\frac{2rM_\mathcal{X}}{N}}, \frac{2M_\mathcal{X}}{(\mu_r - \mu_{r+1})N}\right\}, \\
\mathbb{E}_N[\mathscr{E}_\mathcal{Y}(\mathcal{S}_N, r)] &\leq \min\left\{\sqrt{\frac{2rM_\mathcal{Y}}{N}}, \frac{2M_\mathcal{Y}}{(\lambda_r - \lambda_{r+1})N}\right\},
\end{aligned}$$

   where the expectation is taken with respect to the samples $\mathcal{S}_N = (x_k)_{k=1}^N \overset{iid}{\sim} \gamma$ used to compute $\widehat{F}$, $\widehat{U}_r^{\mathrm{PCA}}$, $\widehat{U}_r^{\mathrm{DIS}}$, and $\widehat{V}_r^{\mathrm{DIS}}$. Here, $M_\mathcal{X}$, $M_\mathcal{Y}$ are non-negative constants depending on the choice of dimension reduction and the function F.

Remark 14. *Not all of Assumptions 8, 10 and 11 are simultaneously needed. We can break down the required assumptions and the corresponding forms of the constants based on the choice of dimension reduction as follows.*

1. *Empirical output PCA, $\widehat{U}_r = \widehat{U}_r^{\mathrm{PCA}}$: Without additional assumptions, we have $K_\mathcal{Y}^{(1)} = 1$ and $M_\mathcal{Y} = 2\mathbb{E}_\gamma[\|(F - \bar{F}) \otimes (F - \bar{F}) - C_\mathcal{Y}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] + 2\|F - \bar{F}\|^4_{L^4_\gamma} + 6\|F - \bar{F}\|^4_{L^2_\gamma}$. Assumption 10 is required for $K_\mathcal{Y}^{(2)}$ to be finite, in which case $K_\mathcal{Y}^{(2)} = K_D$.*

2. *Empirical output DIS, $\widehat{U}_r = \widehat{U}_r^{\mathrm{DIS}}$: This does not require additional assumptions. We have $K_\mathcal{Y}^{(1)} = K_\mathcal{Y}^{(2)} = 1$, and $M_\mathcal{Y} = \mathbb{E}_\gamma[\|D_\mathcal{E} F D_\mathcal{E} F^* - H_\mathcal{Y}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}]$.*

3. *Exact input PCA, $\widehat{V}_r = V_r^{\mathrm{PCA}}$: Since the input PCA is exact, we trivially have $M_\mathcal{X} = 0$. Assumption 8 is required for $K_\mathcal{X}^{(1)}$ and $K_\mathcal{X}^{(2)}$ to be finite, in which case $K_\mathcal{X}^{(1)} = \mathbb{E}_\gamma[\|DF\|^2_{\mathcal{L}(\mathcal{X},\mathcal{Y})}]$ and $K_\mathcal{X}^{(2)} = \mathbb{E}_\gamma[\|D^2F\|^2_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}]\mathbb{E}_\gamma[\|x\|^2_\mathcal{X}] + \mathbb{E}_\gamma[\|DF\|^2_{\mathcal{L}(\mathcal{X},\mathcal{Y})}]$.*

4. *Empirical input DIS, $\widehat{V}_r = \widehat{V}_r^{\mathrm{DIS}}$: Note that we use Assumption 8 as a simple way of ensuring that $(C_\mathcal{X}^{-1}\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are well-defined in $\mathcal{X}$. This can be relaxed by instead directly assuming $(C_\mathcal{X}^{-1}\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are indeed well-defined in $\mathcal{X}$. Without additional assumptions, we have $K_\mathcal{X}^{(1)} = 1$ and $M_\mathcal{X} = \mathbb{E}_\gamma[\|D_\mathcal{E} F^* D_\mathcal{E} F - H_\mathcal{X}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}]$. Assumption 11 is then required for $K_\mathcal{X}^{(2)}$ to be finite, in which case $K_\mathcal{X}^{(2)} = K_H + 1$.*

Remark 15. *The sampling error estimates for $\mathscr{E}_{\mathcal{X}}(\mathcal{S}_N, r)$ and $\mathscr{E}_{\mathcal{Y}}(\mathcal{S}_N, r)$ are the minima of two different bounds, as is typical in the analysis of PCA sampling error (e.g., Reiß and Wahl, 2020). These are often termed the global bound, which has a slow $\mathcal{O}(N^{-1/2})$ convergence rate and a local bound, which has a faster $\mathcal{O}(N^{-1})$ convergence rate but only holds when the eigenvalues are sufficiently well separated (i.e., $\lambda_r - \lambda_{r+1} > 0$ and $\mu_r - \mu_{r+1} > 0$). The constant in the local bound can become very large as r increases. We note that both forms of the bounds are pessimistic, and more refined bounds can be derived by using more sophisticated techniques akin to those presented in Reiß and Wahl (2020) for the analysis of PCA, though these often require additional assumptions such as boundedness or sub-Gaussianity of F.*

Remark 16. *The approximation rates are provided for* ReLU^k *activation under assumptions of additional Sobolev regularity of finite order, i.e., $F \in H^m_\gamma$ with $m \geq 2$. For this reason, the approximation rate suffers from the curse of dimensionality analogous to conventional neural network approximation rates in finite dimensions. Stronger structural or regularity assumptions such as Barron-type conditions or holomorphy may be used to obtain dimension-independent rates for the approximation error and explicit eigenvalue decay rates of the covariance and sensitivity operators. We leave this for future work.*

The rest of this section proceeds as follows. We begin by introducing the conditional expectation, a key tool that we use to analyze the RBNO architecture. We proceed to consider the reconstruction, ridge function, and sampling errors in detail. Lengthy proofs of the results in this section along with additional supporting lemmas are presented in the Appendices for clarity.

3.3 Conditional Expectation and Universal Approximation

The reduced-basis architectures for given bases U_r and V_r are ridge functions—L^2_γ functions that are measurable in the restricted sigma algebra $\sigma(V_r^\dagger)$ (i.e., constant in the orthogonal complement directions of V_r). In L^2_γ, the conditional expectation $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ of a function $F \in L^2_\gamma$ is a projection, and hence $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ is an optimal representation of F in L^2_γ when restricted to the $\sigma(V_r^\dagger)$ measurable functions (Bogachev, 2010). When γ is Gaussian and V_r are orthonormal in the Cameron–Martin space $\mathcal{E}$, the conditional expectation has the representation (Bogachev, 1998, Corollary 3.5.2)

$$\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x) = \mathbb{E}_{z\sim\gamma}[F(Q_r x + (I - Q_r)z)], \tag{16}$$

where $Q_r = V_r V_r^\dagger$ is an orthogonal projection on $\mathcal{E}$. This is used to develop the theory of active subspaces in the finite-dimensional setting in Zahm et al. (2020), but also holds in the infinite-dimensional setting.

For $F \in W^{m,p}(\gamma, \mathcal{Y})$, when $(v_i)_{i=1}^\infty$ is an orthonormal basis of $\mathcal{E}$, the conditional expectations $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ converge to F in $W^{m,p}(\gamma, \mathcal{Y})$ as $r \to \infty$ (Bogachev, 1998, Theorem 5.4.5). Moreover, we can show that the Sobolev derivative can be interchanged with the conditional expectation for $F \in H^m_\gamma$.

Lemma 17 (Interchanging Sobolev differentiation and conditional expectation)**.** *Suppose $F \in H^m_\gamma$ for some $m \in \mathbb{N}$, and let $V_r = (v_i)_{i=1}^r$ be orthonormal in $\mathcal{E}$ such that $(C_{\mathcal{X}}^{-1} v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, the kth Sobolev derivative of the conditional expectation is given by*

$$D^k_{\mathcal{E}} \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D^k_{\mathcal{E}} F(Q_r(\cdot), \ldots, Q_r(\cdot))|\sigma(V_r^\dagger)]$$

for all $k = 1, \ldots, m$, where $D^k_{\mathcal{E}} F(x)(Q_r(\cdot), \ldots, Q_r(\cdot)) : (h_1, \ldots, h_k) \mapsto D^k_{\mathcal{E}} F(x)(Q_r h_1, \ldots, Q_r h_k)$.

The proof of this lemma is provided in Appendix A.4. This result implies that the conditional expectation can be simultaneously used to optimally approximate F and its projected Sobolev derivatives $D^k_{\mathcal{E}} F(Q_r(\cdot), \dots, Q_r(\cdot))$. Thus, in developing our approximation results, we can first approximate $F \in H^m_\gamma$ by its conditional expectation with outputs projected onto an orthonormal basis, and then approximate this projected conditional expectation by a neural network. To this end, the following lemma is useful, with the proof provided in Appendix C.1.

Lemma 18 (Universal approximation of the projected conditional expectation)**.** *Suppose $F \in H^m_\gamma$ for some $m \in \mathbb{N}_0$. Let $U_r = (u_i)_{i=1}^r$ be an orthonormal set in $\mathcal{Y}$ and $V_r = (v_i)_{i=1}^r$ be an orthonormal set in $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1} v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, given $\psi \in \mathbb{A}^m$ and $d_L \in \mathbb{N}$, for any $\epsilon > 0$, there exists a d_L-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ such that the neural operator $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger$ satisfies*

$$\|P_r \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] - \widetilde{F}_\theta\|_{H^m_\gamma} \le \epsilon, \tag{17}$$

where $P_r = U_r U_r^\dagger$ is the orthogonal projection onto the span of U_r.

Lemma 18 immediately allows us to prove the generic universal approximation result of Theorem 12 (see Appendix C.3). Moreover, in further analyzing the H^1_γ case, it allows us to focus on the approximation error committed by the optimal ridge function (the projected conditional expectation), knowing that the subsequent neural network approximation error of the projected conditional expectation can be made arbitrarily small.

Note that for functions F that have additional smoothness, i.e., $F \in H^l_\gamma$ for some $l > m$, their projected conditional expectations also inherit this smoothness. Thus, we can use Theorem 3 to obtain explicit bounds on the size of the neural network needed to achieve a given approximation error ϵ in (17) when using ReLU^k activations. In particular, we can approximate the conditional expectation of $F - b$ for any constant shift $b \in \mathcal{Y}$ at the same approximation rate as F itself. This is summarized in the following result, with the proof provided in Appendix C.2.

Lemma 19 (Approximation of the projected conditional expectation by ReLU^k networks)**.** *Consider the setting of Lemma 18. Assume additionally that $F \in H^l_\gamma$ for some $l \in \mathbb{N}, l > m$ and $\psi = \mathrm{ReLU}^k$ for some $k \in \mathbb{N}, k \ge 2$. Then, there exist constants $\kappa_1 = \kappa_1(r,k,l,m)$ and $\kappa_2 = \kappa_2(r,k,l,m)$ such that for any $\epsilon > 0$ and any shift $b \in \mathcal{Y}$, there exists a ReLU^k neural network $\varphi_{\psi,\theta}$ such that the neural operator $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger$ satisfies*

$$\|\mathbb{E}_\gamma[P_r(F-b)|\sigma(V_r^\dagger)] - \widetilde{F}_\theta\|_{H^m_\gamma} \le \epsilon,$$

where the depth and size of $\varphi_{\psi,\theta}$ satisfy

$$\mathrm{depth}(\varphi_{\psi,\theta}) \le \kappa_1 \left(\log(\epsilon^{-1}\|F\|_{H^l_\gamma} + 1) + 1\right), \quad \mathrm{size}(\varphi_{\psi,\theta}) \le \kappa_2 \left(\left(\epsilon^{-1}\|F\|_{H^l_\gamma}\right)^{\frac{2r}{l-m}} + 1\right).$$

3.4 Decomposition of the Approximation Errors

For RBNOs of the form $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + \widehat{F}$, we can naturally decompose the approximation errors based on orthogonality. This is summarized in the following proposition, the proof of which is provided in Appendix C.4.

Proposition 20. *Suppose $F \in H^1_\gamma$. Let $U_r = (u_i)_{i=1}^r$ be an orthonormal set in $\mathcal{Y}$ and let $V_r = (v_i)_{i=1}^r$ be an orthonormal set in $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1} v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Additionally, let $\varphi \in C^1(\mathbb{R}^r, \mathbb{R}^r)$. Then, the ridge function $\widetilde{F} = U_r \circ \varphi \circ V_r^\dagger + \widehat{F}$ simultaneously satisfies*

$$\|F - \widetilde{F}\|^2_{L^2_\gamma} \le \mathbb{E}_\gamma[\|P_r(F - \widetilde{F})\|^2_{\mathcal{Y}}] + \mathbb{E}_\gamma[\|(I - P_r)(F - \bar{F})\|^2_{\mathcal{Y}}] + \|(I - P_r)(\bar{F} - \widehat{F})\|^2_{\mathcal{Y}} \tag{18}$$

and

$$|F-\widetilde{F}|_{H^1_\gamma}^2 \leq \mathbb{E}_\gamma[\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F})Q_r\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] + \mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I-Q_r)\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] + \mathbb{E}_\gamma[\|(I-P_r)D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2]. \tag{19}$$

Moreover, the terms involving $\widetilde{F}$ can alternatively be written as

$$\mathbb{E}_\gamma[\|P_r(F-\widetilde{F})\|_{\mathcal{Y}}^2] = \mathbb{E}_\gamma[\|U_r^\dagger(F-\widehat{F}) - \varphi\circ V_r^\dagger\|_2^2] \tag{20}$$

and

$$\mathbb{E}_\gamma[\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F})Q_r\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] = \mathbb{E}_\gamma[\|U_r^\dagger D_{\mathcal{E}}F V_r - D\varphi\circ V_r^\dagger\|_F^2] \tag{21}$$

where $\left(\varphi\circ V_r^\dagger\right)(x) = \varphi\left(V_r^\dagger x\right)$, $\left(D\varphi\circ V_r^\dagger\right)(x) = D\varphi\left(V_r^\dagger x\right)$, and $\|\cdot\|_2$ and $\|\cdot\|_F$ denote the standard Euclidean and Frobenius norms in finite dimensions, respectively.

Here, the dimension reduction introduces errors of two types: reconstruction errors and ridge function errors. The reconstruction errors, $\|(I-P_r)(F-\bar{F})\|_{L^2_\gamma}$, $\|(I-P_r)D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}$, and $\|D_{\mathcal{E}}F(I-Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}$ arise from truncating the representations of $F(x)-\bar{F}\in\mathcal{Y}$ in the subspace spanned by U_r, and $D_{\mathcal{E}}F(x)\in\mathrm{HS}(\mathcal{E},\mathcal{Y})$ in the subspaces spanned by U_r on the output side and by V_r on the input side. These are independent of the choice of φ. On the other hand, the ridge function errors $\|P_r(F-\widetilde{F})\|_{L^2_\gamma}$ and $\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F})Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}$ arise from restricting the approximant to depend only on the projected input, and can be analyzed using the conditional expectation. Additionally, sampling errors are present in the mean estimate $\widehat{F}$, and also in constructing the reduced bases.

Recall that the conditional expectation is the optimal ridge function in L^2_γ whose derivative is also an optimal approximation of the projected derivative in $L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))$. As previously discussed, we may further decompose the ridge function error term into the optimal ridge function error committed by the conditional expectation, and an additional error contribution due to the approximation of the conditional expectation by a neural network. This leads to the following error decomposition that forms the basis of the proof for the main result in Theorem 13. We provide the proof of this result in Appendix C.4.

Corollary 21. *Suppose $F\in H^1_\gamma$. Let $U_r=(u_i)_{i=1}^r$ be an orthonormal set in $\mathcal{Y}$ and let $V_r=(v_i)_{i=1}^r$ be an orthonormal set in $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1}v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Additionally, let $\psi\in\mathbb{A}^1$ and $d_L\in\mathbb{N}$. Then, for any $\epsilon>0$, there exists a d_L-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ such that the neural operator $\widetilde{F}_\theta = U_r\circ\varphi_{\psi,\theta}\circ V_r^\dagger + \widehat{F}$ simultaneously satisfies*

$$\|F-\widetilde{F}_\theta\|_{L^2_\gamma}^2 \leq \epsilon + \|P_r(F-\widetilde{F}_r)\|_{L^2_\gamma}^2 + \|(I-P_r)(F-\bar{F})\|_{L^2_\gamma}^2 + \|\bar{F}-\widehat{F}\|_{\mathcal{Y}}^2, \tag{22}$$

and

$$|F-\widetilde{F}_\theta|_{H^1_\gamma}^2 \leq \epsilon + \|D_{\mathcal{E}}F(I-Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 + \|(I-P_r)D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 + \|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_r)Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2, \tag{23}$$

where we have used the shorthand notation $\widetilde{F}_r := \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ for the conditional expectation.

If additionally $F\in H^m_\gamma$ for some $m\in\mathbb{N}, m\geq 2$ and $\psi=\mathrm{ReLU}^k$ for some $k\in\mathbb{N}, k\geq 2$, then there exist constants $\kappa_1=\kappa_1(r,k,m)$ and $\kappa_2=\kappa_2(r,k,m)$ such that $\varphi_{\psi,\theta}$ can be chosen to satisfy

$$\mathrm{depth}(\varphi_{\psi,\theta}) \leq \kappa_1\left(\log(\epsilon^{-\frac{1}{2}}\|F\|_{H^m_\gamma}+1)+1\right), \qquad \mathrm{size}(\varphi_{\psi,\theta}) \leq \kappa_2\left(\left(\epsilon^{-1}\|F\|_{H^m_\gamma}^2\right)^{\frac{r}{m-1}}+1\right).$$

Note that the exponent of ϵ differs from that in Theorem 3 and Lemma 19 by a factor of two since (22) and (23) are stated in terms of the squared errors.

3.5 Reconstruction Errors

3.5.1 Principal Component Analysis

We now proceed to study the reconstruction errors arising from the choice of dimension reduction. We begin by considering the reconstruction errors associated with the PCA bases. It is well known that the PCA bases yield optimal linear reduced representations. We start with the input case, where no sampling errors are involved.

Proposition 22. *The projection given by the exact input PCA minimizes the reconstruction error amongst all rank r orthogonal projections on $\mathcal{X}$, $\mathcal{P}_r(\mathcal{X})$, in the sense that*

$$\mathbb{E}_{x\sim\gamma}[\|(I - Q_r^{\mathrm{PCA}})x\|_{\mathcal{X}}^2] = \min_{Q_r\in\mathcal{P}_r(\mathcal{X})} \mathbb{E}_{x\sim\gamma}[\|(I - Q_r)x\|_{\mathcal{X}}^2] = \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{PCA}}.$$

Proof We reproduce the proof here for completeness. We note that for any $Q_r \in \mathcal{P}_r(\mathcal{X})$, we have

$$\mathbb{E}_{x\sim\gamma}[\|(I - Q_r)x\|_{\mathcal{X}}^2] = \mathrm{Tr}((I - Q_r)C_{\mathcal{X}}(I - Q_r)),$$

which, by Fan's theorem (Lemma 48) is minimized by the projection onto the leading eigenvectors of $C_{\mathcal{X}}$, i.e., $Q_r = Q_r^{\mathrm{PCA}}$. ■

As evident in the proof, the result gives the value of the Hilbert–Schmidt norm of the exact PCA projection $I - Q_r^{\mathrm{PCA}}$ when viewed as a mapping from $\mathcal{E} \to \mathcal{X}$,

$$\|I - Q_r^{\mathrm{PCA}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{X})}^2 = \mathrm{Tr}((I - Q_r^{\mathrm{PCA}})C_{\mathcal{X}}(I - Q_r^{\mathrm{PCA}})) = \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{PCA}}. \tag{24}$$

Although a rank r PCA optimally reconstructs the input x, intuitively, we can see that it may not provide a good reconstruction $D_{\mathcal{E}}F(I - Q_r^{\mathrm{PCA}})$ of $D_{\mathcal{E}}F$ since the PCA on the input does not incorporate any information about the mapping F. As an example, when F is a map depending only on the $(r+1)$th basis, a truncated input PCA of rank r will always yield a projected derivative $D_{\mathcal{E}}FQ_r$ of zero. However, when measuring the reconstruction error of $D_{\mathcal{E}}F(I - Q_r^{\mathrm{PCA}})$ in the Hilbert–Schmidt norm $\mathrm{HS}(\mathcal{E},\mathcal{Y})$, smaller weights are placed on the input directions with small variance. This allows one to place a bound on the reconstruction error of the derivative.

Proposition 23. *Suppose $F \in C^1(\mathcal{X},\mathcal{Y})$ and $DF \in L^2(\gamma,\mathcal{L}(\mathcal{X},\mathcal{Y}))$. Then the derivative reconstruction error for the exact input PCA is bounded by*

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I - Q_r^{\mathrm{PCA}})\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] \le \mathbb{E}_\gamma[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2] \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{PCA}}. \tag{25}$$

Proof The approach taken here is analogous to that in the proof of Proposition 3.1 of Zahm et al. (2020), which specifically assumes that F is Lipschitz continuous. Here, we will use the fact that $I - Q_r^{\mathrm{PCA}} \in \mathrm{HS}(\mathcal{E},\mathcal{X})$ to write

$$\|D_{\mathcal{E}}F(I - Q_r^{\mathrm{PCA}})\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \le \|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}\|I - Q_r^{\mathrm{PCA}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{X})}.$$

Therefore,

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I - Q_r^{\mathrm{PCA}})\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] \le \mathbb{E}_\gamma[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2]\|I - Q_r^{\mathrm{PCA}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{X})}^2,$$

along with (24) gives the bound (25). Note that when F is Lipschitz continuous with Lipschitz constant L, we have $\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})} \leq L$, and we recover the result of Zahm et al. (2020). ■

Since we consider a centered (mean-shifted) PCA for the output, the output PCA basis optimally reconstructs the mean-shifted output $F - \bar{F}$. On the other hand, when sampling errors are present, the empirical PCA basis becomes suboptimal, where the gap in the error bound is termed the excess risk. As discussed in Reiß and Wahl (2020); Bhattacharya et al. (2021); Lanthaler (2023), the excess risk can be bounded using the Hilbert–Schmidt distance between $C_{\mathcal{Y}}$ and its estimator $\widehat{C}_{\mathcal{Y}}$. This is summarized in the following proposition.

Proposition 24. *Suppose $F \in L^2_\gamma$. Then, the exact output PCA basis minimizes the reconstruction error amongst all rank $r \in \mathbb{N}$ orthogonal projections $\mathcal{P}_r(\mathcal{Y})$ in the sense that*

$$\mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})(F - \bar{F})\|_{\mathcal{Y}}^2] = \min_{P_r \in \mathcal{P}_r(\mathcal{Y})} \mathbb{E}_\gamma[\|(I - P_r)(F - \bar{F})\|_{\mathcal{Y}}^2] = \sum_{i=r+1}^{\infty} \lambda_i^{\mathrm{PCA}},$$

while the reconstruction error for the empirical output PCA is bounded by

$$\mathbb{E}_\gamma[\|(I - \widehat{P}_r^{\mathrm{PCA}})(F - \bar{F})\|_{\mathcal{Y}}^2] \leq \sum_{i=r+1}^{\infty} \lambda_i^{\mathrm{PCA}} + \min\left\{ \sqrt{2r}\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\mathrm{PCA}} - \lambda_{r+1}^{\mathrm{PCA}}} \right\}.$$

Proof The reconstruction error for the exact output PCA is analogous to the input case shown in Proposition 22. The excess risk corresponding to the empirical PCA with sample-based estimator $\widehat{C}_{\mathcal{Y}}$ is a consequence of Lemma 50. ■

The output PCA basis is constructed on the outputs of the map F, and hence is aware of the sensitivity of the map F in the output space. That is, directions in which F changes significantly are expected to be captured by the output PCA basis. However, it is still difficult to derive similar bounds for the derivative reconstruction error using output PCA in terms of its eigenvalues. For example, we can see that output PCA error is such that

$$\begin{aligned}
&\mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] \\
&\quad = \mathrm{Tr}((I - P_r^{\mathrm{PCA}})H_{\mathcal{Y}}(I - P_r^{\mathrm{PCA}})) \\
&\quad = \mathrm{Tr}((I - P_r^{\mathrm{PCA}})C_{\mathcal{Y}}(I - P_r^{\mathrm{PCA}})) + \mathrm{Tr}((I - P_r^{\mathrm{PCA}})(H_{\mathcal{Y}} - C_{\mathcal{Y}})(I - P_r^{\mathrm{PCA}})) \\
&\quad = \sum_{i=r+1}^{\infty} \lambda_i^{\mathrm{PCA}} + \mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] - \mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})(F - \bar{F})\|_{\mathcal{Y}}^2],
\end{aligned}$$

where the additional error term

$$\mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2] - \mathbb{E}_\gamma[\|(I - P_r^{\mathrm{PCA}})(F - \bar{F})\|_{\mathcal{Y}}^2] \geq 0$$

due to the Poincaré inequality (Theorem 63).

Instead, we can look towards Assumption 10 to provide a bound on the derivative reconstruction error in terms of the eigenvalues of the output PCA basis. Note that the Poincaré inequality again suggests that generally, $K_D \geq 1$ for the inverse inequality constant in Assumption 10. In particular, K_D quantifies the extent to which the output PCA basis captures the variation in derivatives. Intuitively, for operators that are largely monotonic, the deviations $F - \bar{F}$ do capture

the derivatives well, while for operators with highly oscillatory behavior this may not be the case. For example, polynomial functions have K_D that can be bounded by the degree of the polynomial, as shown in Proposition 54 (Appendix A.5). We also provide a non-polynomial example in Section 4 and computationally estimate K_D for our PDE-based numerical examples in Section 5.

Using Assumption 10, we can directly obtain the following result.

Proposition 25. *Suppose $F \in H^1_\gamma$ satisfies Assumption 10. Then, the derivative reconstruction error for the empirical output PCA is bounded by*

$$\begin{aligned}&\mathbb{E}_\gamma[\|(I - \widehat{P}_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \\ &\quad\le K_D\left(\sum_{i=r+1}^{\infty}\lambda_i^{\mathrm{PCA}} + \min\left\{\sqrt{2r}\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}}{\lambda_r^{\mathrm{PCA}} - \lambda_{r+1}^{\mathrm{PCA}}}\right\}\right). \qquad (26)\end{aligned}$$

Proof We recognize that $\|(I - \widehat{P}_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} = \sum_{i=1}^{\infty}\|D_{\mathcal{E}}F^*(I - \widehat{P}_r^{\mathrm{PCA}})u_i\|^2_{\mathcal{E}}$ for any $(u_i)_{i=1}^{\infty}$ that is an orthonormal basis of $\mathcal{Y}$. By extending the empirical output PCA basis, we obtain a basis $(\widehat{u}_i^{\mathrm{PCA}})_{i=1}^{\infty}$ of $\mathcal{Y}$, which we can use to evaluate the Hilbert–Schmidt norm,

$$\|(I - \widehat{P}_r^{\mathrm{PCA}})D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} = \sum_{i=r+1}^{\infty}\|D_{\mathcal{E}}F^*\widehat{u}_i\|^2_{\mathcal{E}} = \sum_{i=r+1}^{\infty}\left\langle D_{\mathcal{E}}FD_{\mathcal{E}}F^*\widehat{u}_i^{\mathrm{PCA}}, \widehat{u}_i^{\mathrm{PCA}}\right\rangle_{\mathcal{Y}}.$$

Taking an expectation and applying Assumption 10 yields

$$\sum_{i=r+1}^{\infty}\mathbb{E}_\gamma\left[\left\langle D_{\mathcal{E}}FD_{\mathcal{E}}F^*\widehat{u}_i^{\mathrm{PCA}}, \widehat{u}_i^{\mathrm{PCA}}\right\rangle_{\mathcal{Y}}\right] = \sum_{i=r+1}^{\infty}\left\langle \widehat{u}_i^{\mathrm{PCA}}, H_{\mathcal{Y}}\widehat{u}_i^{\mathrm{PCA}}\right\rangle_{\mathcal{Y}} \le K_D\sum_{i=r+1}^{\infty}\left\langle \widehat{u}_i^{\mathrm{PCA}}, C_{\mathcal{Y}}\widehat{u}_i^{\mathrm{PCA}}\right\rangle_{\mathcal{Y}}.$$

Finally, the desired bound follows from Proposition 24. ■

3.5.2 Derivative-Informed Subspaces

We now consider the reconstruction properties for projections defined using DIS. Since they effectively amount to a PCA on the derivative operators, we have similar optimal reconstruction properties of the derivative. The following proposition extends the results of Zahm et al. (2020) and Baptista et al. (2022) to the infinite-dimensional setting. It also includes the effect of sampling errors, which is considered for the input DIS (active subspace) in the finite-dimensional setting by Lam et al. (2020); Cui and Tong (2022). Again, the sampling errors are analyzed in Section 3.7.

Proposition 26. *Suppose $F \in H^1_\gamma$ and assume that for the exact and empirical input DIS bases, $(C_{\mathcal{X}}^{-1}v_i^{\mathrm{DIS}})_{i=1}^r$ and $(C_{\mathcal{X}}^{-1}\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, projections onto the exact derivative-informed subspaces minimize the derivative reconstruction errors. That is,*

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I - Q_r^{\mathrm{DIS}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \min_{Q_r\in\mathcal{P}_r(\mathcal{E})}\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I - Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \sum_{i=r+1}^{\infty}\mu_i^{\mathrm{DIS}},$$

$$\mathbb{E}_\gamma[\|(I - P_r^{\mathrm{DIS}})D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \min_{P_r\in\mathcal{P}_r(\mathcal{Y})}\mathbb{E}_\gamma[\|(I - P_r)D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \sum_{i=r+1}^{\infty}\lambda_i^{\mathrm{DIS}}.$$

Moreover, in the presence of sampling errors, the derivative reconstruction errors for the empirical DIS are bounded by

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I-\widehat{Q}_r^{\mathrm{DIS}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \leq \sum_{i=r+1}^{\infty}\mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{X}}-\widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}}-\widehat{H}_{\mathcal{X}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}}{\mu_r^{\mathrm{DIS}}-\mu_{r+1}^{\mathrm{DIS}}}\right\}, \tag{27}$$

$$\mathbb{E}_\gamma[\|(I-\widehat{P}_r^{\mathrm{DIS}})D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \leq \sum_{i=r+1}^{\infty}\lambda_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{Y}}-\widehat{H}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|H_{\mathcal{Y}}-\widehat{H}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}}{\lambda_r^{\mathrm{DIS}}-\lambda_{r+1}^{\mathrm{DIS}}}\right\}. \tag{28}$$

Proof We start with the input side. For any projection $Q_r \in \mathcal{P}_r(\mathcal{E})$,

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I-Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \mathbb{E}_\gamma[\mathrm{Tr}((I-Q_r)D_{\mathcal{E}}^*FD_{\mathcal{E}}F(I-Q_r))] = \mathrm{Tr}((I-Q_r)H_{\mathcal{X}}(I-Q_r)),$$

where the interchange of trace with expectation is valid, since each term in the infinite sum for the trace is non-negative, and $D_{\mathcal{E}}^*FD_{\mathcal{E}}F$ is assumed to be integrable. By Lemma 49, the trace is minimized by Q_r^{DIS} with value

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(I-Q_r^{\mathrm{DIS}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \sum_{i=r+1}^{\infty}\mu_i^{\mathrm{DIS}}.$$

The case for the output projection is analogous, since for $P_r \in \mathcal{P}_r(\mathcal{Y})$,

$$\mathbb{E}_\gamma[\|(I-P_r)D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \mathbb{E}_\gamma[\mathrm{Tr}((I-P_r)D_{\mathcal{E}}FD_{\mathcal{E}}F^*(I-P_r))] = \mathrm{Tr}((I-P_r)H_{\mathcal{Y}}(I-P_r))$$

In the presence of sampling errors, Lemma 50 yields the desired bounds for both the input and output subspaces. ■

We can use the Poincaré inequality (Theorem 63) to obtain a reconstruction error bound of the output itself using the output DIS, analogous to the output PCA. We note that this bound is, in general, not optimal due to the use of the Poincaré inequality. In fact, the trailing eigenvalue sum for the output DIS cannot be smaller than that of the output PCA, where the latter is the optimal reconstruction error for the output itself.

Proposition 27. *Suppose $F \in H^1_\gamma$. Then, the reconstruction error for the empirical output DIS is bounded by*

$$\mathbb{E}_\gamma[\|(I-\widehat{P}_r^{\mathrm{DIS}})(F-\bar{F})\|^2_{\mathcal{Y}}] \leq \sum_{i=r+1}^{\infty}\lambda_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{Y}}-\widehat{H}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|H_{\mathcal{Y}}-\widehat{H}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}}{\lambda_r^{\mathrm{DIS}}-\lambda_{r+1}^{\mathrm{DIS}}}\right\}. \tag{29}$$

Proof Since for any $P_r \in \mathcal{P}_r(\mathcal{Y})$, the function $(I-P_r)F$ has derivative $(I-P_r)D_{\mathcal{E}}F$, and mean $(I-P_r)\bar{F}$, we can apply the Poincaré inequality (Theorem 63) to obtain

$$\mathbb{E}_\gamma[\|(I-P_r)(F-\bar{F})\|^2_{\mathcal{Y}}] \leq \mathbb{E}_\gamma[\|(I-P_r)D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}].$$

The bound (28) yields the desired result. Note that if we instead use the exact DIS, $P_r = P_r^{\mathrm{DIS}}$, the bound is simply

$$\mathbb{E}_\gamma[\|(I-P_r^{\mathrm{DIS}})(F-\bar{F})\|^2_{\mathcal{Y}}] \leq \sum_{i=r+1}^{\infty}\lambda_i^{\mathrm{DIS}}.$$

■

3.6 Ridge Function Errors

We now consider the ridge function errors that arise due to the projection of the input. This is studied by Zahm et al. (2020) for the approximation error in L^2_γ for finite-dimensional input and output spaces, and subsequently used to derive estimates on DIPNet approximations (O'Leary-Roseberry, Villa, et al., 2022; O'Leary-Roseberry, Du, et al., 2022). In this section, we provide extensions to the case of infinite-dimensional inputs and outputs, and also provide bounds on the derivative error.

3.6.1 Ridge Function Errors of the Output Values

As previously discussed, given $V_r = (v_i)_{i=1}^r$, a set of orthonormal vectors in $\mathcal{E}$ and its encoding operator $V_r^\dagger = (\langle v_1, \cdot\rangle_\mathcal{E}, \dots, \langle v_r, \cdot\rangle_\mathcal{E})$, the conditional expectation $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ is an optimal representation of F when restricted to the sigma algebra $\sigma(V_r^\dagger)$. Our strategy is to study how the conditional expectation $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ approximates F, since we have already shown that its projection $P_r\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ onto the output subspace spanned by U_r can in turn be approximated arbitrarily well by an RBNO (Lemma 18).

To this end, we know that the conditional expectation satisfies a subspace Poincaré inequality, which is shown by Zahm et al. (2020) in the finite-dimensional context. We will verify that an analogous version holds in the context of separable Hilbert spaces. The proof follows from the Poincaré inequality itself, and is given in Appendix C.5.

Theorem 28 (Subspace Poincaré inequality)**.** *Suppose $F \in H^1_\gamma$. Then, given any orthonormal set $V_r = (v_i)_{i=1}^r$ in $\mathcal{E}$ for which $(C_\mathcal{X}^{-1}v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$,*

$$\mathbb{E}_\gamma\left[\|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_\mathcal{Y}^2\right] \leq \mathbb{E}_\gamma\left[\|D_\mathcal{E}F(I-Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right]. \tag{30}$$

Substituting the bounds from Proposition 23 and Proposition 26 on the derivative reconstruction error, we can explicitly write out the bounds for the PCA and DIS bases. This result is analogous to Proposition 3.1 of Zahm et al. (2020).

Corollary 29 (Ridge function error in L^2_γ)**.** *Suppose $F \in H^1_\gamma$. Additionally, assume $(C_\mathcal{X}^{-1}v_i^{\mathrm{DIS}})_{i=1}^r$ and $(C_\mathcal{X}^{-1}\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are well-defined in $\mathcal{X}$ for the exact and empirical input DIS bases. Then,*

$$\mathbb{E}_\gamma\left[\|F - \mathbb{E}_\gamma[F|\sigma((V_r^{\mathrm{DIS}})^\dagger)]\|_\mathcal{Y}^2\right] \leq \sum_{i=r+1}^\infty \mu_i^{\mathrm{DIS}}$$

when using the exact input DIS and

$$\mathbb{E}_\gamma\left[\|F - \mathbb{E}_\gamma[F|\sigma((\widehat{V}_r^{\mathrm{DIS}})^\dagger)]\|_\mathcal{Y}^2\right] \leq \sum_{i=r+1}^\infty \mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_\mathcal{X} - \widehat{H}_\mathcal{X}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_\mathcal{X} - \widehat{H}_\mathcal{X}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\}$$

when using the empirical input DIS.

If additionally $F \in C^1(\mathcal{X},\mathcal{Y})$ and $DF \in L^2(\gamma, \mathcal{L}(\mathcal{X},\mathcal{Y}))$, then

$$\mathbb{E}_\gamma\left[\|F - \mathbb{E}_\gamma[F|\sigma((V_r^{\mathrm{PCA}})^\dagger)]\|_\mathcal{Y}^2\right] \leq \mathbb{E}_\gamma\left[\|DF\|^2_{\mathcal{L}(\mathcal{X},\mathcal{Y})}\right] \sum_{i=r+1}^\infty \mu_i^{\mathrm{PCA}}$$

when using the exact input PCA.

3.6.2 Ridge Function Errors of the Derivatives

We next consider the ridge function error of the derivative term $\mathbb{E}_\gamma[\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F})Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}]$. Recall that $F \in H^2_\gamma$, the Sobolev derivative $D_{\mathcal{E}}F$ is also a mapping in $H^1(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))$. Thus, the projected derivative $D_{\mathcal{E}}F(x)Q_r$ has an optimal ridge function representation

$$\mathbb{E}_\gamma[D_{\mathcal{E}}FQ_r|\sigma(V_r^\dagger)](x) = \mathbb{E}_{z\sim\gamma}[D_{\mathcal{E}}F(Q_rx + (I-Q_r)z)Q_r] = \mathbb{E}_\gamma[D_{\mathcal{E}}F|\sigma(V_r^\dagger)](x)Q_r,$$

which coincides with the derivative of the conditional expectation, $D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$. Using this fact, we can apply the Subspace Poincaré inequality on the projected Sobolev derivative, which bounds the derivative ridge function error by the Hessian acting on the orthogonal complement of the projection Q_r.

Proposition 30 (Subspace Poincaré inequality for the derivative)**.** *Suppose $F \in H^2_\gamma$. Then, given any orthonormal set $V_r = (v_i)_{i=1}^r$ in $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1}v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$,*

$$\mathbb{E}_\gamma\left[\|D_{\mathcal{E}}FQ_r - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right] \leq \sum_{i=1}^\infty \mathbb{E}_\gamma[\|D^2_{\mathcal{E}}F(w_i,\cdot)(I-Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}], \tag{31}$$

where $(w_i)_{i=1}^\infty$ is any orthonormal basis of $\mathcal{E}$ and $D^2_{\mathcal{E}}F(w_i,\cdot)$ is the linear operator $h \mapsto D^2_{\mathcal{E}}F(w_i,h)$.

Proof Lemma 17 implies that $D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D_{\mathcal{E}}F|\sigma(V_r^\dagger)]Q_r$. Therefore,

$$\begin{aligned}\mathbb{E}_\gamma\left[\|D_{\mathcal{E}}FQ_r - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right] &= \mathbb{E}_\gamma\left[\|D_{\mathcal{E}}FQ_r - \mathbb{E}_\gamma[D_{\mathcal{E}}F|\sigma(V_r^\dagger)]Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right]\\ &\leq \mathbb{E}_\gamma\left[\|D_{\mathcal{E}}F - \mathbb{E}_\gamma[D_{\mathcal{E}}F|\sigma(V_r^\dagger)]\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right],\end{aligned}$$

where we have made use of $\|Q_r\|_{\mathcal{L}(\mathcal{E},\mathcal{E})} = 1$. By our assumptions, $D_{\mathcal{E}}F \in H^1(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))$, so we can apply the subspace Poincaré inequality

$$\mathbb{E}_\gamma\left[\|D_{\mathcal{E}}FQ_r - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right] \leq \mathbb{E}_\gamma[\|D^2_{\mathcal{E}}F(\cdot,(I-Q_r)(\cdot))\|^2_{\mathrm{HS}_2(\mathcal{E},\mathcal{Y})}],$$

where $D^2_{\mathcal{E}}F(x)(\cdot,(I-Q_r)(\cdot)) : (h_1,h_2) \mapsto D^2_{\mathcal{E}}F(x)(h_1,(I-Q_r)h_2)$ takes values in $\mathrm{HS}_2(\mathcal{E},\mathcal{Y})$. Moreover, we can write the $\mathrm{HS}_2(\mathcal{E},\mathcal{Y})$ norm as

$$\|D^2_{\mathcal{E}}F(\cdot,(I-Q_r)(\cdot))\|^2_{\mathrm{HS}_2(\mathcal{E},\mathcal{Y})} = \sum_{i=1}^\infty \|D^2_{\mathcal{E}}F(x)(w_i,\cdot)(I-Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})},$$

where $D^2_{\mathcal{E}}F(x)(w_i,\cdot)(I-Q_r) : h \mapsto D^2_{\mathcal{E}}F(x)(w_i,(I-Q_r)h)$. Since the sum consists of non-negative terms, interchanging expectation with summation yields (31).

■

In the case of Q_r coming from the input PCA, we can further bound the Hessian action by the trailing input PCA eigenvalues at the cost of a (potentially large) constant, analogous to that of the derivative reconstruction error.

Proposition 31 (Derivative ridge function error bound with input PCA)**.** *Suppose $F \in H^2_\gamma$ satisfies Assumption 8. Then, for any orthogonal projection $P_r \in \mathcal{P}_r(\mathcal{Y})$, the exact input PCA satisfies*

$$\mathbb{E}_\gamma\left[\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((V_r^{\mathrm{PCA}})^\dagger)])Q_r^{\mathrm{PCA}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right] \leq \mathbb{E}_\gamma[\|x\|^2_{\mathcal{X}}]\mathbb{E}_\gamma\left[\|D^2F\|^2_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}\right]\sum_{i=r+1}^\infty \mu_i^{\mathrm{PCA}}. \tag{32}$$

Proof Since $\|P_r\|_{\mathcal{L}(\mathcal{Y},\mathcal{Y})} = 1$, we can ignore the output projection by writing

$$\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((V_r^{\mathrm{PCA}})^\dagger)])Q_r^{\mathrm{PCA}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \le \|(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((V_r^{\mathrm{PCA}})^\dagger)])Q_r^{\mathrm{PCA}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}. \tag{33}$$

Based on (31), we inspect the Hessian action term $\|D^2_{\mathcal{E}}F(x)(w_i,\cdot)(I - Q_r^{\mathrm{PCA}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}$ for some orthonormal basis $(w_i)_{i=1}^\infty$ of $\mathcal{E}$. By Assumption 8, we have that $D^2F \in L^2(\gamma;\mathcal{L}_2(\mathcal{X},\mathcal{Y}))$, and hence

$$\|D^2_{\mathcal{E}}F(x)(w_i,\cdot)(I - Q_r^{\mathrm{PCA}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \le \|D^2F(x)(w_i,\cdot)\|^2_{\mathcal{L}(\mathcal{X},\mathcal{Y})}\|I - Q_r^{\mathrm{PCA}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{X})},$$

where $\|I - Q_r^{\mathrm{PCA}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{X})} = \sum_{i=r+1}^\infty \mu_i^{\mathrm{PCA}}$ due to (24). Moreover, we know

$$\|D^2F(x)(w_i,\cdot)\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})} = \sup_{\|\hat{x}\|_{\mathcal{X}}\neq 0} \frac{\|D^2F(x)(w_i,\hat{x})\|_{\mathcal{Y}}}{\|\hat{x}\|_{\mathcal{X}}} \le \|D^2F(x)\|_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}\|w_i\|_{\mathcal{X}}.$$

Let $w_i = v_i^{\mathrm{PCA}}$. Due to our definition of v_i^{PCA} being orthonormal in $\mathcal{E}$, we have

$$\|w_i\|^2_{\mathcal{X}} = \|v_i^{\mathrm{PCA}}\|^2_{\mathcal{X}} = \|C_{\mathcal{X}}^{-1/2}C_{\mathcal{X}}^{1/2}v_i^{\mathrm{PCA}}\|^2_{\mathcal{X}} = \mu_i^{\mathrm{PCA}}\|v_i^{\mathrm{PCA}}\|^2_{\mathcal{E}} = \mu_i^{\mathrm{PCA}}.$$

Taking expectations with respect to γ and summing over w_i, we obtain

$$\sum_{i=1}^\infty \mathbb{E}_{x\sim\gamma}[\|D^2F(x)(w_i,\cdot)(I - Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \le \mathbb{E}_{x\sim\gamma}[\|x\|^2_{\mathcal{X}}]\mathbb{E}_{x\sim\gamma}[\|D^2F(x)\|^2_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}] \sum_{i=r+1}^\infty \mu_i^{\mathrm{PCA}}, \tag{34}$$

where we have additionally used the fact that $\sum_{i=1}^\infty \mu_i^{\mathrm{PCA}} = \mathbb{E}_{x\sim\gamma}[\|x\|^2_{\mathcal{X}}]$. Applying the subspace Poincaré inequality (31) to (33) and substituting the bound (34) then yields (32). ■

The case for the derivative-informed subspace is less straightforward. Unlike the PCA basis, the DIS basis does not guarantee an optimal reconstruction of the input x (i.e., not necessarily small $\|I - Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{X})}$), and we cannot make use of an analogous argument. However, similar to the case with output PCA, we recognize that the derivative-informed subspace can also result in a bounded Hessian projection error when the derivatives $D_{\mathcal{E}}F$ are able to represent the Hessian $D^2_{\mathcal{E}}F$ on average. This is the subject of Assumption 11, which directly translates to a bound on the derivative ridge function error in terms of the trailing eigenvalues of the DIS problem.

Proposition 32 (Derivative ridge function error bound with DIS)**.** *Suppose $F \in H^2_\gamma$ satisfies Assumption 11 and $(C_{\mathcal{X}}^{-1}\hat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, for any orthogonal projection $P_r \in \mathcal{P}_r(\mathcal{Y})$, the empirical input DIS satisfies*

$$\mathbb{E}_\gamma\left[\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((\widehat{V}_r^{\mathrm{DIS}})^\dagger)])\widehat{Q}_r^{\mathrm{DIS}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right]$$
$$\le K_H\left(\sum_{i=r+1}^\infty \mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\}\right). \tag{35}$$

Proof As in the computations for the input PCA basis, we have

$$\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((\widehat{V}_r^{\mathrm{DIS}})^\dagger)])\widehat{Q}_r^{\mathrm{DIS}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \le \|(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((\widehat{V}_r^{\mathrm{DIS}})^\dagger)])\widehat{Q}_r^{\mathrm{DIS}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}.$$

We then use the subspace Poincaré inequality (31), which can be alternatively written as

$$\mathbb{E}_\gamma\left[\|(D_{\mathcal{E}}F - D_{\mathcal{E}}\mathbb{E}_\gamma[F|\sigma((\widehat{V}_r^{\mathrm{DIS}})^\dagger)])\widehat{Q}_r^{\mathrm{DIS}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}\right] \le \sum_{i=r+1}^\infty \mathbb{E}_\gamma[\|D^2_{\mathcal{E}}F(\cdot,\hat{v}_i^{\mathrm{DIS}})\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}].$$

Assumption 11 is sufficient to give

$$\sum_{i=r+1}^{\infty} \mathbb{E}_\gamma[\|D^2_{\mathcal{E}} F(\cdot, \widehat{v}_i^{\text{DIS}})\|^2_{\text{HS}(\mathcal{E},\mathcal{Y})}] \leq K_H \sum_{i=r+1}^{\infty} \mathbb{E}_\gamma[\|D_{\mathcal{E}} F \widehat{v}_i^{\text{DIS}}\|^2_{\mathcal{Y}}] = K_H \mathbb{E}_\gamma[\|D_{\mathcal{E}} F (I - \widehat{Q}_r^{\text{DIS}})\|^2_{\text{HS}(\mathcal{E},\mathcal{Y})}].$$

Proposition 26 then yields (35).

■

Here, the inverse inequality constant K_H captures how well the derivatives $D_{\mathcal{E}} F$ represent the Hessian $D^2_{\mathcal{E}} F$ on average. Similar to the derivative inverse inequality, we again expect Assumption 11 to hold when F is largely monotonic. In particular, polynomial mappings of degree n also satisfy the inequality (13) with $K_H = n - 1$. In Section 4, we also show that solution operators for a class of semilinear elliptic PDEs with monotone nonlinearity satisfy Assumption 11.

3.7 Sampling Errors

We now discuss approaches for dealing with sampling errors associated with the estimators of the mean $\widehat{F}$, covariance $\widehat{C}_{\mathcal{Y}}$, and sensitivity operators $\widehat{H}_{\mathcal{X}}$ and $\widehat{H}_{\mathcal{Y}}$. Under mild moment assumptions, one can prove a bound on the expected sampling errors. To this end, we largely follow the approaches taken in Reiß and Wahl (2020); Bhattacharya et al. (2021), where the authors analyzed the sampling errors for the construction of the PCA. The following useful result is presented in the proof of Bhattacharya et al. (2021, Lemma B.2), which we summarize as a standalone lemma and present its proof in Appendix C.6.1.

Lemma 33 (Monte Carlo error for vector-valued random variables)**.** *Let z be a random variable taking values in a separable Hilbert space $\mathcal{H}$ with distribution ν and mean $\bar{z} = \mathbb{E}_{z\sim\nu}[z]$. Suppose that z has a bounded second moment, $\mathbb{E}_{z\sim\nu}[\|z\|^2_{\mathcal{H}}] < \infty$. Then, given $N \in \mathbb{N}$ i.i.d. samples $(z_k)_{k=1}^N \overset{iid}{\sim} \nu$, the sample estimator $\hat{z} = \frac{1}{N}\sum_{k=1}^N z_k$ has an expected square error of*

$$\mathbb{E}_N[\|\hat{z} - \bar{z}\|^2_{\mathcal{H}}] = \frac{\mathbb{E}_{z\sim\nu}[\|z - \bar{z}\|^2_{\mathcal{H}}]}{N},$$

where the expectation $\mathbb{E}_N$ is taken with respect to the random samples $(z_k)_{k=1}^N$.

This allows us to bound all the sampling errors in expectation. In particular, the usual Monte Carlo error of the mean estimator $\widehat{F}$ follows as an immediate corollary.

Corollary 34. *Suppose $F \in L^2_\gamma$. Then, given $N \in \mathbb{N}$ i.i.d. samples $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$, the sample estimator $\widehat{F} = \frac{1}{N}\sum_{k=1}^N F(x_k)$ has an expected square error of*

$$\mathbb{E}_N[\|\widehat{F} - \bar{F}\|^2_{\mathcal{Y}}] = \frac{\|F - \bar{F}\|^2_{L^2_\gamma}}{N}$$

where the expectation $\mathbb{E}_N$ is taken with respect to the random samples $(x_k)_{k=1}^N$.

Additionally, under the assumptions that the random variable z has bounded fourth moments, we can obtain a bound on the error of the sample estimator $\hat{z}$ in terms of its fourth moment. The proof is again provided in Appendix C.6.1.

Lemma 35. *Let z be a random variable taking values in a separable Hilbert space $\mathcal{H}$ with distribution ν and mean $\bar{z} = \mathbb{E}_{z\sim\nu}[z]$. Suppose that z has a bounded fourth moment, $\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^4] < \infty$. Then, given $N \in \mathbb{N}$ i.i.d. samples $(z_k)_{k=1}^N \overset{iid}{\sim} \nu$, the sample estimator $\widehat{z} = \frac{1}{N}\sum_{k=1}^N z_k$ has a fourth-moment error of*

$$\mathbb{E}_N[\|\hat{z} - \bar{z}\|_{\mathcal{H}}^4] \leq \frac{\mathbb{E}_{z\sim\nu}[\|z-\bar{z}\|_{\mathcal{H}}^4] + 3\mathbb{E}_{z\sim\nu}[\|z-\bar{z}\|_{\mathcal{H}}^2]^2}{N^2},$$

where the expectation $\mathbb{E}_N$ is taken with respect to the random samples $(z_k)_{k=1}^N$.

We use the above results to obtain bounds on the sampling errors for the empirical PCA and DIS. In particular, the empirical output PCA errors for the uncentered case are considered in Bhattacharya et al. (2021). In the centered (mean-shifted) case, we additionally need to account for the error in the mean estimate, $\bar{F} - \widehat{F}$. The result is an estimate similar to that in Bhattacharya et al. (2021, Lemma B.2).

Proposition 36. *Suppose $F \in L_\gamma^4$. Then, there exists a constant $M_{C_\mathcal{Y}}$ such that the sampling error for the empirical output PCA based on $N \in \mathbb{N}$ i.i.d. samples of the input, $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$, satisfies the bound*

$$\mathbb{E}_N[\|C_\mathcal{Y} - \widehat{C}_\mathcal{Y}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}^2] \leq \frac{M_{C_\mathcal{Y}}}{N},$$

where the expectation $\mathbb{E}_N$ is taken with respect to the random samples $(x_k)_{k=1}^N$.

We can perform a similar calculation for the sampling errors of $\widehat{H}_\mathcal{X}$ and $\widehat{H}_\mathcal{Y}$, this time under the assumption that the Sobolev derivatives $\|D_\mathcal{E}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}$ have bounded fourth moments.

Proposition 37. *Suppose $F \in W_\gamma^{1,4}$. Then, there exist constants $M_{H_\mathcal{X}}$ and $M_{H_\mathcal{Y}}$ such that the sampling error for the empirical sensitivity operators*

$$\widehat{H}_\mathcal{X} = \frac{1}{N}\sum_{k=1}^N D_\mathcal{E}F(x_k)^* D_\mathcal{E}F(x_k) \quad \text{and} \quad \widehat{H}_\mathcal{Y} = \frac{1}{N}\sum_{k=1}^N D_\mathcal{E}F(x_k) D_\mathcal{E}F(x_k)^*$$

based on $N \in \mathbb{N}$ i.i.d. samples of the input $(x_k)_{k=1}^N \overset{iid}{\sim} \gamma$, satisfy the bounds

$$\mathbb{E}_N[\|H_\mathcal{X} - \widehat{H}_\mathcal{X}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2] \leq \frac{M_{H_\mathcal{X}}}{N}$$

for the input, and

$$\mathbb{E}_N[\|H_\mathcal{Y} - \widehat{H}_\mathcal{Y}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}^2] \leq \frac{M_{H_\mathcal{Y}}}{N}$$

for the output. The expectation $\mathbb{E}_N$ is taken with respect to the random samples $(x_k)_{k=1}^N$.

The proofs for both sampling error results, Propositions 36 and 37, are given in Appendix C.6.2.

Remark 38. *The above results can be extended to high-probability bounds for the sampling errors using Markov's inequality, though these bounds are typically very conservative. Tighter bounds can be obtained using a vector-valued Bernstein inequality under additional assumptions such as sub-Gaussianity of F. We do not present these results here for the sake of brevity.*

4. Analytic Examples

In this section, we present two examples of operators on infinite-dimensional spaces and illustrate how the assumptions in Section 3.1 can be verified. We begin with a simple example of a linear PDE with a nonlinear transformation in the source term, for which we have explicit expressions for the derivative actions that can be used to verify the desired moment and inverse inequalities from Assumptions 8 to 11. We then proceed to consider a more interesting class of operators defined by solutions of semilinear elliptic PDEs involving monotonically increasing nonlinearities. In this case, although we cannot establish the derivative inverse inequality, we are able to show that the remaining moment bounds and the Hessian inverse inequality hold, allowing us to apply the error bounds when using DIS for output dimension reduction and either DIS or PCA for input dimension reduction. The proofs of the results in this section are presented in Appendices D.1 and D.2.

4.1 Coefficient Map of a Linear PDE with Nonlinear Transformation

Consider the linear PDE on the interval $\Omega = (0, \pi)$,

$$-\Delta y = f \quad \text{in } \Omega,$$

with homogeneous Dirichlet boundary conditions. Let f be defined by an expansion in the Fourier sine basis,

$$f(x)(s) = \sum_{k=1}^{\infty} g(x^{(k)}) \sin(ks),$$

where $x = (x^{(k)})_{k=1}^{\infty} \in \ell^2$ is the coefficient vector and $g : \mathbb{R} \to \mathbb{R}$ is a nonlinearity that acts componentwise on x. Here, we are using $x^{(k)}$ to denote the kth component of the vector $x \in \ell^2$ and $s \in \Omega$ to denote the spatial variable to avoid notational clashes. Then, the solution of the PDE is given by the Fourier sine series expansion

$$y(s) = \sum_{k=1}^{\infty} \frac{g(x^{(k)})}{k^2} \sin(ks).$$

We will make the following assumptions on the nonlinearity g.

Assumption 39. *Let $m \geq 2$ be an integer. Suppose $g \in C^m(\mathbb{R})$ is a nonzero odd function with $g(0) = 0$ and $g'(0) = 1$. Moreover, suppose g has bounded derivatives up to order m, i.e., there exists $M > 0$ such that $|g^{(k)}(t)| \leq M$ for all $t \in \mathbb{R}$ and $k = 1, 2, \ldots, m$.*

We then define the input-output map as the mapping from $\mathcal{X} = \ell^2$ to $\mathcal{Y} = \ell^2$ given by

$$F(x) = \left(\frac{g(x^{(k)})}{k^2} \right)_{k=1}^{\infty}, \tag{36}$$

That is, F can be understood as a mapping from the coefficients of the source term (before the nonlinear transformation) to the coefficients of the solution y. Here, Lipschitz continuity of g, which is implied by the boundedness of g', ensures that F is well-defined. Moreover, continuity of $g^{(n)}$, $n = 1, \ldots, m$ ensures that $F \in C^m(\mathcal{X}, \mathcal{Y})$, with derivatives given by

$$D^n F(x)(v_1, \ldots, v_n) = \left(\frac{g^{(n)}(x^{(k)})}{k^2} v_1^{(k)} \cdots v_n^{(k)} \right)_{k=1}^{\infty}. \tag{37}$$

We now consider a Gaussian measure $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ on $\mathcal{X}$ with a diagonal, non-degenerate covariance operator $C_{\mathcal{X}}$. That is, $C_{\mathcal{X}} v = (\sigma_k^2 v^{(k)})_{k=1}^{\infty}$ with $\sigma_k > 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} \sigma_k^2 < \infty$. We can verify that F satisfies the desired differentiability properties and moment bounds.

Proposition 40 (Moment bounds for the coefficient map)**.** *Let $F : \mathcal{X} \to \mathcal{Y}$ be defined as in* (36) *with g satisfying Assumption 39 for some $m \in \mathbb{N}, m \geq 2$. Additionally, let $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ be a Gaussian measure on $\mathcal{X}$ with a diagonal covariance operator $C_{\mathcal{X}} v = (\sigma_k^2 v^{(k)})_{k=1}^{\infty}$, where $\sigma_k > 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} \sigma_k^2 < \infty$. Then, F has the following properties.*

1. *(Differentiability) $F \in C^m(\mathcal{X}, \mathcal{Y})$.*
2. *(Fréchet derivative bounds) $D^n F \in L^p(\gamma, \mathcal{L}_n(\mathcal{X}, \mathcal{Y}))$ for all $n = 1, \ldots, m$ and $p \in [1, \infty)$.*
3. *(Sobolev derivative bounds) $F \in W^{m,p}(\gamma, \mathcal{Y})$ for all $p \in [1, \infty)$. A fortiori, $F \in H^m_\gamma \cap W^{1,4}_\gamma$.*

Moreover, since both the mapping F and the covariance operator $C_{\mathcal{X}}$ are diagonal, we can explicitly compute the expectations for the output and its derivatives, allowing us to verify the inverse inequalities.

Proposition 41 (Inverse inequalities for the coefficient map)**.** *Let $F : \mathcal{X} \to \mathcal{Y}$ and $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ be defined as in Proposition 40. Then, F satisfies the derivative and Hessian inverse inequalities.*

4.2 Semilinear Elliptic PDEs with Monotonically Increasing Nonlinearity

Next, we consider mappings defined by the solution of semilinear elliptic PDEs involving monotonically increasing nonlinearities. In this case, the standard theory of monotone operators and the implicit function theorem can be used to obtain the moment bounds and the Hessian inverse inequality, allowing us to apply the error bounds when using DIS for output dimension reduction and either DIS or PCA for input dimension reduction. To simplify the calculations, we will focus on the case where the PDE domain is one-dimensional, e.g., $\Omega = (0, 1)$, though we believe that the results can be extended to higher-dimensional domains $\Omega \subset \mathbb{R}^d$ with $d \leq 3$ by using appropriate Sobolev embeddings. Again, to avoid notational clashes, we will use $s \in \Omega$ to denote the spatial variable and $d\Omega$ to denote the spatial integration measure.

Let $\Omega = (0, 1)$ and consider the semilinear elliptic PDE,

$$-\Delta y + g(y) = f \quad \text{in } \Omega, \tag{38}$$

with homogeneous Dirichlet boundary conditions. We seek solutions $y \in H_0^1(\Omega)$ given inputs $f \in H^{-1}(\Omega)$, where $H_0^1(\Omega)$ is the usual Sobolev space equipped with the norm $\|v\|_{H_0^1(\Omega)} = \left(\int_\Omega |\nabla v|^2 d\Omega\right)^{1/2}$ and $H^{-1}(\Omega)$ is the topological dual of $H_0^1(\Omega)$. Here, $g : \mathbb{R} \to \mathbb{R}$ is a nonlinearity that acts pointwise on y, understood as a Nemytskii operator from $H_0^1(\Omega)$ to $H^{-1}(\Omega)$. We will make the following assumptions on g.

Assumption 42 (Monotonically increasing nonlinearity with polynomial growth)**.** *Let $m \geq 2$ be an integer. Suppose that the nonlinearity $g \in C^m(\mathbb{R})$ is a monotonically increasing function and $g(0) = 0$. Moreover, suppose that g and its derivatives have at most polynomial growth. That is, there exist $M > 0$ and $q \geq 0$ such that $|g^{(k)}(t)| \leq M(1 + |t|^q)$ for all $t \in \mathbb{R}$ and $k = 0, 1, \ldots, m$.*

Under this assumption, the solution operator $S : H^{-1}(\Omega) \ni f \to S(f) = y \in H_0^1(\Omega)$ is a well-defined operator that is continuously m-times differentiable, a result which follows from conventional theory for semilinear elliptic PDEs. We present the well-posedness and differentiability results in Propositions 64 to 66, which can be found in Appendix D.2.

We will consider a mapping F based on this solution operator S. To this end, we take $\mathcal{X} := L^2(\Omega)$ as the input space and $\mathcal{Y} := H_0^1(\Omega)$ as the output space, along with a centered Gaussian measure $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ on $\mathcal{X}$ with a non-degenerate covariance operator $C_{\mathcal{X}}$. We then define the

mapping $F : \mathcal{X} \to \mathcal{Y}$ as $F(x) := S(\iota_{\mathcal{X}} x)$, where $\iota_{\mathcal{X}} : \mathcal{X} \to \mathcal{Y}'$ is the canonical embedding of $L^2(\Omega)$ into $H^{-1}(\Omega)$, i.e., $\iota_{\mathcal{X}} x = \langle x, \cdot \rangle_{L^2(\Omega)} \in H^{-1}(\Omega) = \mathcal{Y}'$. Recall that this embedding operator is compact, and we have $\|\iota_{\mathcal{X}} x\|_{H^{-1}(\Omega)} \leq \kappa_p \|x\|_{L^2(\Omega)}$ for all $x \in \mathcal{X}$, where κ_p is the Poincaré constant. Thus, F inherits the regularity properties of S, yielding the following result.

Proposition 43. *Let $F = S \circ \iota_{\mathcal{X}}$ be the mapping from $\mathcal{X} = L^2(\Omega)$ to $\mathcal{Y} = H_0^1(\Omega)$ defined above, where $S : H^{-1}(\Omega) \to H_0^1(\Omega)$ is the solution operator of the semilinear elliptic PDE (38) with the nonlinearity g satisfying Assumption 42 with $m \in \mathbb{N}, m \geq 2$. Let $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ be a centered Gaussian measure on $\mathcal{X}$ with a non-degenerate covariance operator $C_{\mathcal{X}}$. Then, F has the following properties:*

1. *(Differentiability) $F \in C^m(\mathcal{X}, \mathcal{Y})$.*
2. *(Fréchet derivative bounds) $D^n F \in L^p(\gamma, \mathcal{L}_n(\mathcal{X}, \mathcal{Y}))$ for all $n = 1, \ldots, m$ and $p \in [1, \infty)$.*
3. *(Sobolev derivative bounds) $F \in W^{m,p}(\gamma, \mathcal{Y})$ for all $p \in [1, \infty)$. A fortiori, $F \in H^m_\gamma \cap W^{1,4}_\gamma$.*

Moreover, we can show that F satisfies the Hessian inverse inequality, which together with the moment bounds is sufficient to apply the result for using DIS for input dimension reduction.

Proposition 44. *Let $F : \mathcal{X} \to \mathcal{Y}$ and $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ be defined as in Proposition 43. Then, F satisfies the Hessian inverse inequality.*

5. Numerical Experiments

In this section, we present several numerical experiments to validate the performance of RBNOs in learning an operator and its derivative. Here, we focus on operators arising from the solution of PDEs given inputs distributed as Gaussian random fields. We compare the generalization errors in terms of the L^2_γ norm and H^1_γ (semi)norm across different dimension reduction strategies, ranks, architectural choices, and training sample sizes. The generalization errors are also decomposed into their reconstruction error and ridge function error (neural network error) components. For the reconstruction errors, we compare the reconstruction errors based on the exact covariance and sensitivity operators (using large sample sizes as a proxy), as well as the excess risk associated with the sample estimation. For the neural network errors, we make comparisons for neural networks constructed using the different basis pairs, and consider both H^1_γ Sobolev training, which explicitly encourages derivative consistency, as well as conventional L^2_γ training, which only accounts for the output accuracy in training.

These numerical experiments seek to explore both agreement with our theory and the practical settings that are not covered by the theoretical analysis. In particular, error sources such as the statistical errors in the finite training set for the neural network and optimization errors associated with the nonconvex training problem are not analyzed, but will contribute to the overall generalization error.

5.1 PDE Problems

For the numerical experiments, we consider elliptic PDE problems posed on 1D and 2D domains Ω, where discretization errors can be controlled with modest discretization dimensions. This ensures that the discretized derivative operators (i.e., Jacobian matrices) can be computed in full for the purpose of generating test data. Nevertheless, the set of PDE problems we consider features nonlinear PDEs and non-trivial input-output dependence.

Here, we will let x denote the input function, y denote the solution, and instead will use $s \in \mathbb{R}^d, d = 1, 2$ to denote the spatial variable. Additionally, we will use ∇ to denote the gradient and Δ to denote the Laplacian, both with respect to the spatial variable s. For the inputs, we will use Gaussian random fields $\gamma = \mathcal{N}(0, C_{\mathcal{X}})$ over $\mathcal{X} = L^2(\Omega)$, with $C_{\mathcal{X}}$ taking the form of the inverse of an elliptic PDE operator, $C_{\mathcal{X}} = (-a_\Delta \Delta + a_I)^{-\alpha}$, where $a_\Delta > 0, a_I > 0$, and $\alpha > 0$ control the pointwise variance, correlation structure, and smoothness of the random field.

Semilinear elliptic PDE Our first 1D example is the semilinear elliptic PDE posed on the unit interval $\Omega = (0, 1)$:

$$\begin{aligned} -\nabla \cdot (c_1(s)\nabla y(s)) + c_2 y(s)^3 &= x(s), \qquad s \in \Omega, \\ y(0) &= 0, \\ \frac{\partial y}{\partial s}(1) &= 0, \end{aligned}$$

where $c_1(s) = 0.001 + 0.1\chi_{(0.5,1)}(s)$ and $c_2 = 1$ are the diffusion and reaction coefficients, and $x \sim \mathcal{N}(0, (-2\Delta + 10)^{-1})$.

Steady Burgers equation For a second 1D example, we consider the steady Burgers equation posed on the unit interval $\Omega = (0, 1)$:

$$\begin{aligned} -\nabla \cdot (c_1 \nabla y(s)) + y(s)\nabla y(s) &= c_2(s)x(s), \qquad s \in \Omega, \\ y(0) &= 0, \\ y(1) &= 0, \end{aligned}$$

where $c_1 = 0.01$ is the viscosity and the source term is a profile $c_2(s) = (40/\pi)\exp(-(s-0.4)^2/0.05^2)$ centered at $s = 0.4$ and scaled by the random field $x \sim \mathcal{N}(0, (-10\Delta + 20)^{-1})$.

Linear elasticity We also consider a 2D problem arising from linear elasticity with a random Young's modulus. To avoid overloading notation, we use c_λ and c_μ to denote the Lamé parameters (as opposed to the commonly used λ and μ). These are given as

$$c_\lambda = \frac{c_E c_\nu}{(1 + c_\nu)(1 - 2c_\nu)}, \quad c_\mu = \frac{c_E}{2(1 + c_\nu)},$$

where c_E is the Young's modulus and c_ν is the Poisson's ratio. We then introduce the stress tensor $\mathbf{T}$ as a function of the displacement $y = (y_1, y_2)$ such that

$$\mathbf{T} = c_\lambda \nabla \cdot y \mathbf{I} + c_\mu \left(\nabla y + \nabla y^T \right),$$

where $\mathbf{I}$ is the identity tensor on $\mathbb{R}^2$. We pose the PDE on the domain $\Omega = (0, 1)^2 \setminus \Omega_R$, where Ω_R is a disk of radius $R = 0.2$ centered at $(0.5, 0.5)$. The governing equations are then given by

$$\begin{aligned} -\nabla \cdot \mathbf{T} &= 0 \quad \text{in } \Omega, \\ y &= 0 \quad \text{on } \Gamma_L, \\ \mathbf{T}n &= \tau \quad \text{on } \Gamma_R, \\ \mathbf{T}n &= 0 \quad \text{on } \Gamma_T \cup \Gamma_B \cup \partial\Omega_R, \end{aligned}$$

where Γ_L is the left boundary, Γ_R is the right boundary, Γ_T is the top boundary, Γ_B is the bottom boundary, and $\partial\Omega_R$ is the boundary of the disk Ω_R. Additionally, τ is the traction vector on the right boundary and n is the outward normal vector. The input variable $x(s)$ is a Gaussian random

field defining the Young's modulus $c_E(s) = 1 + 40\exp(x(s))$ where $x \sim \mathcal{N}(0, (-0.2\Delta + 2)^{-2})$. Moreover, we fix $c_\nu = 0.3$, and take

$$\tau(s) = \left(\frac{1}{0.1\sqrt{2\pi}} \exp\left(-\frac{(s_2 - 0.5)^2}{2 \cdot 0.1^2} \right), 0 \right).$$

This corresponds to a tensile load with a Gaussian profile applied to the right boundary.

5.2 Data Generation and Basis Computation

We use the finite element method to discretize the PDEs and generate the training and test data. This is implemented using FEniCS (Logg et al., 2012) for the finite element solvers and hIPPYlib/hiPPYflow (Villa et al., 2021; O'Leary-Roseberry and Villa, 2021) for training data generation. In particular, for each input sample $x_k \sim \gamma$, the PDE is solved to obtain the output $y_k = F(x_k)$. Moreover, given discretizations of the input and output spaces with dimensions $d_\mathcal{X}$ and $d_\mathcal{Y}$, the derivative of the solution operator, $DF(x_k)$, corresponds to a Jacobian matrix of size $d_\mathcal{Y} \times d_\mathcal{X}$. This can be computed using direct and adjoint sensitivity methods. That is, we can obtain a Jacobian-vector product (JVP) $DF(x_k)v$ or a vector-Jacobian product (VJP) $DF(x_k)^T v$ for any vector $v \in \mathbb{R}^{d_\mathcal{X}}$ or $\mathbb{R}^{d_\mathcal{Y}}$ by solving the linearized state PDE or its adjoint, respectively.

For the 1D examples, we use a uniform mesh with $N_{\text{el}} = 256$ linear elements. In this case, the discretized dimensions of the input and output spaces are $d_\mathcal{X} = d_\mathcal{Y} = 257$. For the 2D example, we use an unstructured mesh with linear triangular elements, where the discretized dimensions of the input and output spaces are $d_\mathcal{X} = 407$ and $d_\mathcal{Y} = 814$. In either case, we have chosen the discretization dimension to be sufficiently large such that numerical errors in solving the PDE are unimportant, yet sufficiently small so that we can compute the Jacobian matrix, $DF(x) \in \mathbb{R}^{d_\mathcal{Y} \times d_\mathcal{X}}$, in full by direct sensitivity methods. That is, we generate training and testing data tuples of the form $(x_k, y_k, DF(x_k))$ for $k = 1, \ldots, N$, where $x_k \sim \gamma$. For each training run, the training data are used to estimate the covariance operator $\widehat{C}_\mathcal{Y}$ or the sensitivity operators $\widehat{H}_\mathcal{X}$ and $\widehat{H}_\mathcal{Y}$, which are then used to solve the eigenvalue problem to obtain the basis functions.

We note that in practice, for high discretization dimensions, randomized methods such as those described in Saibaba et al. (2015) can be used to efficiently solve the PCA and DIS eigenvalue problems. This requires only the Jacobian-vector products and vector-Jacobian products on a small number of random vectors, rather than the full Jacobian matrix itself. Similarly, for Sobolev training, one only needs to compute the reduced Jacobian matrix, $U_r^\dagger DF(x_k) V_r$, corresponding to the action of the derivative operator on the reduced bases U_r and V_r obtained from the PCA/DIS procedures. Moreover, by reusing factorizations of the linearized PDE solves for multiple JVPs or VJPs, the computational cost of generating the derivative data can be significantly reduced to be comparable to that of generating the output value data. We refer to O'Leary-Roseberry, Villa, et al. (2022); O'Leary-Roseberry et al. (2024) for more detailed descriptions of practical procedures for computing DIS and derivative training data.

While not considered here, we also note that the dimension reduction and Sobolev training framework can in principle be applied to time-dependent PDEs, such as in Chen et al. (2025); Go and Chen (2025b). In these cases, the computational efficiency of such approaches depends on the class of time-dependent PDEs considered; parabolic PDEs tend to be amenable to linear dimension reduction, while hyperbolic and advection-dominated PDEs tend to have slow decay of the Kolmogorov n-width, rendering linear dimension reduction less effective. Moreover, modifications to the approximation theory may be required depending on the problem formulation (e.g., learning the state at some final time versus learning a time-dependent state) and the representation

Table 2: Hyperparameters used for the latent neural networks in the numerical experiments for the three PDE problems considered. Note that the width is expressed as a multiple of the rank r.

Problem	Depth d_L	Width d_W	Activation ψ	Initial learning rate
Semilinear elliptic	6	$2r$	softplus	0.001
Steady Burgers	4	$2r$	softplus	0.00025
Linear elasticity	4	$2r$	softplus	0.001

used to approximate the latent dynamics (e.g., using a recurrent neural network versus explicitly parametrizing the neural network by time as in POD-DL-ROMs and DeepONets).

5.3 Neural Network Architecture and Training

In the numerical experiments, we use the Sobolev training formulation (5) to train the neural networks using both the output value and derivative data. The L^2_γ and H^1_γ components of the loss function are normalized by scaling each data point by its norm, i.e., $a_{0,k} = \|U_r^\dagger(F(x_k) - b)\|_2^2$ and $a_{1,k} = \|U_r^\dagger D_{\mathcal{E}} F(x_k) V_r\|_F^2$, as described in Section 2.4. We have found that this approach works well to prevent the neural network from overfitting to outliers in the derivative data. Recall that for conventional L^2_γ training, we simply omit the derivative term, adopting (7) as the training objective with the same normalization $a_{0,k}$ for the output value data.

We consider fully connected feedforward neural networks with a constant width d_W for the hidden layers and a depth of d_L. These are trained using the Adam optimizer with a constant batch size of 25 for 3,000 epochs, using a piecewise constant learning rate schedule that is successively halved five times starting at the 2,250th epoch. The hyperparameters, including d_W, d_L, and initial learning rates are selected independently for each PDE problem, and are summarized in Table 2.

5.4 Numerical Results

5.4.1 Verifying the Derivative Inverse Inequality

While we have theoretically shown that the Hessian inverse inequality can hold for nontrivial examples such as semilinear elliptic PDEs, we only demonstrated the derivative inverse inequality for a simpler example with diagonal derivative operators. For this reason, we begin by empirically verifying the derivative inverse inequality for the three PDE examples considered. Though it is theoretically possible to verify the Hessian inverse inequality, we omit this calculation due to the significant computational burdens associated with evaluating the second derivative tensors.

Recall that the derivative inverse inequality states that there exists a constant K_D such that for all $u \in \mathcal{Y}$, $\langle u, H_{\mathcal{Y}} u\rangle_{\mathcal{Y}} \leq K_D \langle u, C_{\mathcal{Y}} u\rangle_{\mathcal{Y}}$. Here, it suffices to verify the inequality for $u \in \mathcal{Y}$ that satisfy $\langle u, C_{\mathcal{Y}} u\rangle_{\mathcal{Y}} > 0$ without loss of generality. Indeed, if $\langle u, C_{\mathcal{Y}} u\rangle_{\mathcal{Y}} = \mathbb{E}_\gamma[\langle u, F - \bar{F}\rangle_{\mathcal{Y}}^2] = 0$, then $f(x) = \langle u, F(x) - \bar{F}\rangle_{\mathcal{Y}} = 0$ as an element of $L^2(\gamma, \mathbb{R})$. Since $f \in H^1(\gamma, \mathbb{R})$, this in turn implies that $D_{\mathcal{E}} f(x) = D_{\mathcal{E}} F(x)^* u = 0$ as an element of $L^2(\gamma, \mathcal{E})$, and hence $\langle u, H_{\mathcal{Y}} u\rangle_{\mathcal{Y}} = 0$.

Thus, with a fixed discretization, both $H_{\mathcal{Y}}$ and $C_{\mathcal{Y}}$ are finite-dimensional matrices, and the derivative inverse inequality will always hold with a finite constant K_D. In order to assess the validity of the derivative inverse inequality for the underlying infinite-dimensional operators, we verify that K_D is bounded as the discretization dimension increases. To do so, we compute K_D for a sequence of discretizations with increasing dimensions and check that K_D remains bounded. Numerically, we can estimate K_D by solving the eigenvalue problem $C_{\mathcal{Y}}^{\dagger/2} H_{\mathcal{Y}} C_{\mathcal{Y}}^{\dagger/2} u = \lambda u$, where

$C_{\mathcal{Y}}^{\dagger/2} = \sum_{j=1}^{r} (\lambda_j^{\text{PCA}})^{-1/2} u_j^{\text{PCA}} \otimes u_j^{\text{PCA}}$ is the square root of the pseudo-inverse of $C_{\mathcal{Y}}$ with r being the rank of $C_{\mathcal{Y}}$, and taking K_D to be the largest eigenvalue $\lambda_{\max}$.

The estimates of K_D for the three PDE examples are shown in Figure 1. The operators $H_{\mathcal{Y}}$ and $C_{\mathcal{Y}}$ are estimated using 20,000 training samples to ensure that the estimates of $H_{\mathcal{Y}}$ and $C_{\mathcal{Y}}$ are accurate. In all three examples, we find that the estimates of K_D do not grow significantly with the discretization dimension. However, we also find that the estimates of K_D for the steady Burgers equation example are much larger than those for the semilinear elliptic PDE and linear elasticity examples, and fluctuate more with respect to the discretization dimension. This is likely due to the sensitivity of the eigenvalue problem to numerical errors since the eigenvalues of $C_{\mathcal{Y}}$ decay rapidly to zero. Overall, we take these results as suggestive but not conclusive evidence that the derivative inverse inequality holds for the examples considered.

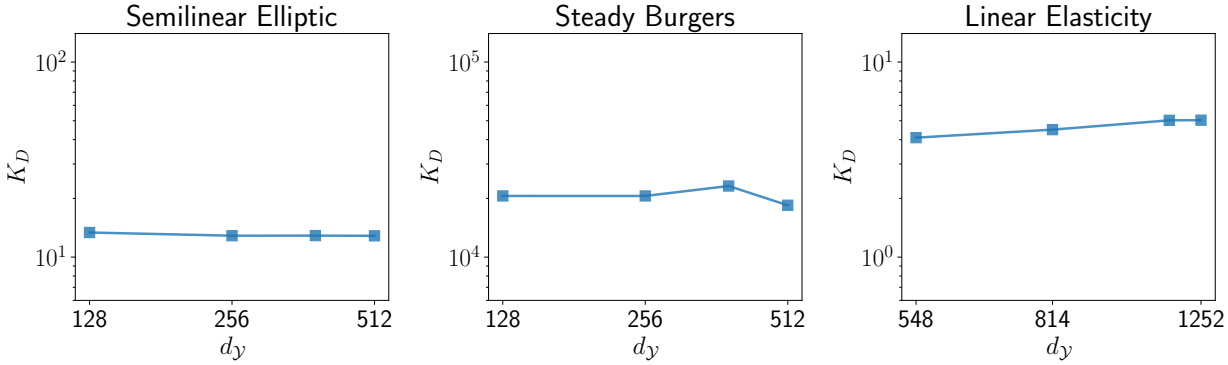


Figure 1: Estimates of K_D for the three PDE examples across a sequence of discretizations with increasing dimensions. The operators $H_{\mathcal{Y}}$ and $C_{\mathcal{Y}}$ are estimated using 20,000 training samples.

5.4.2 Comparison of Reconstruction Errors

We begin by presenting a comparison of the different choices of dimension reduction strategies in reconstructing the output and the derivatives. To this end, we compute the PCA/DIS bases using a large sample size of $N = 20{,}000$, which we take as a proxy for the exact PCA/DIS. The resulting eigenvalues are shown in Figure 2 for the PCA/DIS bases corresponding to the three PDE problems: semilinear elliptic, steady Burgers, and linear elasticity. Additionally, we present samples of the inputs and outputs, along with a selection of the first few basis vectors in each case. These are given in Figure 3 for the 1D examples (semilinear elliptic and steady Burgers PDEs) and in Figure 4 for the 2D linear elasticity problem.

In the spectra of the covariance/sensitivity operators in Figure 2, we observe that the eigenvalues decay rapidly at an algebraic rate (i.e., $\lambda_i \sim i^{-\beta}$ for some $\beta > 0$) in all cases. On the input side, the PCA eigenvalues simply correspond to the eigenvalues of the covariance operator (second-order/fourth-order elliptic PDE operators). These decay rates are independent of the input-output map. On the other hand, the DIS eigenvalues tend to decay at a faster rate than the PCA eigenvalues, reflecting the sensitivity information about the underlying map. This can be seen in the input basis vectors shown in Figure 3. Whereas PCA basis vectors are global sinusoidal functions, the DIS basis vectors are localized to the regions of low diffusion coefficient in the semilinear elliptic example and near the Gaussian peak in the steady Burgers equation example. On the output side, the spectra and corresponding basis vectors are similar between the PCA and DIS in the examples considered. This reflects the fact that both the output PCA and DIS take into account the

underlying mapping. Moreover, we also observe that the DIS eigenvalues are larger than the PCA eigenvalues.

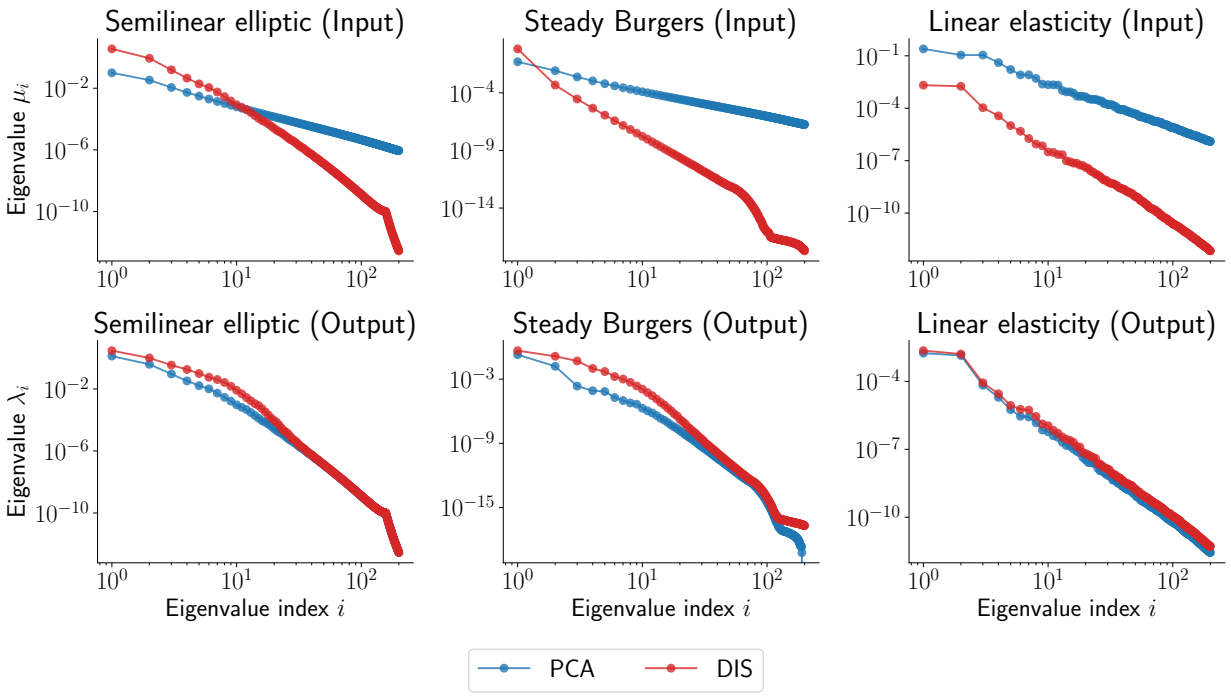


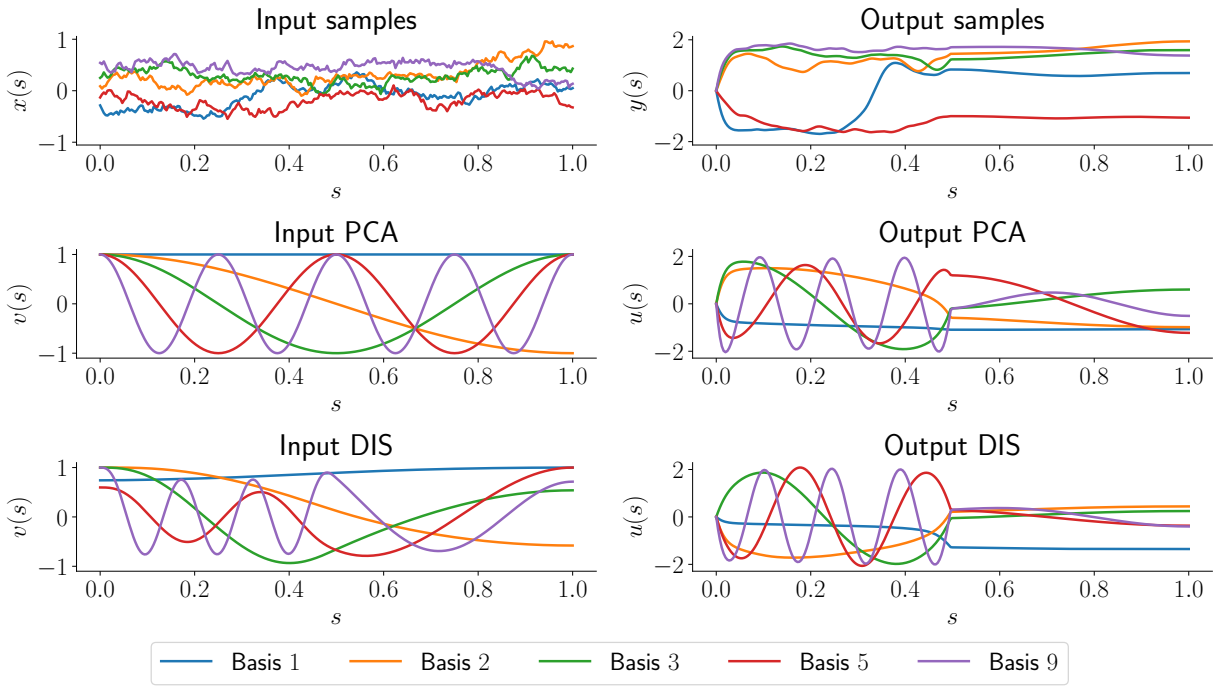
Figure 2: First 200 eigenvalues corresponding to the PCA and DIS bases for the three PDE problems considered. These are computed using a large sample size of $N = 20{,}000$, which we take as a proxy for the exact PCA/DIS.

In Figure 5, we present the reconstruction errors for the output (by the output PCA/DIS) and its derivative (by output PCA/DIS and input PCA/DIS) for the three PDE problems considered in this work. That is, we plot the reconstruction errors,

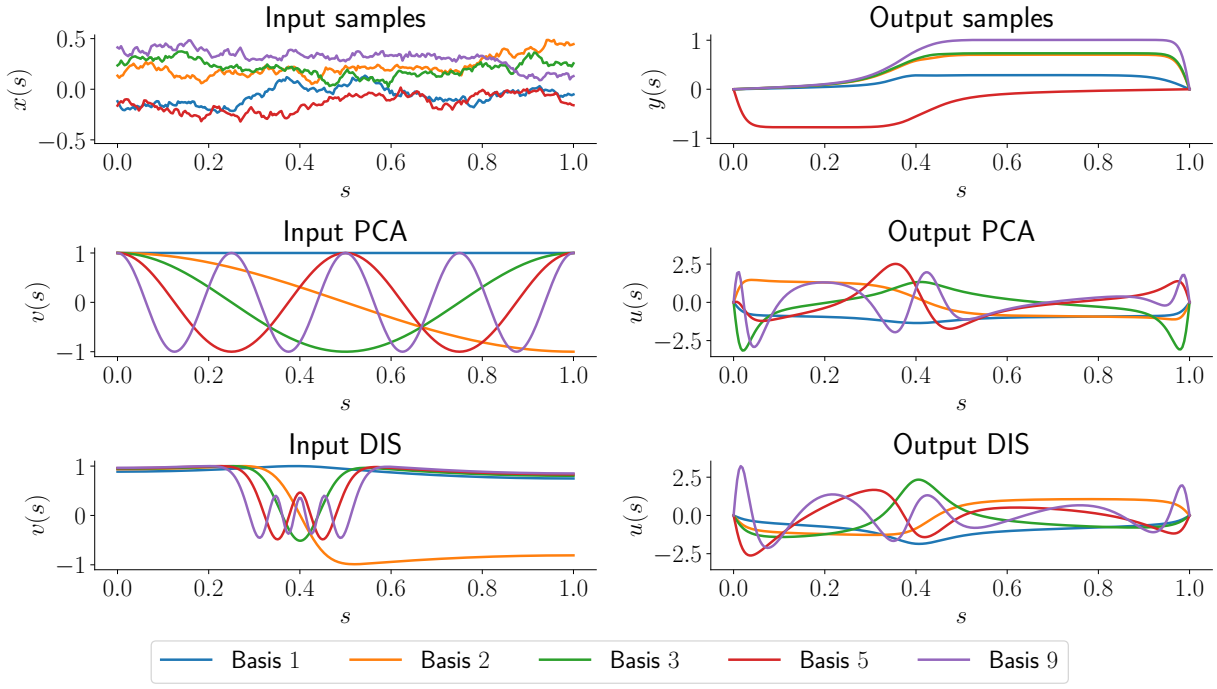
$$\frac{\|(I-P_r)(F-\bar{F})\|_{L^2_\gamma}^2}{\|F\|_{L^2_\gamma}^2}, \quad \frac{\|(I-P_r)D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}{\|D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}, \quad \text{and} \quad \frac{\|D_{\mathcal{E}}F(I-Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}{\|D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2},$$

for different ranks r, where each is normalized by the second moment of the output or the derivative. Here, the errors are computed using a test set of $N_{\text{test}} = 22{,}000$ output and derivative samples. The corresponding reconstruction error bounds of Section 3.5 are also plotted for reference. Specifically, for the output reconstruction by output PCA/DIS and the derivative reconstruction by input/output DIS, the bounds are simply the trailing eigenvalue sums and are computed and plotted exactly in Figure 5. On the other hand, for the derivative reconstruction by input/output PCA, the bounds are additionally scaled by a constant ($\mathbb{E}_\gamma[\|DF\|^2_{\mathcal{L}(\mathcal{X},\mathcal{Y})}]$ and K_D, respectively). Since these constants can be large and may obscure the visualization, we do not explicitly plot the bounds using these constants. Instead, we plot the lines $K_1 \sum_{i=r+1}^{\infty} \lambda_i^{\text{PCA}}$ and $K_2 \sum_{i=r+1}^{\infty} \mu_i^{\text{PCA}}$ for some prescribed values of K_1, K_2 to illustrate the decay rate of the reconstruction error bound.

For the output reconstruction, output PCA and DIS perform comparably, with significant reductions in the reconstruction error as the rank r increases. The two differ only slightly in the steady Burgers example, where PCA errors are noticeably smaller for the lower ranks. Similarly, the two output bases are also comparable when reconstructing the derivative, where the reconstruction

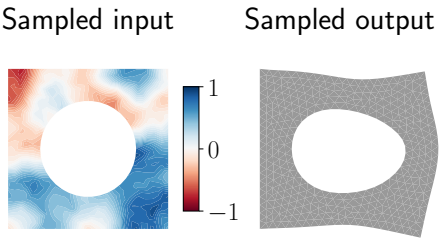


(a) Semilinear elliptic PDE

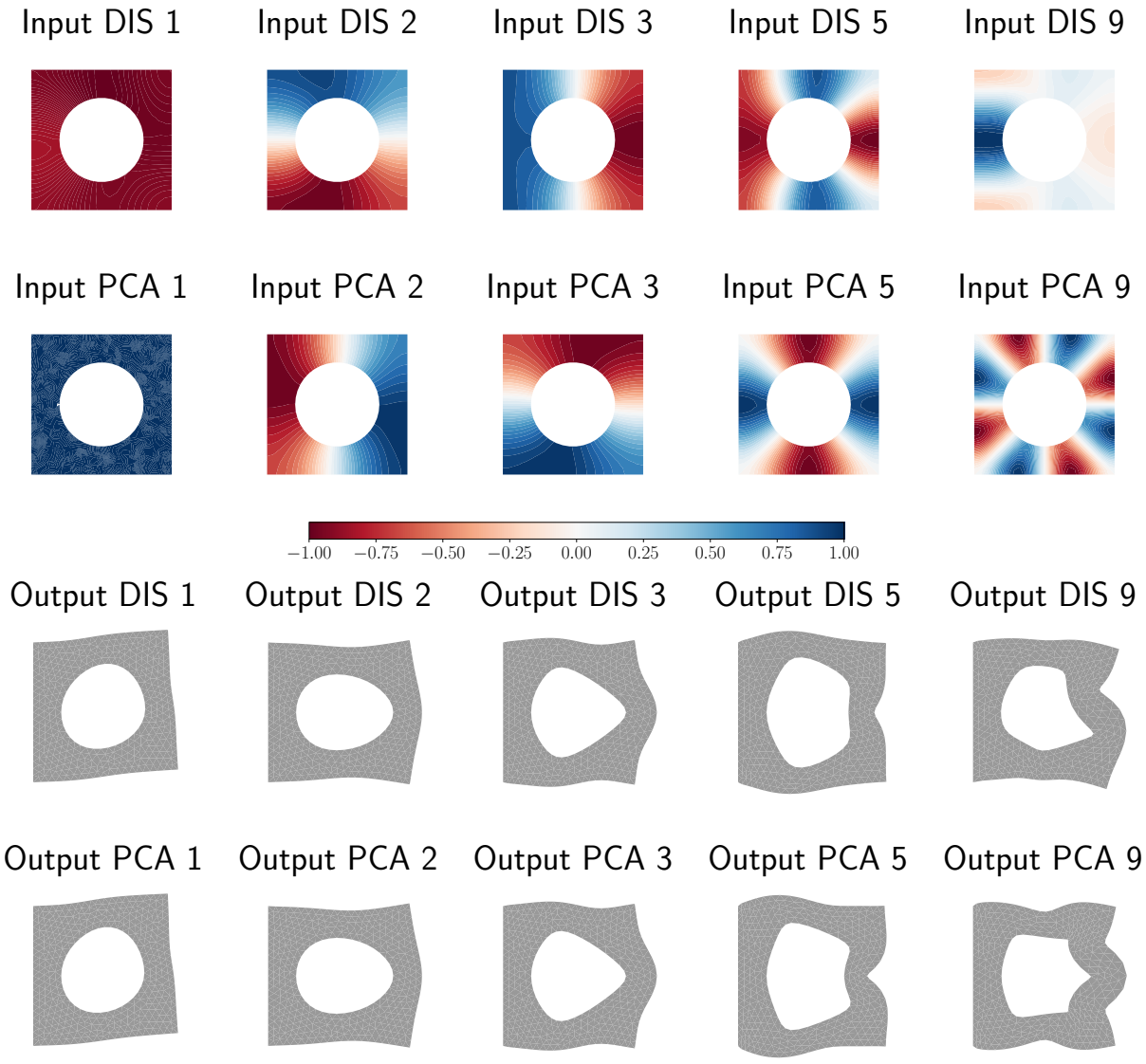


(b) Steady Burgers equation

Figure 3: Samples of input-output pairs along with the 1st, 2nd, 3rd, 5th, and 9th PCA/DIS basis vectors for the 1D PDE examples. Top: semilinear elliptic PDE. Bottom: steady Burgers equation. Note the localization of the input DIS basis vectors to the region $s \leq 0.5$ in the semilinear elliptic PDE example where the diffusion coefficient is smallest, and to the region near $s = 0.4$ that is the center of the Gaussian profile scaling the source in the Burgers example.

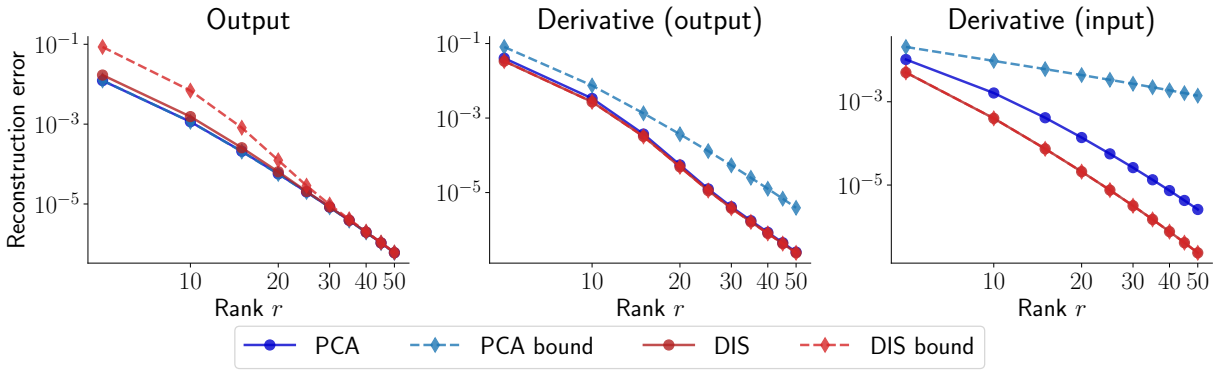


(a) Sample input and output

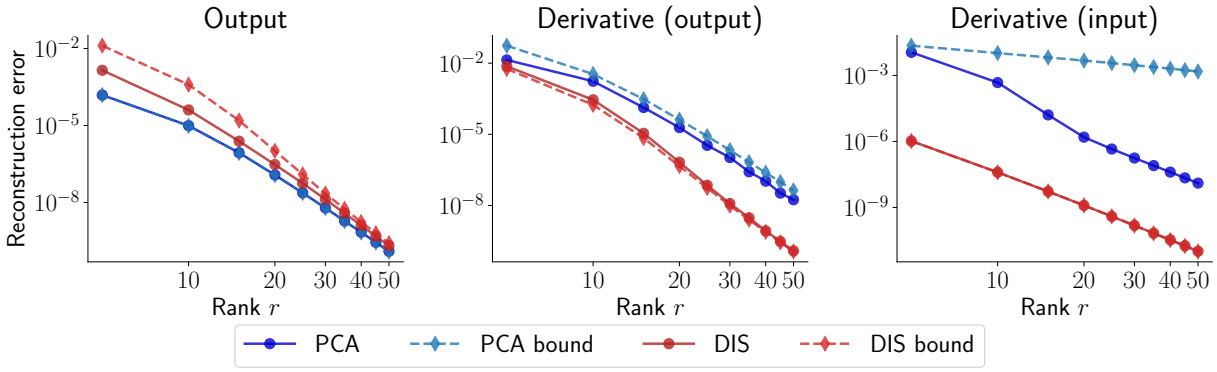


(b) Input and output bases

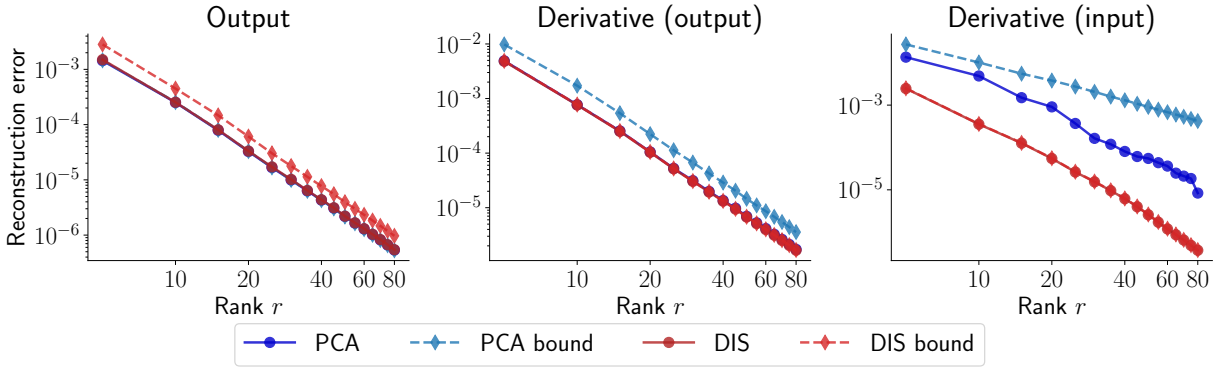
Figure 4: Top: a sample of an input-output pair for the linear elasticity problem. Bottom: the 1st, 2nd, 3rd, 5th, and 9th PCA/DIS basis vectors for the input and output of the linear elasticity problem.



(a) Semilinear elliptic PDE



(b) Steady Burgers equation



(c) Linear elasticity equation

Figure 5: Reconstruction errors (normalized by the second moments of the test data) in the output and its derivatives for the three PDE problems. Left column: reconstruction error of the output by output PCA/DIS. Center column: reconstruction error of the derivative by the output PCA/DIS. Right column: reconstruction error of the derivative by the input PCA/DIS. Bounds for the reconstruction error in terms of the trailing eigenvalues are shown in the dashed lines. Here, true bounds are shown for the output PCA/DIS in output reconstruction and input/output DIS for derivative reconstruction cases. For the reconstruction of the derivatives by input/output PCA basis, we do not explicitly compute the constant, but instead the plot shows the line corresponding to $K_1 \sum_{i=r+1}^{\infty} \lambda_i^{\text{PCA}}$ and $K_2 \sum_{i=r+1}^{\infty} \mu_i^{\text{PCA}}$ for some prescribed K_1, K_2 to illustrate the decay rate of the reconstruction error bound.

errors using the PCA and DIS bases are almost identical for the semilinear elliptic PDE and linear elasticity examples. However, for the steady Burgers example, the DIS reconstruction errors are consistently smaller than those of PCA, though both errors do decay rapidly. On the other hand, when reconstructing the derivative using the input PCA/DIS, the DIS bases produce significantly lower errors than the PCA bases. As previously mentioned, this is to be expected, since the input PCA basis does not incorporate any information about the underlying map and hence cannot be expected to accurately capture its derivatives.

We also make a few remarks on the reconstruction error bounds. First, we note that for the cases of output reconstruction by the output PCA and derivative reconstruction by the input/output DIS, the bounds are, in fact, equalities, and match the observed errors. In the case of the output reconstruction by the output DIS, the bounds, though slightly conservative, do provide a good approximation of the observed errors, especially for the higher ranks. For the derivative reconstruction by the output PCA, the bound required the additional assumption of Assumption 10, postulating the existence of a constant that relates the map's derivative $D_{\mathcal{E}}F$ to its output F. While this is only numerically verified for the examples, the reconstruction errors do appear to decay at a rate consistent with a bound of the form $K_D \sum_{i=r+1}^{\infty} \lambda_i^{\mathrm{PCA}}$, suggesting that the assumption may be reasonable. Finally, for the derivative reconstruction bounds of the input PCA, we observe that the $K \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{PCA}}$ bounds are overly conservative for the 1D example problems. The observed errors instead decay at a faster rate that tracks with the DIS rate, suggesting that the PCA bases may be serendipitously aligned with the sensitivity of the operator in these cases. On the other hand, for the 2D linear elasticity problem, the observed errors align more closely with the $K \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{PCA}}$ bound, which is a slower decay than the DIS.

5.4.3 Excess Risks in Reconstruction Errors

Next, we consider the excess risks associated with the sampling errors in the empirical PCA/DIS. That is, we compute the empirical PCA and DIS using smaller sample sizes ranging from $N = 80$ to $N = 640$, and then compare the excess in reconstruction errors relative to the proxy of the exact PCA/DIS, which we compute with $N = 20{,}000$. That is, for each empirical output PCA $\widehat{U}_r^{\mathrm{PCA}}$, we evaluate the excess risk in the output reconstruction error,

$$\frac{\|(I - \widehat{P}_r^{\mathrm{PCA}})(F - \bar{F})\|_{L^2_\gamma}^2 - \|(I - P_r^{\mathrm{PCA}})(F - \bar{F})\|_{L^2_\gamma}^2}{\|F\|_{L^2_\gamma}^2},$$

while for the empirical output/input DIS, $\widehat{U}_r^{\mathrm{DIS}}$ and $\widehat{V}_r^{\mathrm{DIS}}$, we evaluate the excess risk in the derivative reconstruction error,

$$\frac{\|(I - \widehat{P}_r^{\mathrm{DIS}})D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 - \|(I - P_r^{\mathrm{DIS}})D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}{\|D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}$$

and

$$\frac{\|D_{\mathcal{E}}F(I - \widehat{Q}_r^{\mathrm{DIS}})\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 - \|D_{\mathcal{E}}F(I - Q_r^{\mathrm{DIS}})\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2}{\|D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2},$$

all normalized by the second moment of the output or the derivative, respectively. Here, the errors are again estimated using the test set of $N_{\mathrm{test}} = 22{,}000$ function and derivative samples. This is repeated over 10 independent runs to obtain results on the average excess risk. We present these in Figure 6, Figure 7, and Figure 8 for the semilinear elliptic, steady Burgers, and linear elasticity problems, respectively.

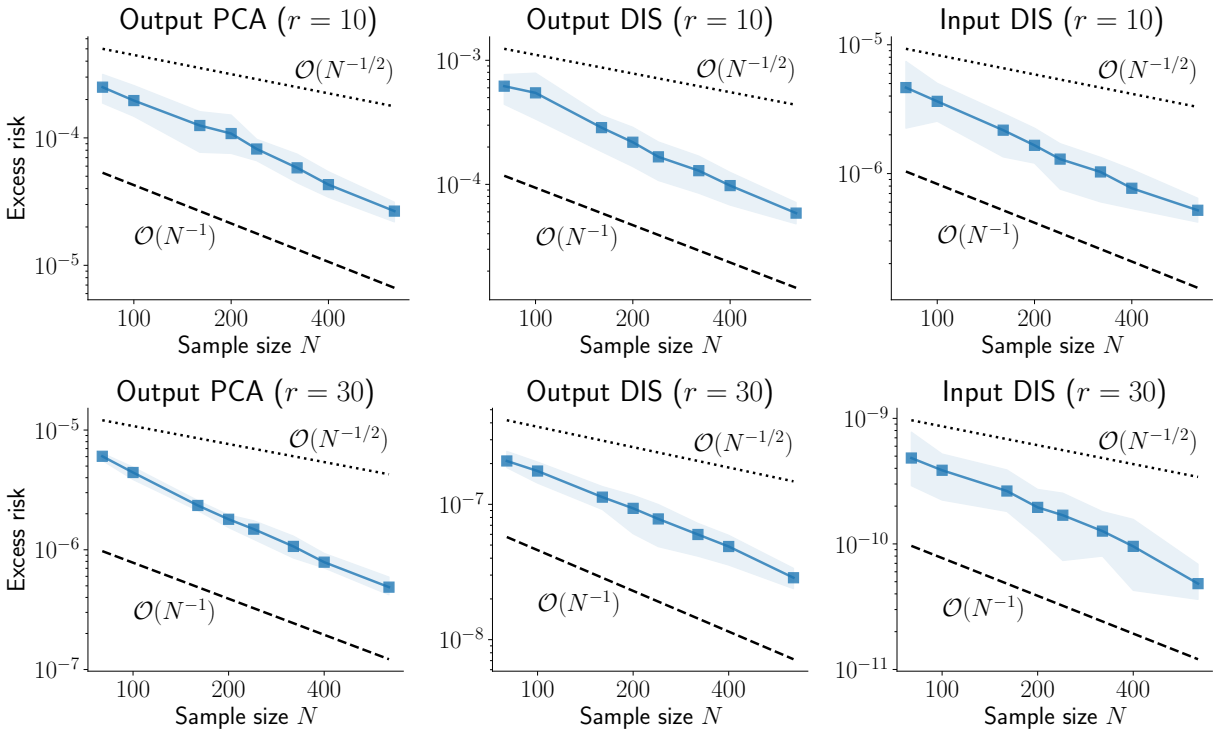


Figure 6: Excess risks in the output and derivative reconstruction errors for the semilinear elliptic PDE problem for ranks $r = 10$ and $r = 30$. Values are normalized by the second moments of the test data. For reference, the global $N^{-1/2}$ rate is shown in the dotted line while the local N^{-1} rate is shown in the dashed line. The filled regions correspond to the 10%–90% quantile ranges of the 10 independent runs.

The excess risks corresponding to the semilinear elliptic and linear elasticity examples both appear to follow the local rate of $\mathcal{O}(N^{-1})$ for both PCA and DIS and for both ranks considered. Moreover, the excess risks decrease as the ranks increase, implying that the inverse dependency on the spectral gap in the bounds of Propositions 25 and 26 is not strongly felt in these examples. With the rapid decay of eigenvalues, spectral gaps for the larger ranks are orders of magnitude smaller than those of the smaller ranks and should otherwise have a great impact on the excess risk bound. This suggests that the particular form of the excess risk bound is pessimistic in these scenarios, and that more care is needed in deriving sharper estimates, such as those in Blanchard et al. (2006); Reiß and Wahl (2020).

The results for the steady Burgers example are less conclusive. The $\mathcal{O}(N^{-1})$ rate is observed in some cases, while the error appears to follow more closely the rate of $\mathcal{O}(N^{-1/2})$ for the output DIS with $r = 10$ and the input DIS for both $r = 10$ and $r = 30$. However, we also note that the resulting trends are likely polluted with sampling errors in computing both the PCA/DIS used as the reference, as well as in the estimation of the excess risks themselves by finite test data and finitely many runs. In particular, the excess risks for the input DIS are on the order of 10^{-10} using just $N = 80$, and hence small errors in the computations can lead to significant variations in the estimated excess risks. For this reason, one should be careful in drawing conclusions from the steady Burgers example.

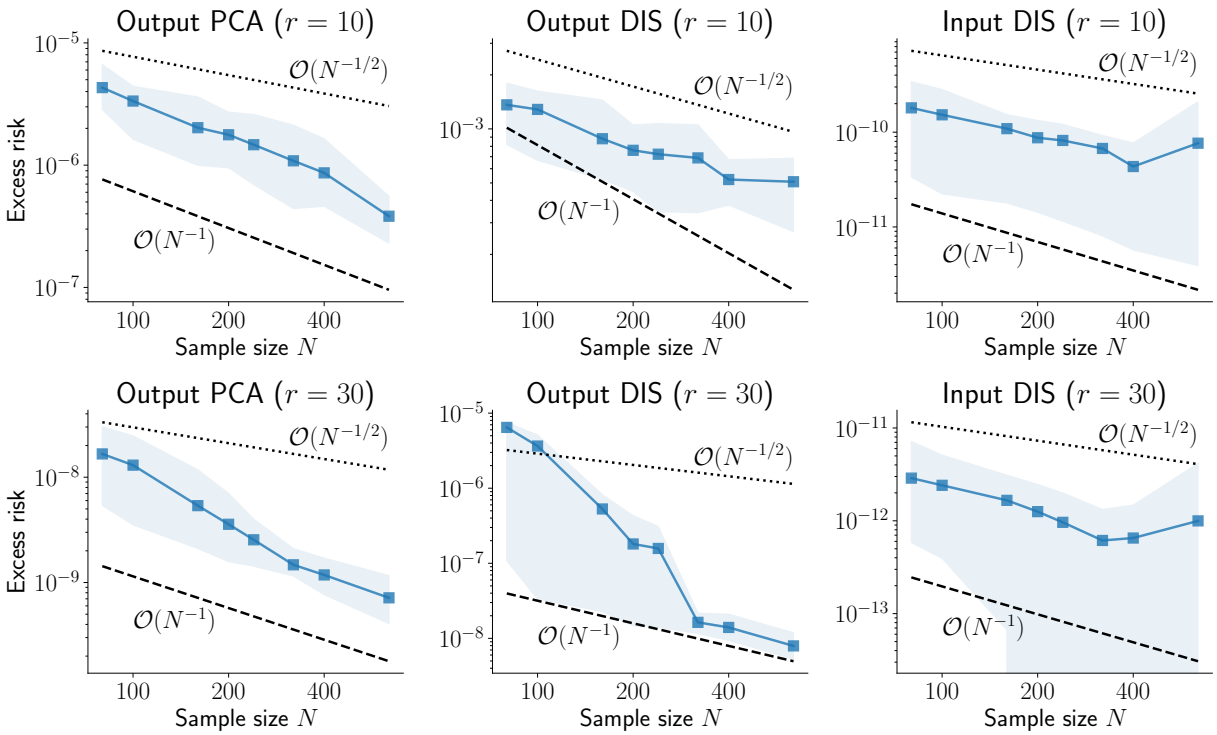


Figure 7: Excess risks in the output and derivative reconstruction errors for the steady Burgers problem for ranks $r = 10$ and $r = 30$. Values are normalized by the second moments of the test data. For reference, the global $N^{-1/2}$ rate is shown in the dotted line while the local N^{-1} rate is shown in the dashed line. The filled regions correspond to the 10%–90% quantile ranges of the 10 independent runs.

5.4.4 Neural Network Generalization Errors

We now present experiments in which we attempt to learn the operators and their derivatives using the RBNO architecture. For these experiments, we train RBNOs via the Sobolev training (H^1_γ training) procedure for the four possible pairs of input/output bases across a range of different ranks. For each rank, we consider training sample sizes of $N = 640$ up to $N = 16{,}000$. We then compute the errors in predicting the output and its derivative, i.e.,

$$\frac{\|F - \widetilde{F}_\theta\|^2_{L^2_\gamma}}{\|F\|^2_{L^2_\gamma}} \quad \text{and} \quad \frac{\|D_{\mathcal{E}} F - D_{\mathcal{E}} \widetilde{F}_\theta\|^2_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))}}{\|D_{\mathcal{E}} F\|^2_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))}}.$$

The end-to-end neural operator generalization errors are estimated on a test dataset consisting of $N_{\text{test}} = 4{,}000$ function and derivative samples. These error results are presented in two different ways. For fixed ranks, we plot the generalization errors as a function of the number of training samples, while for fixed sample sizes, we plot the generalization errors as a function of the rank. The results are shown in Figure 9, Figure 10, and Figure 11 for the semilinear elliptic, steady Burgers, and linear elasticity problems, respectively. Note that the labels for the different input and output basis pairs adopt the convention of *input basis-output basis*, e.g., DIS-PCA corresponds to the case of input DIS and output PCA.

We begin by making a few general observations. First, for any fixed rank r, both the output and derivative generalization errors decrease as the training sample size N increases, often continuing

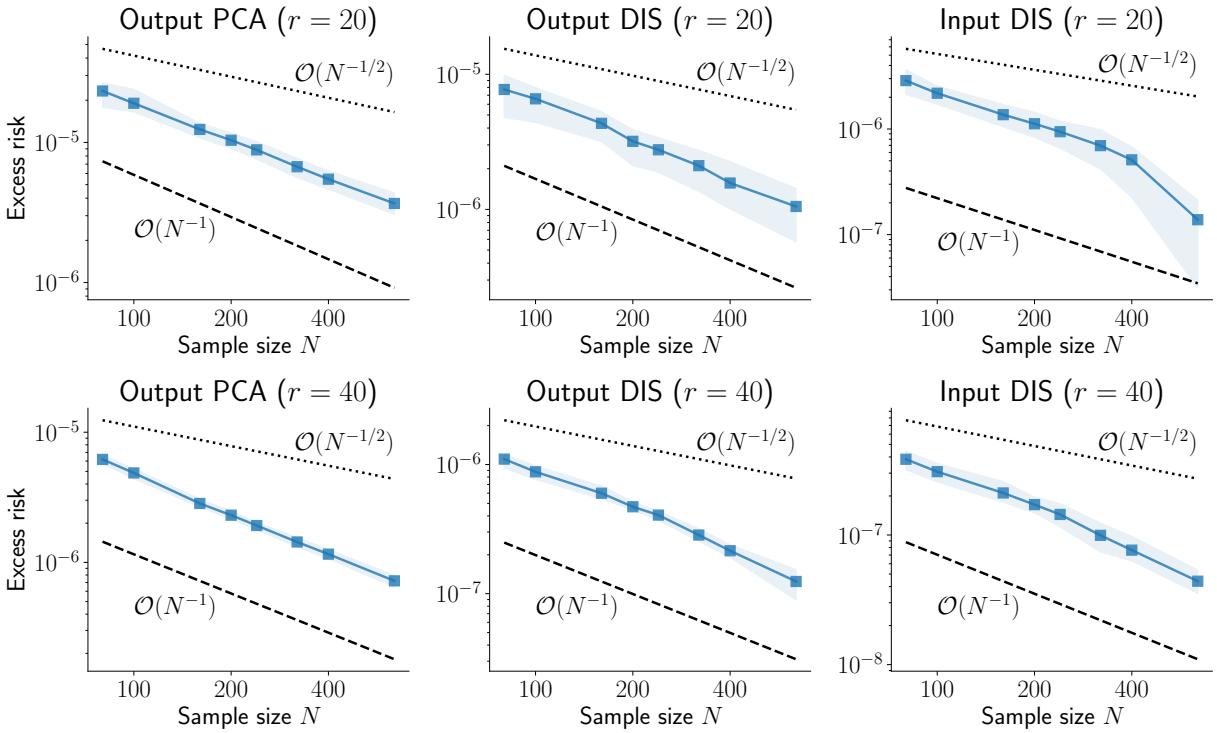


Figure 8: Excess risks in the output and derivative reconstruction errors for the linear elasticity problem for ranks $r = 20$ and $r = 40$. Values are normalized by the second moments of the test data. For reference, the global $N^{-1/2}$ rate is shown in the dotted line while the local N^{-1} rate is shown in the dashed line. The filled regions correspond to the 10%–90% quantile ranges of the 10 independent runs.

until a plateau is reached (i.e., a floor value in the error where further increasing the sample size does not lead to a significant decrease in the error). This likely corresponds to the limit of either the ridge function approximation (i.e., truncated inputs/outputs) or the approximation power of the neural network itself, as the plateau tends to occur later for larger ranks and the corresponding floor values in error are lower. We also note that, often, the output errors reach the plateau long before the derivative errors, while the derivative errors continue to improve with increasing sample sizes. This is especially noticeable in the steady Burgers example, where a fast decrease in the derivative errors with respect to the training sample size begins *after* the output errors begin to plateau. The observed behavior may suggest that the neural networks tend to first learn the output values before they learn the derivatives.

We also recognize that, unlike the reconstruction errors, the generalization errors of the neural operators given a fixed training sample size do not decrease monotonically with respect to rank, as is typical of bias-variance trade-offs in neural network training. For smaller ranks, the generalization errors can be made small with fewer training samples, but ultimately reach the limit of the truncated representation. On the other hand, the ridge functions can be made more expressive with a larger rank, but require more training samples to learn an accurate neural network representation. Thus, for a fixed number of training samples, there is an optimal rank that achieves the smallest generalization error for the output and derivative predictions. As seen in Figures 9 to 11, the optimal ranks are larger when the training sample sizes are larger. Moreover, the optimal ranks for approximating the output tend to be larger than the optimal ranks for approximating

the derivatives, which is consistent with the observation that the derivatives are more difficult to learn.

Finally, we discuss the effect of the choice of dimension reduction. In particular, the choice of input dimension reduction, which determines the optimal ridge function error, has the greatest impact on the generalization errors. In most cases, the input DIS significantly outperforms the input PCA in terms of the errors for both the output and the derivatives. This suggests that despite the fact that the DIS basis does not explicitly incorporate the second-derivative information required to bound the derivative ridge function errors (as in Proposition 32), it remains an effective choice of dimension reduction for learning the derivatives in practice. The differences in errors between the input DIS and PCA do, however, tend to decrease as the rank increases. For large ranks, the generalization errors are likely dominated by the neural network's ability to learn the underlying mapping rather than the dimension reduction.

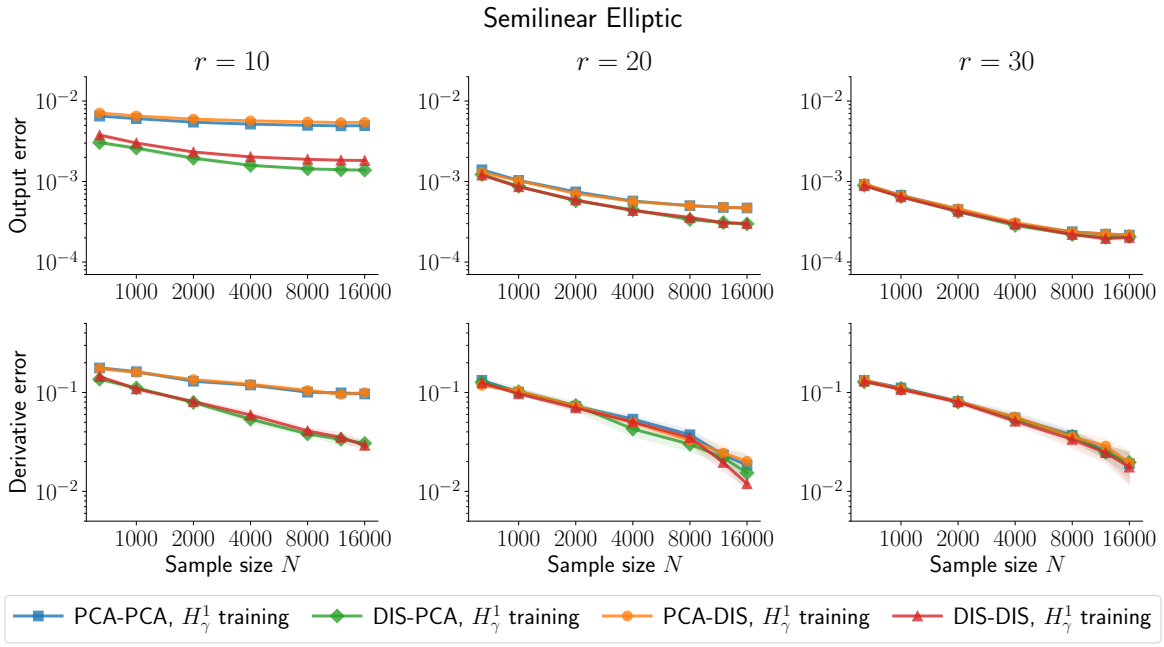
On the other hand, the choice of output dimension reduction only modestly affects the generalization errors, since it is only involved in the reconstruction of the output/derivatives. Across the examples considered, this difference is only significant for the steady Burgers example at low ranks ($r = 10, 15$) and for sufficiently large training sets, where the training errors can be made sufficiently small such that the difference between the reconstruction errors is apparent. In such cases, the two strategies perform as expected—the PCA leads to smaller errors in the output while the DIS leads to smaller derivative errors.

5.4.5 Comparing Sobolev Training to Conventional Training

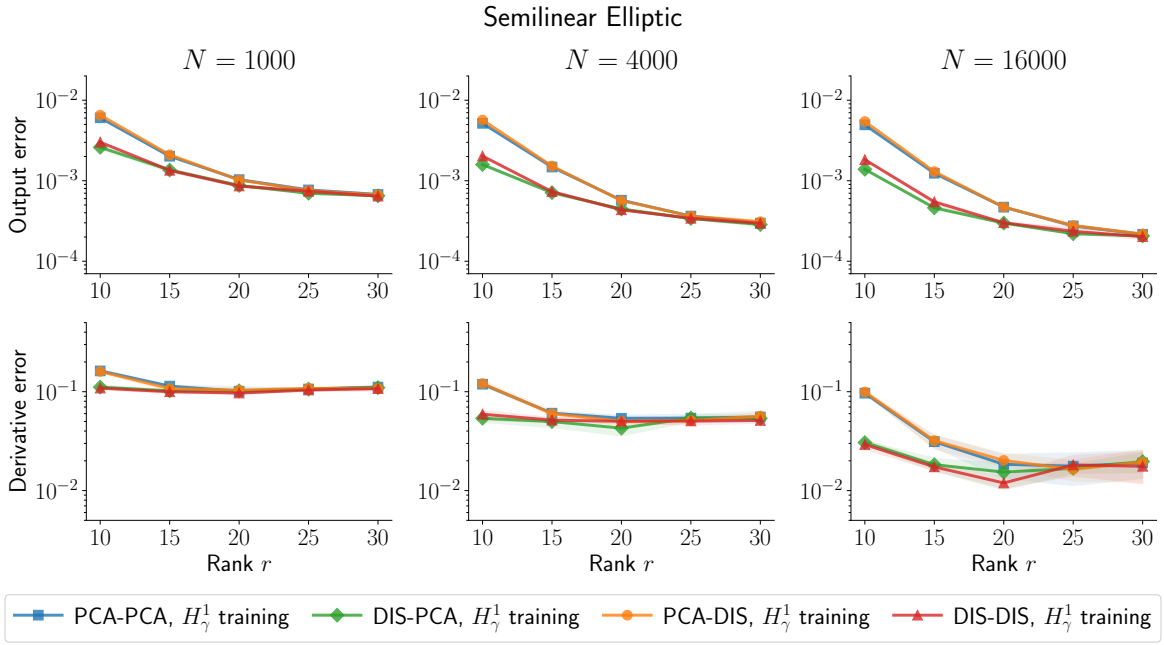
Lastly, we compare Sobolev training (H^1_γ training) to conventional training (L^2_γ training) for the RBNOs, where the same input and output bases are used for both training procedures. Recall that in conventional L^2_γ training, we only use the input-output pairs $(x_k, U_r^\dagger F(x_k))$ as our training data, while in H^1_γ Sobolev training, we use the augmented input-output-derivative tuple $(x_k, U_r^\dagger F(x_k), U_r^\dagger D_{\mathcal{E}} F(x_k) V_r)$ as our training data. For clarity, we present only the comparisons for the PCA-PCA and DIS-DIS basis pairs, and note that the differences between the two training procedures show similar trends for the other two basis pairs (PCA-DIS and DIS-PCA). The resulting generalization errors are compared in Figure 12, Figure 13, and Figure 14 for the semilinear elliptic, steady Burgers, and linear elasticity problems, respectively.

Overall, we observe that the generalization errors for both the output and derivatives tend to be smaller with H^1_γ training than with L^2_γ training, especially for smaller training sample sizes and for the derivative predictions. In many cases, H^1_γ training can yield over an order of magnitude smaller errors. This is intuitive, since for each sample of the input, H^1_γ training incorporates additional training data in the form of the projected derivative $U_r^\dagger D_{\mathcal{E}} F(x_k) V_r$, which mitigates overfitting and explicitly encourages derivative consistency. As the number of training samples increases, the generalization errors for the L^2_γ training tend to improve, and in some cases, the errors can be comparable to those of the H^1_γ training, especially for the output predictions. This suggests that with sufficiently many training input-output pairs, the neural network can eventually learn the underlying mappings and their derivatives even without the derivative data for the example problems considered here.

These results are consistent with the observations reported in the literature on derivative-informed operator learning, and we refer to the works of O'Leary-Roseberry et al. (2024); Cao et al. (2025); Luo et al. (2025); Cao et al. (2026) for more detailed comparisons between H^1_γ and L^2_γ training for a range of operator learning and downstream tasks involving more complex PDE operators. In particular, these works show that by leveraging the reduced-basis architecture and efficient strategies for computing the projected derivatives, H^1_γ Sobolev training can be conducted

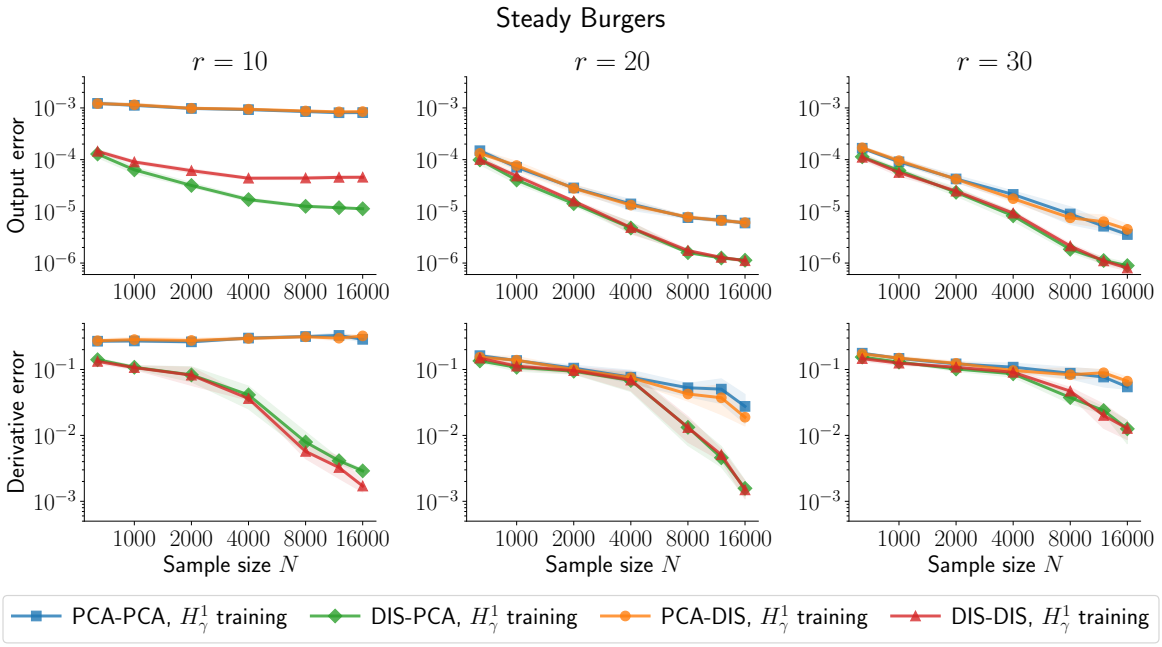


(a) Generalization errors for the semilinear elliptic PDE as a function of the number of training samples for fixed ranks $r = 10$, 20, and 30.

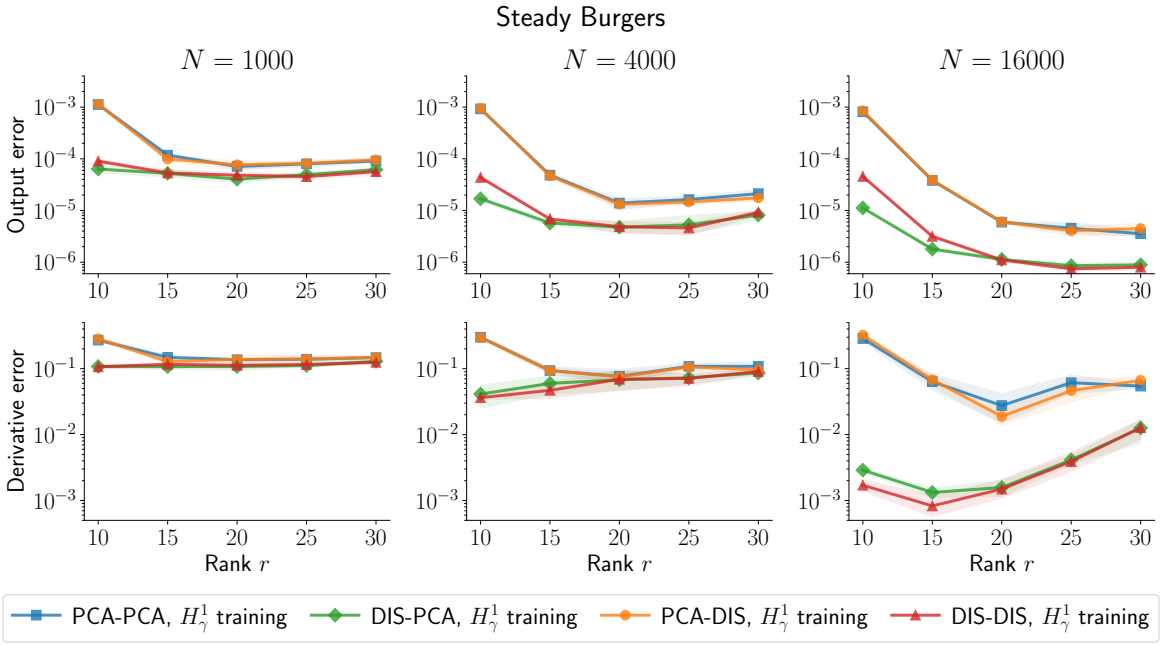


(b) Generalization error for the semilinear elliptic PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 9: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the semilinear elliptic PDE normalized by the second moments of the test data. The neural networks are trained via Sobolev training (H^1_γ training) using different basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.

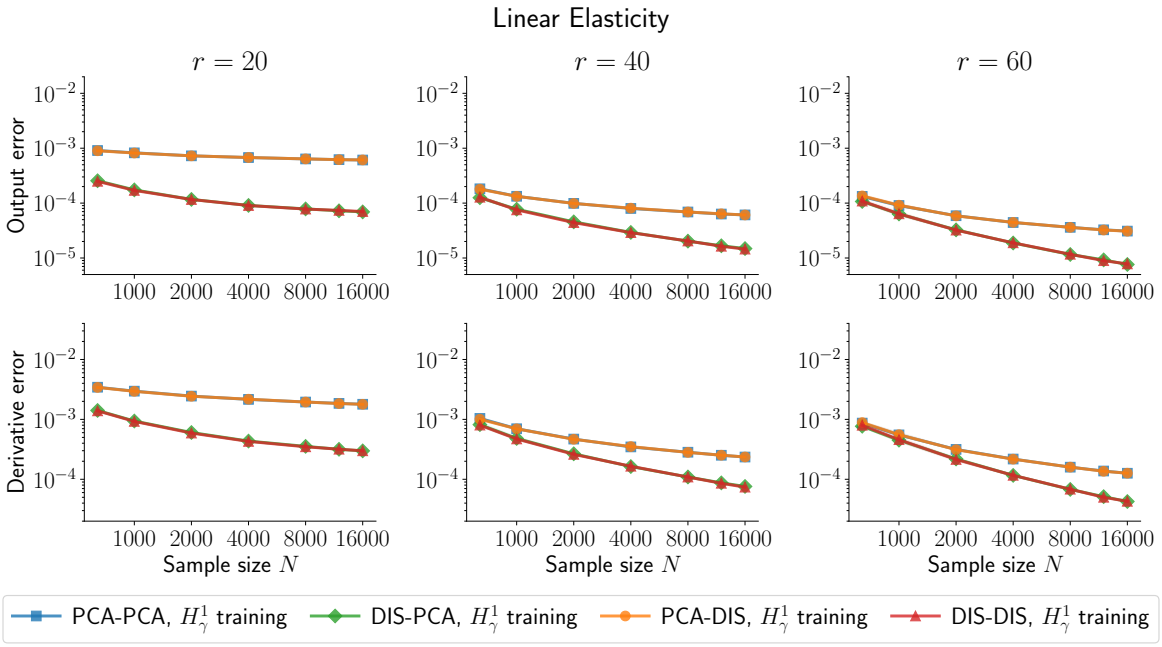


(a) Generalization errors for the steady Burgers PDE as a function of the number of training samples for fixed ranks $r = 10$, 20, and 30.

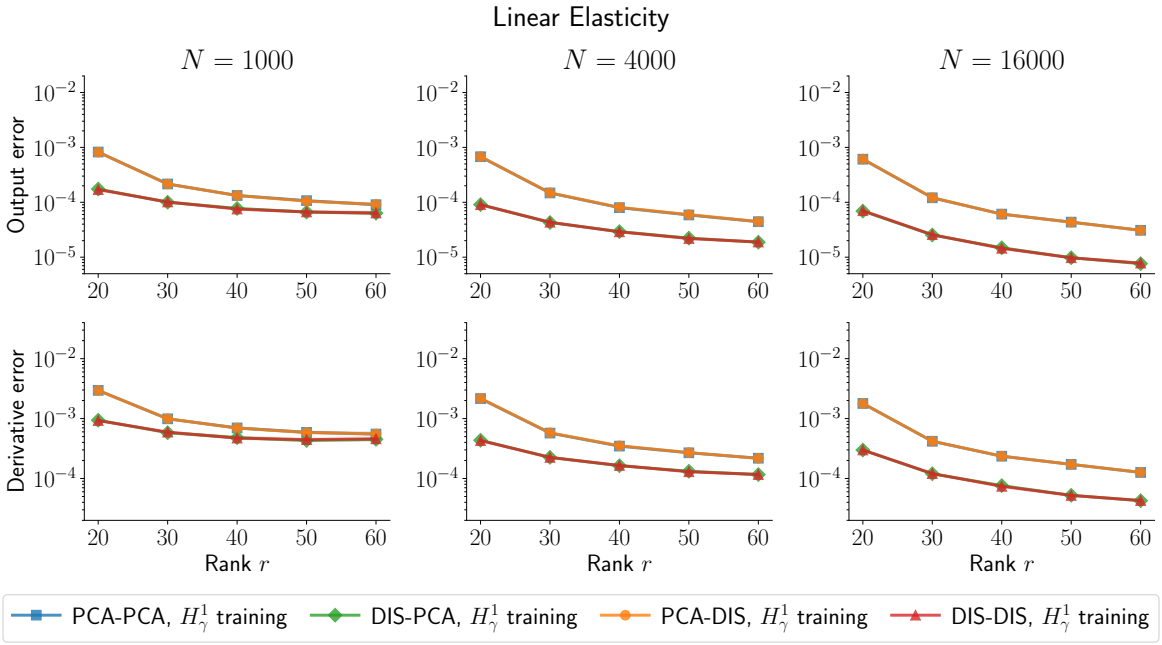


(b) Generalization error for the steady Burgers PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 10: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the steady Burgers PDE normalized by the second moments of the test data. The neural networks are trained via Sobolev training (H^1_γ training) using different basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.



(a) Generalization errors for the linear elasticity PDE as a function of the number of training samples for fixed ranks $r = 20$, 40, and 60.



(b) Generalization error for the linear elasticity PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 11: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the linear elasticity PDE normalized by the second moments of the test data. The neural networks are trained via Sobolev training (H^1_γ training) using different basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.

with only a modest increase in computational cost compared to conventional L^2_γ training, while yielding significant improvements in generalization errors for both the output and derivatives.

When comparing the different choices of bases for the L^2_γ training, we again observe that input DIS tends to outperform PCA for both the output and derivative predictions. This suggests that the benefits of DIS for learning in the H^1_γ norm exist even when using L^2_γ training. We note that similar observations have also been reported in the literature of active subspaces for surrogate modeling in terms of L^2_γ accuracy (Zahm et al., 2020; O'Leary-Roseberry, Villa, et al., 2022).

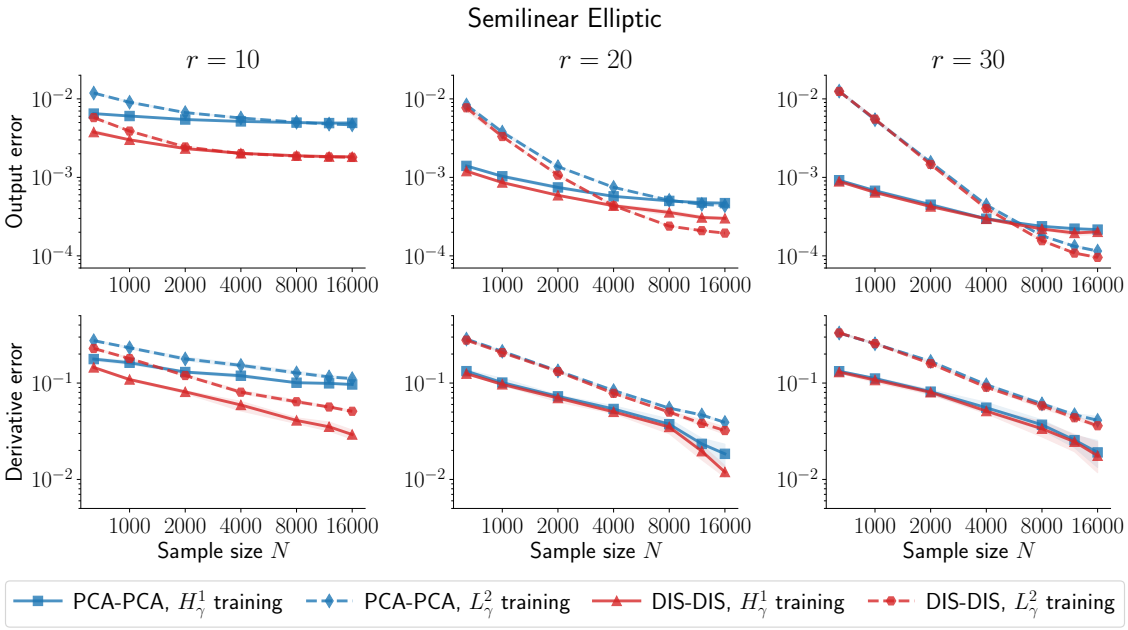
6. Conclusions

In this work, we considered the analysis of derivative-informed operator surrogates that use reduced-basis representations of the input and output spaces and a neural network that is constructed to approximate the finite-dimensional mapping between the truncated inputs and outputs. The main contributions of the present work are H^1_γ approximation error bounds developed for different choices of dimension reduction—namely, PCA and DIS.

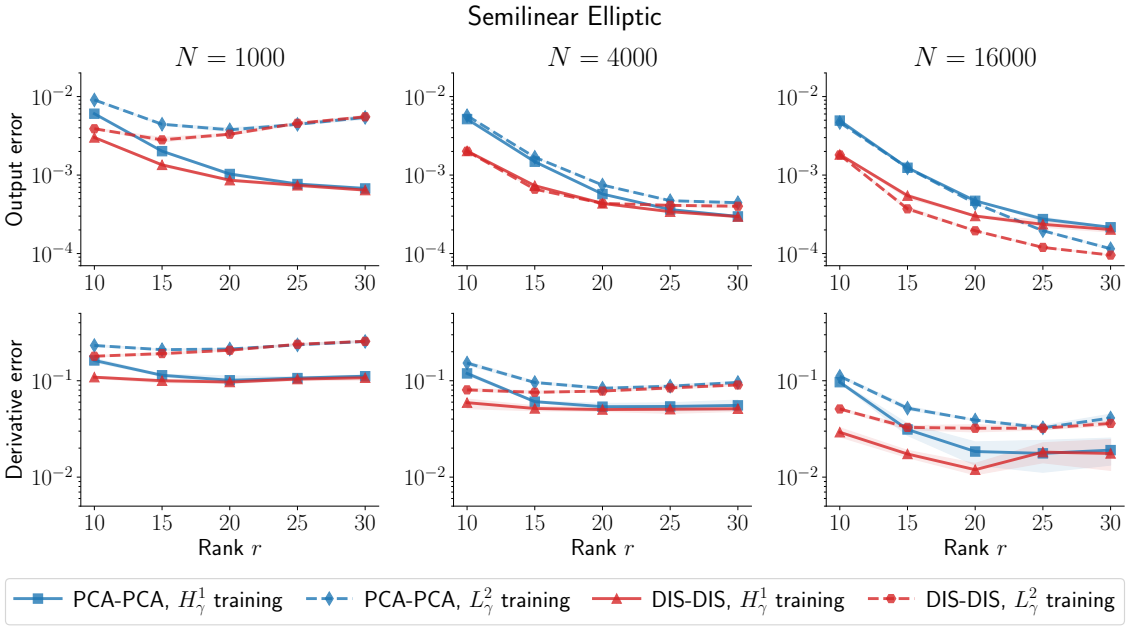
To this end, we first presented a universal approximation theorem for generic orthonormal bases, which shows that approximation of operators in H^m_γ for any $m \geq 0$ to arbitrary accuracy using the reduced-basis architecture is possible. To provide more insight on how the dimension reduction strategy affects the approximation errors, we proceeded with a basis-specific analysis, decomposing the error into dimension reduction error and neural network approximation error. The dimension reduction errors were considered in detail for the PCA and DIS bases, including the statistical errors associated with the sample-based estimators of the PCA/DIS operators. The neural network approximation error was analyzed using the universal approximation results for Sobolev norms from Hornik (1991), which we extended to accommodate deep architectures and non-saturating activation functions while providing approximation rates in terms of the size of the network when specifically using ReLU^k activation functions. The different results were combined to provide an end-to-end error bound. We note that, of the various combinations, the mildest assumptions were required to bound the H^1_γ approximation error when using PCA for input dimension reduction and DIS for output dimension reduction. In developing these results, we also filled gaps in existing results for the derivative-informed subspaces in the L^2_γ case, especially in the separable Hilbert space setting.

We also presented numerical results to validate our theoretical predictions, comparing the reconstruction errors, excess risks, as well as the end-to-end errors of the resulting neural operators in both L^2_γ and H^1_γ. For the reconstruction errors, we observed that the PCA/DIS eigenvalue decay does indeed translate to appropriate bounds in the reconstruction errors. In the case of empirical PCA/DIS, we observed that the excess risks of the reconstructions typically decrease with sample size N at a rate consistent with the local regime of $\mathcal{O}(N^{-1})$ as opposed to the global regime of $\mathcal{O}(N^{-1/2})$. We did not, however, observe the effect of the inverse dependence on the spectral gap $(\lambda_r - \lambda_{r+1})$ in our numerical examples. This suggests that our approach for bounding the excess risk is pessimistic, and that more refined approaches such as those taken in Blanchard et al. (2006); Reiß and Wahl (2020) may be necessary to provide sharper and more practical bounds.

When used to construct operator surrogates, we observed that the choice of dimension reduction (PCA vs DIS) is especially important for the performance of the neural operator in both output and derivative learning. This has important practical implications—an effective choice of dimension reduction allows for smaller rank and hence smaller networks, which in turn requires fewer training samples and lower training times. This is especially true for the derivative error, which often required more training samples to learn accurately compared to the output itself. In this regard, the DIS significantly outperformed the PCA for input dimension reduction, since it explicitly accounts

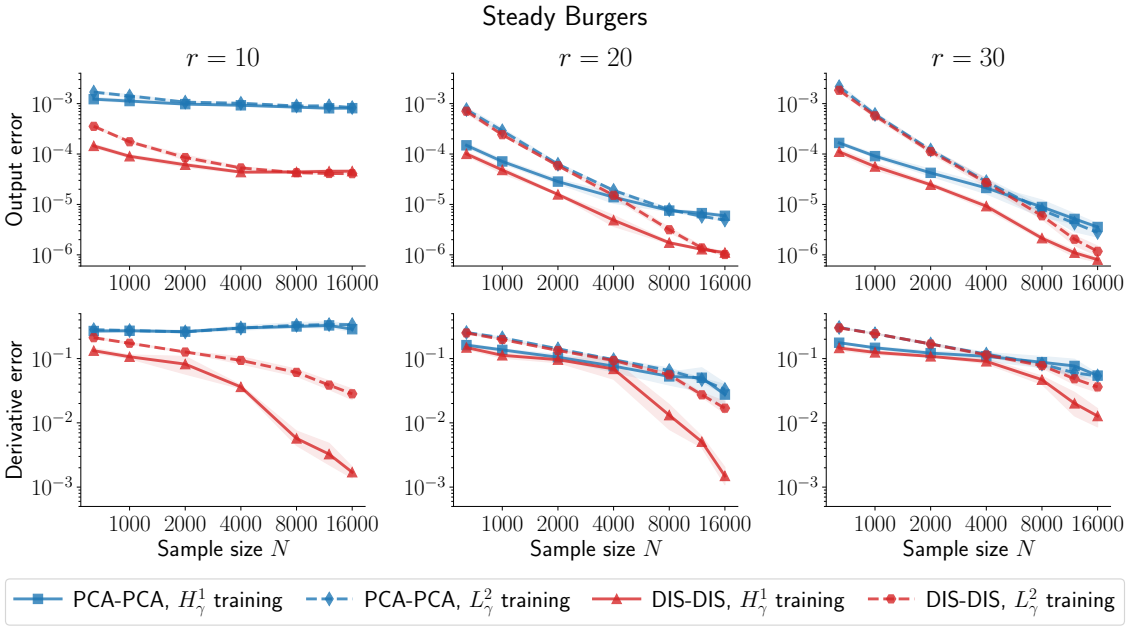


(a) Generalization errors for the semilinear elliptic PDE as a function of the number of training samples for fixed ranks $r = 10$, 20, and 30.

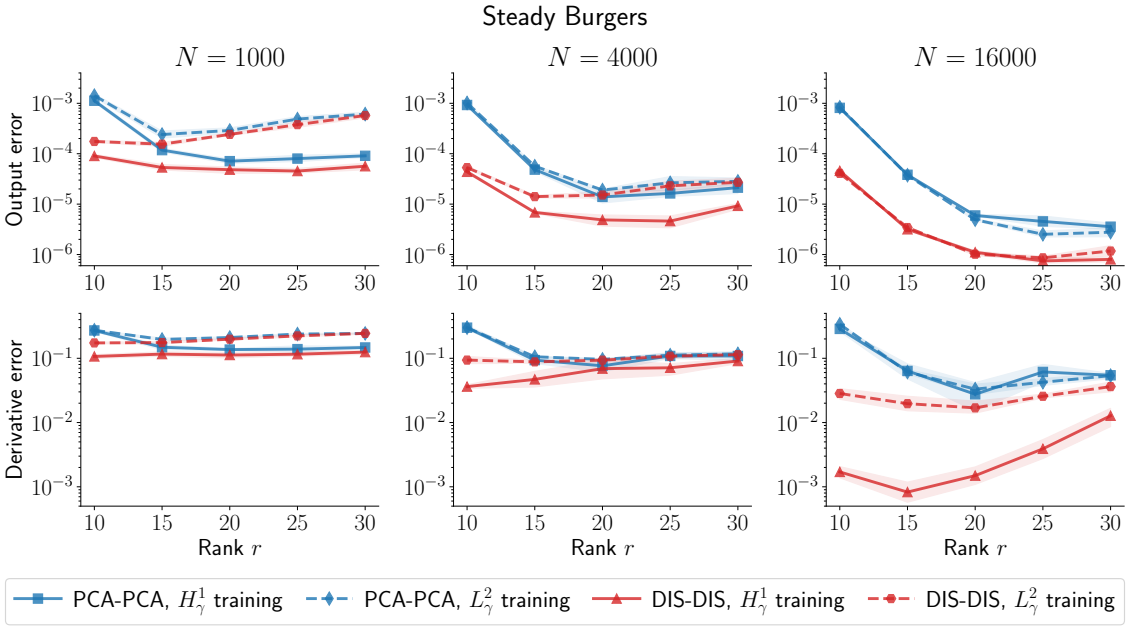


(b) Generalization error for the semilinear elliptic PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 12: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the semilinear elliptic PDE normalized by the second moments of the test data. The neural networks are trained via either Sobolev training (H^1_γ training) or conventional training (L^2_γ training) using the PCA-PCA and DIS-DIS basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.

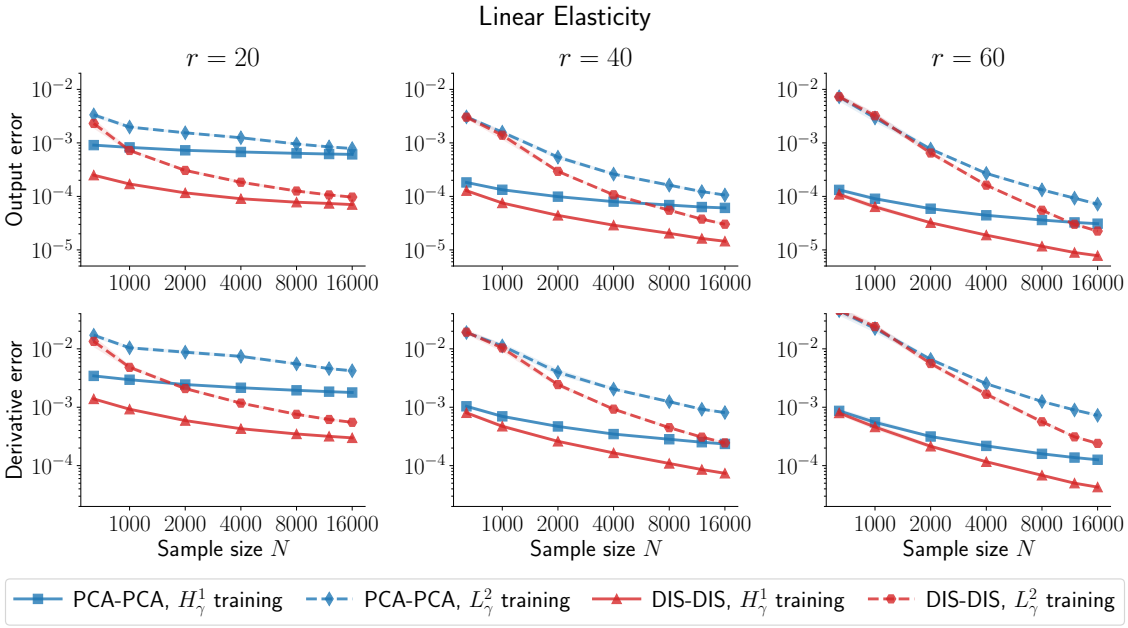


(a) Generalization errors for the steady Burgers PDE as a function of the number of training samples for fixed ranks $r = 10$, 20, and 30.

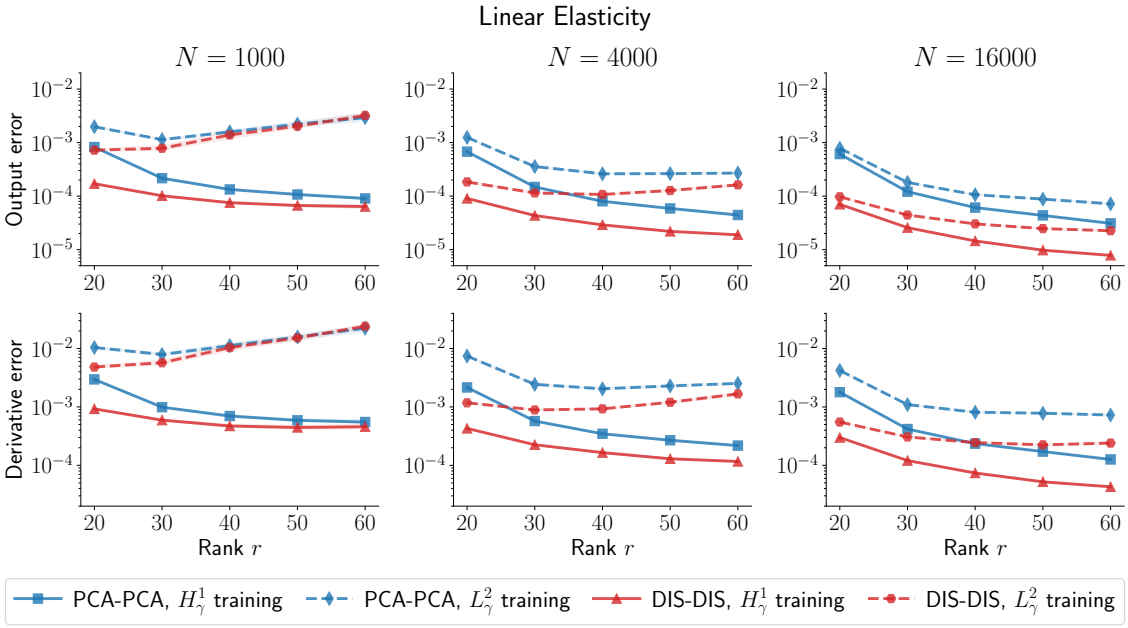


(b) Generalization error for the steady Burgers PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 13: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the steady Burgers PDE normalized by the second moments of the test data. The neural networks are trained via either Sobolev training (H^1_γ training) or conventional training (L^2_γ training) using the PCA-PCA and DIS-DIS basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.



(a) Generalization errors for the linear elasticity PDE as a function of the number of training samples for fixed ranks $r = 20$, 40, and 60.



(b) Generalization error for the linear elasticity PDE as a function of the rank for a fixed number of training samples, $N = 1{,}000$, 4,000, and 16,000.

Figure 14: Neural network generalization errors for the output ($\|\cdot\|^2_{L^2_\gamma}$) and derivatives ($|\cdot|^2_{H^1_\gamma}$) of the linear elasticity PDE normalized by the second moments of the test data. The neural networks are trained via either Sobolev training (H^1_γ training) or conventional training (L^2_γ training) using the PCA-PCA and DIS-DIS basis pairs for dimension reduction. Results are presented as both a function of training sample size (top) and as a function of rank (bottom). Solid lines correspond to averages over 10 independent runs, while the filled regions correspond to the 10%–90% quantile ranges across these runs.

for the sensitivity of the underlying map, while the PCA and DIS provided similar performance when used for output dimension reduction for the examples considered.

We note that several of our results for dimension reduction required additional assumptions. For example, Assumptions 10 and 11 were needed in order to bound higher-order derivatives by lower-order ones. Although we have provided analytical examples to show that these are satisfied for certain classes of operators, it would be of interest to identify alternative properties of operators that can be used to ascertain the reconstruction/ridge function bounds for the derivatives. One possible example is holomorphy (Lopushansky, 2023; Marcati et al., 2025). It is known that holomorphic operators are amenable to linear dimension reduction (Lanthaler et al., 2022; Lanthaler, 2023; Schwab and Zech, 2023; Adcock et al., 2024a; Herrmann et al., 2024; Adcock et al., 2025), possibly allowing one to derive favorable approximation rates in the infinite-dimensional Sobolev norms in terms of both the latent dimension r and the size of the neural network.

Moreover, the error bounds presented in this work for the dimension reduction relied on the spectral decay of the PCA/DIS operators. This is commonly observed for elliptic PDE problems (such as those shown in the numerical examples), where explicit estimates may be available under the aforementioned holomorphy assumptions. However, this may not be the case in settings such as advection-dominated or hyperbolic PDE problems. Thus, although linear dimension reduction such as PCA/DIS is often preferred for its statistical efficiency, one may need to consider nonlinear dimension reduction strategies in such cases, for which the incorporation of derivative information is a possible direction for future work.

We also note that neither the statistical learning errors nor the training (optimization) errors of the neural network are treated theoretically in this work. However, both are important for the understanding of derivative-informed operator learning, since many performance benefits arise due to the inclusion of derivative information in the *Sobolev training* process itself. In this regard, an analysis of generalization errors and out-of-distribution errors for Sobolev training, extending the works of, say, Lara Benitez et al. (2024); Reinhardt et al. (2026), is of great interest. Overall, we believe operator learning with derivative information remains open to further theoretical and numerical explorations, with many interesting outstanding research questions.

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Appendix A. Useful Supporting Results

A.1 Cameron–Martin space

We note some useful facts about the Cameron–Martin space.

Lemma 45. *Suppose $v_i \in \mathcal{X}$ are eigenvectors of the non-degenerate covariance operator $C_{\mathcal{X}}$ such that $C_{\mathcal{X}} v_i = \lambda_i v_i$ for $i \in \mathbb{N}$. Then $\tilde{v}_i = C_{\mathcal{X}}^{1/2} v_i$ is an orthonormal basis for the Cameron–Martin space $\mathcal{E}$.*

Proof Consider any $h \in \mathcal{E}$. Then $h = C_{\mathcal{X}}^{1/2} w$ for some $w \in \mathcal{X}$. Moreover, $(v_i)_{i=1}^{\infty}$ is an orthonormal basis for $\mathcal{X}$ such that $\varepsilon_n := w - \sum_{i=1}^{n} v_i \langle v_i, w \rangle_{\mathcal{X}} \to 0$. Now note that

$$\begin{aligned}
\varepsilon_n &= C_{\mathcal{X}}^{-1/2} h - \sum_{i=1}^{n} C_{\mathcal{X}}^{-1/2} C_{\mathcal{X}}^{1/2} v_i \left\langle C_{\mathcal{X}}^{-1/2} C_{\mathcal{X}}^{1/2} v_i, C_{\mathcal{X}}^{-1/2} h \right\rangle_{\mathcal{X}} \\
&= C_{\mathcal{X}}^{-1/2} \left(h - \sum_{i=1}^{n} \tilde{v}_i \langle \tilde{v}_i, h \rangle_{\mathcal{E}} \right) \\
&=: C_{\mathcal{X}}^{-1/2} \tilde{\varepsilon}_n,
\end{aligned}$$

where $\tilde{\varepsilon}_n = h - \sum_{i=1}^{n} \tilde{v}_i \langle \tilde{v}_i, h \rangle_{\mathcal{E}}$ is the complementary component of the projection onto the span of $(\tilde{v}_i)_{i=1}^{n}$ in $\mathcal{E}$. It remains to see that $\|\tilde{\varepsilon}_n\|_{\mathcal{E}}^2 = \langle \tilde{\varepsilon}_n, \tilde{\varepsilon}_n \rangle_{\mathcal{E}} = \langle \varepsilon_n, \varepsilon_n \rangle_{\mathcal{X}} \to 0$. ■

Lemma 46. *The inclusion operator $\iota : \mathcal{E} \ni w \mapsto w \in \mathcal{X}$ is Hilbert–Schmidt.*

Proof We use the basis generated by the covariance $(C_{\mathcal{X}}^{1/2} v_i)_{i=1}^{\infty} = (\lambda_i^{1/2} v_i)_{i=1}^{\infty}$ to evaluate the Hilbert–Schmidt norm,

$$\|\iota\|_{\mathrm{HS}(\mathcal{E},\mathcal{X})}^2 = \sum_{i=1}^{\infty} \left\langle \lambda_i^{1/2} v_i, \lambda_i^{1/2} v_i \right\rangle_{\mathcal{X}} = \sum_{i=1}^{\infty} \lambda_i = \mathrm{Tr}(C_{\mathcal{X}}) < \infty.$$

■

A.2 Convergence in Sobolev Norms

Here, we provide a supporting result for the convergence of functions in the Sobolev norm $H_\gamma^1 = H^1(\gamma, \mathcal{Y})$.

Lemma 47. *Let $F \in H_\gamma^1$ and let $(F_n)_{n=1}^{\infty}$ be a sequence of H_γ^1 functions such that $F_n \to F$ in $L^2(\gamma, \mathcal{Y})$ and $D_{\mathcal{E}} F_n \to \Psi$ for some $\Psi \in L^2(\gamma, \mathrm{HS}(\mathcal{E}, \mathcal{Y}))$. Then, $F_n \to F$ in H_γ^1 and $D_{\mathcal{E}} F = \Psi$.*

Proof Recall that for any sequence $(\varphi_n)_{n=1}^{\infty}$ in $\mathcal{FC}^{\infty}$ that converges to F in $L_\gamma^2 := L^2(\gamma, \mathcal{Y})$ and is Cauchy in H_γ^1, we have $\varphi_n \to F$ in H_γ^1 (see Bogachev, 1998, Section 5.2). In particular, this means $\lim_{n\to\infty} D_{\mathcal{E}} \varphi_n = D_{\mathcal{E}} F$. It remains to construct the sequence $\varphi_n \in \mathcal{FC}^{\infty}$ using $F_n \in H_\gamma^1$. For $k \in \mathbb{N}$, we will use a subsequence F_{n_k} such that $\|F_{n_k} - F\|_{L_\gamma^2} \leq 1/k$ and $\|D_{\mathcal{E}} F_{n_k} - \Psi\|_{L_\gamma^2} \leq 1/k$. Moreover, by the density of $\mathcal{FC}^{\infty}$ in H_γ^1, we can find φ_{n_k} such that $\|\varphi_{n_k} - F_{n_k}\|_{H_\gamma^1} \leq 1/k$. Thus we have

$$\|F - \varphi_{n_k}\|_{L_\gamma^2} \leq \|F - F_{n_k}\|_{L_\gamma^2} + \|F_{n_k} - \varphi_{n_k}\|_{L_\gamma^2} \leq 2/k.$$

Moreover,

$$\|D_{\mathcal{E}} \varphi_{n_k} - \Psi\|_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))} \leq \|D_{\mathcal{E}} \varphi_{n_k} - D_{\mathcal{E}} F_{n_k}\|_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))} + \|D_{\mathcal{E}} F_{n_k} - \Psi\|_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathcal{Y}))} \leq 2/k.$$

That is, $\varphi_{n_k} \to F$ in L_γ^2 and $D_{\mathcal{E}} \varphi_{n_k}$ converges to Ψ in $L^2(\gamma, \mathrm{HS}(\mathcal{E}, \mathcal{Y}))$. By the uniqueness of the limit, this means $\varphi_{n_k} \to F$ in H_γ^1 and $D_{\mathcal{E}} F = \Psi$. ■

A.3 Trace Minimization and Eigenvalue Problems

In this section, we state Fan's theorem, which is a result on the maximization of quadratic forms over orthonormal sets. This is related to trace minimization and is used extensively for the reconstruction error results.

Lemma 48 (Fan's theorem, as stated in Bhattacharya et al., 2021)**.** *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle_{\mathcal{H}})$ be a separable Hilbert space and $A : \mathcal{H} \to \mathcal{H}$ be a non-negative, self-adjoint, compact operator. Denote by $\lambda_1 \geq \lambda_2 \geq \dots$ the eigenvalues of A, and for any $r \in \mathbb{N}$ (where $\mathbb{N} = \{1, 2, \dots\}$), let S_r be the set of collections of r orthonormal elements of $\mathcal{H}$. Then,*

$$\max_{(u_i)_{i=1}^r \in S_r} \sum_{i=1}^r \langle Au_i, u_i\rangle_{\mathcal{H}} = \sum_{i=1}^r \lambda_i.$$

Fan's theorem asserts that of all orthogonal projections, projections onto dominant eigenvectors of a non-negative, self-adjoint, compact operator A minimize the trace of $(I - P_r)A(I - P_r)$, which shows up in the analysis of the reconstruction errors. This is summarized in the following lemma.

Lemma 49 (Trace minimization)**.** *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle_{\mathcal{H}})$ be a separable Hilbert space and $A : \mathcal{H} \to \mathcal{H}$ be a non-negative, self-adjoint, trace-class operator with eigenpairs $(\lambda_i, u_i)_{i=1}^\infty$ given in order of decreasing eigenvalues. Let also $\mathcal{P}_r(\mathcal{H})$ denote the class of rank $r \in \mathbb{N}$ orthogonal projections in $\mathcal{H}$. Then,*

$$\min_{P_r \in \mathcal{P}_r(\mathcal{H})} \operatorname{Tr}((I - P_r)A(I - P_r)) = \sum_{i=r+1}^\infty \lambda_i,$$

and the minimum is attained by $P_r = \sum_{i=1}^r u_i \otimes u_i = \sum_{i=1}^r u_i \langle u_i, \cdot\rangle_{\mathcal{H}}$.

Proof Note that for trace-class operator A and $I - \widetilde{P}_r \in \mathcal{L}(\mathcal{H}, \mathcal{H})$, their composition is also trace-class. By the cyclic property of the trace, we then have

$$\operatorname{Tr}((I - \widetilde{P}_r)A(I - \widetilde{P}_r)) = \operatorname{Tr}(A(I - \widetilde{P}_r)) = \operatorname{Tr}(A) - \operatorname{Tr}(\widetilde{P}_r A \widetilde{P}_r).$$

For any $\widetilde{P}_r \in \mathcal{P}_r(\mathcal{H})$, the projection can be written as $\widetilde{P}_r = \sum_{i=1}^r \tilde{u}_i \otimes \tilde{u}_i$ for some orthonormal basis of $\mathcal{H}$, $(\tilde{u}_i)_{i=1}^\infty$, where $(\tilde{u}_i)_{i=1}^r$ spans the range of $\widetilde{P}_r$. Then, using the same basis to compute the second trace, we have

$$\operatorname{Tr}(\widetilde{P}_r A \widetilde{P}_r) = \sum_{i=1}^r \langle \tilde{u}_i, A\tilde{u}_i\rangle_{\mathcal{H}},$$

which by Lemma 48, has a maximal value of $\sum_{i=1}^r \lambda_i$. Moreover, this is attained by $(u_i)_{i=1}^r$ coming from the eigenvectors of A with the largest eigenvalues. Thus, the minimum of $\operatorname{Tr}((I-\widetilde{P}_r)A(I-\widetilde{P}_r))$ is given by

$$\operatorname{Tr}(A) - \sum_{i=1}^r \langle u_i, Au_i\rangle_{\mathcal{H}} = \sum_{i=1}^\infty \lambda_i - \sum_{i=1}^r \lambda_i = \sum_{i=r+1}^\infty \lambda_i,$$

which is attained by the projection $P_r = \sum_{i=1}^r u_i \otimes u_i$. ■

When the eigenvalue problem is solved for an approximation $\widehat{A}$ (e.g., an empirical estimate of A), the corresponding projection is suboptimal. The excess error can be bounded by the difference between A and $\widehat{A}$ as measured in the $\mathrm{HS}(\mathcal{H}, \mathcal{H})$ norm.

Lemma 50. *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle_{\mathcal{H}})$ be a separable Hilbert space. Suppose $A \in \mathcal{L}(\mathcal{H},\mathcal{H})$ and $\widehat{A} \in \mathcal{L}(\mathcal{H},\mathcal{H})$ are non-negative, self-adjoint, and trace-class, with eigenpairs $(\lambda_i, u_i)_{i=1}^{\infty}$ and $(\widehat{\lambda}_i, \widehat{u}_i)_{i=1}^{\infty}$, respectively, both given in order of decreasing eigenvalues. Then, given $r \in \mathbb{N}$, the projection $\widehat{P}_r = \sum_{i=1}^{r} \widehat{u}_i \otimes \widehat{u}_i$ satisfies*

$$\mathrm{Tr}(A(I - \widehat{P}_r)) \leq \sum_{i=r+1}^{\infty} \lambda_i + \sqrt{2r}\|A - \widehat{A}\|_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \tag{39}$$

and

$$\mathrm{Tr}(A(I - \widehat{P}_r)) \leq \sum_{i=r+1}^{\infty} \lambda_i + \frac{2\|A - \widehat{A}\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}}{\lambda_r - \lambda_{r+1}}. \tag{40}$$

Proof The proof follows that presented in Reiß and Wahl (2020) in the context of covariance estimation, noting that it applies generally for self-adjoint trace-class operators. We reproduce the proof here for completeness. We start by writing

$$\mathrm{Tr}(A(I - \widehat{P}_r)) = \mathrm{Tr}(A(I - P_r)) + \mathrm{Tr}(A(P_r - \widehat{P}_r)) = \sum_{i=r+1}^{\infty} \lambda_i + \mathrm{Tr}(A(P_r - \widehat{P}_r)),$$

where $P_r = \sum_{i=1}^{r} u_i \otimes u_i$ comes from the exact optimal projection. We now seek to bound the excess error

$$\mathscr{E}_r = \mathrm{Tr}(A(P_r - \widehat{P}_r)), \tag{41}$$

often referred to as the excess risk in covariance estimation. By Lemma 48, $\widehat{P}_r$ maximizes $\mathrm{Tr}(\widehat{A}\widehat{P}_r)$ amongst rank r orthogonal projections. Therefore, we know that $\mathrm{Tr}(\widehat{A}\widehat{P}_r) \geq \mathrm{Tr}(\widehat{A}P_r)$ and so $\mathrm{Tr}(\widehat{A}(P_r - \widehat{P}_r)) \leq 0$. Subtracting this from the excess error yields

$$\mathscr{E}_r = \mathrm{Tr}(A(P_r - \widehat{P}_r)) \leq \mathrm{Tr}((A - \widehat{A})(P_r - \widehat{P}_r)).$$

Using the Cauchy–Schwarz inequality, we have

$$\mathscr{E}_r \leq \mathrm{Tr}((A - \widehat{A})(P_r - \widehat{P}_r)) = \left\langle A - \widehat{A}, P_r - \widehat{P}_r \right\rangle_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \leq \|A - \widehat{A}\|_{\mathrm{HS}(\mathcal{H},\mathcal{H})}\|P_r - \widehat{P}_r\|_{\mathrm{HS}(\mathcal{H},\mathcal{H})}. \tag{42}$$

Expanding the projection error gives

$$\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = \|P_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} + \|\widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} - 2\sum_{i=1}^{\infty} \left\langle P_r e_i, \widehat{P}_r e_i \right\rangle_{\mathcal{H}}.$$

where $\|P_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = r$ and $\|\widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = r$. Moreover, using $e_i = u_i$, we have $\langle P_r u_i, \widehat{P}_r u_i\rangle_{\mathcal{H}} = \langle u_i, \widehat{P}_r u_i\rangle_{\mathcal{H}} \geq 0$ for $i \leq r$ and $\langle P_r u_i, \widehat{P}_r u_i\rangle_{\mathcal{H}} = 0$ for $i > r$. Thus, $\|P_r - \widehat{P}_r\|_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \leq \sqrt{2r}$, giving (39).

For (40), we first note that the bound is trivially infinite if $\lambda_{r+1} = \lambda_r$, so we proceed assuming $\lambda_r > \lambda_{r+1}$. We claim that the excess error upper bounds the projection error as

$$\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \leq \frac{2\mathscr{E}_r}{\lambda_r - \lambda_{r+1}}. \tag{43}$$

Combining this claim with (42), we have

$$\mathscr{E}_r^2 \leq \|A - \widehat{A}\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \leq \frac{2\mathscr{E}_r}{\lambda_r - \lambda_{r+1}}\|A - \widehat{A}\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}.$$

Dividing through by $\mathscr{E}_r$ gives (40).

It remains then to prove the claim in (43). To this end, we begin by introducing the notation $A_i = u_i \otimes u_i$, $P_r^\perp = I - P_r$, and $\widehat{P}_r^\perp = I - \widehat{P}_r$. We also have the identities $P_r - \widehat{P}_r = \widehat{P}_r^\perp - P_r^\perp$ and $\mathrm{Tr}(P_r\widehat{P}_r^\perp) = \mathrm{Tr}(\widehat{P}_r P_r^\perp)$. This latter identity follows from the fact

$$\mathrm{Tr}(P_r\widehat{P}_r^\perp) = \mathrm{Tr}(P_r(\widehat{P}_r^\perp - P_r^\perp)) = \mathrm{Tr}(-P_r(\widehat{P}_r - P_r)) = \mathrm{Tr}((I-P_r)(\widehat{P}_r - P_r)) = \mathrm{Tr}(P_r^\perp \widehat{P}_r),$$

since $\mathrm{Tr}(\widehat{P}_r - P_r) = 0$. Now, expanding the norm $\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}$, we have

$$\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = \mathrm{Tr}((P_r - \widehat{P}_r)(P_r - \widehat{P}_r)) = \mathrm{Tr}((P_r - \widehat{P}_r)(\widehat{P}_r^\perp - P_r^\perp)) = 2\mathrm{Tr}(P_r\widehat{P}_r^\perp)$$

where we have used the fact that $\mathrm{Tr}(P_rP_r^\perp) = \mathrm{Tr}(\widehat{P}_r\widehat{P}_r^\perp) = 0$ and $\mathrm{Tr}(P_r\widehat{P}_r^\perp) = \mathrm{Tr}(\widehat{P}_rP_r^\perp)$. Moreover, since $\mathrm{Tr}(P_r\widehat{P}_r^\perp) = \mathrm{Tr}(P_r\widehat{P}_r^\perp\widehat{P}_r^\perp P_r) = \|P_r\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}$, we can further expand this as

$$\|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = 2\|P_r\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = 2\sum_{i=1}^r \|A_i\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}. \tag{44}$$

On the other hand, the excess error is given by

$$\begin{aligned}
\mathscr{E}_r &= \sum_{i=1}^\infty \lambda_i \left\langle A_i, P_r - \widehat{P}_r \right\rangle_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \\
&= \sum_{i=1}^r \lambda_i \left\langle A_i, \widehat{P}_r^\perp - P_r^\perp \right\rangle_{\mathrm{HS}(\mathcal{H},\mathcal{H})} - \sum_{j=r+1}^\infty \lambda_j \left\langle A_j, \widehat{P}_r - P_r \right\rangle_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \\
&= \sum_{i=1}^r \lambda_i \mathrm{Tr}(A_i\widehat{P}_r^\perp) - \sum_{j=r+1}^\infty \lambda_j \mathrm{Tr}(A_j\widehat{P}_r).
\end{aligned}$$

Moreover, we have $\beta\mathrm{Tr}(P_r\widehat{P}_r^\perp) - \beta\mathrm{Tr}(\widehat{P}_rP_r^\perp) = 0$ for any $\beta \in \mathbb{R}$. Thus, subtracting this from the sum yields

$$\begin{aligned}
\mathscr{E}_r &= \sum_{i=1}^r (\lambda_i - \beta)\mathrm{Tr}(A_i\widehat{P}_r^\perp) + \sum_{j=r+1}^\infty (\beta - \lambda_j)\mathrm{Tr}(A_j\widehat{P}_r) \\
&= \sum_{i=1}^r (\lambda_i - \beta)\|A_i\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} + \sum_{j=r+1}^\infty (\beta - \lambda_j)\|A_j\widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})}
\end{aligned}$$

Taking $\beta = \lambda_{r+1}$ and comparing the resulting expression with (44), we see that

$$\frac{2\mathscr{E}_r}{\lambda_r - \lambda_{r+1}} \geq 2\sum_{i=1}^r \frac{\lambda_i - \lambda_{r+1}}{\lambda_r - \lambda_{r+1}}\|A_i\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} \geq 2\sum_{i=1}^r \|A_i\widehat{P}_r^\perp\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})} = \|P_r - \widehat{P}_r\|^2_{\mathrm{HS}(\mathcal{H},\mathcal{H})},$$

since $\lambda_i - \lambda_{r+1} \geq \lambda_r - \lambda_{r+1} \geq 0$ for $i \leq r$ and $\lambda_{r+1} - \lambda_j \geq 0$ for $j \geq r+1$. This proves the claim. ■

A.4 Interchanging Differentiation and Integration

For the analysis of the H^1_γ approximation properties of conditional expectations, we need to consider the interchange of (Sobolev) differentiation and (conditional) expectation. We first look at the sufficient conditions for interchanging Fréchet differentiation and integration. A general result is difficult to find, so we state and prove a version that is sufficient for our purposes.

Theorem 51 (Interchanging Fréchet differentiation and integration)**.** *Let $\mathcal{X}_1$, $\mathcal{X}_2$ and $\mathcal{Y}$ be separable Banach spaces, $F : \mathcal{X}_1 \times \mathcal{X}_2 \to \mathcal{Y}$ be a continuously (Fréchet) differentiable function, and ν be a probability measure on $\mathcal{X}_2$. Suppose that F additionally satisfies the following assumptions*

1. *The function $F(x_1, \cdot) : \mathcal{X}_2 \to \mathcal{Y}$ is integrable with respect to ν for every $x_1 \in \mathcal{X}_1$*
2. *The partial derivative $\partial_{x_1} F(x_1, \cdot) : \mathcal{X}_2 \to \mathcal{L}(\mathcal{X}_1, \mathcal{Y})$ is integrable with respect to ν for every $x_1 \in \mathcal{X}_1$*
3. *There exists an open set $\mathcal{U} \subset \mathcal{X}_1$ and a function $g \in L^1(\mathcal{X}_2, \nu; \mathbb{R})$ such that*

$$\|\partial_{x_1} F(x_1, x_2)\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})} \leq g(x_2)$$

for every $x_1 \in \mathcal{U}$ and $x_2 \in \mathcal{X}_2$.

Then, for all $x_1 \in \mathcal{U}$,

$$D_{x_1} \mathbb{E}_{x_2 \sim \nu}[F(x_1, x_2)] = \mathbb{E}_{x_2 \sim \nu}[\partial_{x_1} F(x_1, x_2)].$$

Proof Let $F_1(x_1) := \mathbb{E}_{x_2 \sim \nu}[F(x_1, x_2)]$ and $\Psi(x_1) := \mathbb{E}_{x_2 \sim \nu}[\partial_{x_1} F(x_1, x_2)]$, where both are well-defined by the integrability assumptions. In particular, at any $x_1 \in \mathcal{X}_1$, $\Psi(x_1)$ is a linear operator on $\mathcal{X}_1$, and for any $h \in \mathcal{X}_1$,

$$\|\Psi(x_1)h\|_{\mathcal{Y}} = \|\mathbb{E}_{x_2 \sim \nu}[\partial_{x_1} F(x_1, x_2)h]\|_{\mathcal{Y}} \leq \mathbb{E}_{x_2 \sim \nu}[\|\partial_{x_1} F(x_1, x_2)\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})}]\|h\|_{\mathcal{X}_1}.$$

Thus, integrability implies $\mathbb{E}_{x_2 \sim \nu}[\|\partial_{x_1} F(x_1, x_2)\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})}] < \infty$, such that $\Psi(x_1) \in \mathcal{L}(\mathcal{X}_1, \mathcal{Y})$ for any $x_1 \in \mathcal{X}_1$. Consider now the residual

$$\frac{\|R(x_1, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} := \frac{\|F_1(x_1 + h) - F_1(x_1) - \Psi(x_1)h\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}},$$

as $h \to 0$. Combining the terms inside the expectation, we have

$$\frac{\|R(x_1, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} = \frac{\|\mathbb{E}_{x_2 \sim \nu}[F(x_1 + h, x_2) - F(x_1, x_2) - \partial_{x_1} F(x_1, x_2)h]\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}}.$$

Again, by Bochner integrability, we can bring the norm inside the expectation

$$\frac{\|R(x_1, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} \leq \mathbb{E}_{x_2 \sim \nu}\left[\frac{\|F(x_1 + h, x_2) - F(x_1, x_2) - \partial_{x_1} F(x_1, x_2)h\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}}\right].$$

Since F is differentiable, the integrand

$$\frac{\|R_1(x_1, x_2, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} := \frac{\|F(x_1 + h, x_2) - F(x_1, x_2) - \partial_{x_1} F(x_1, x_2)h\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} \to 0$$

as $h \to 0$ for every $x_1 \in \mathcal{X}_1$ and $x_2 \in \mathcal{X}_2$. It remains to bound this by an L^1 function to apply the dominated convergence theorem. The triangle inequality and the mean value theorem give

$$\frac{\|R_1(x_1, x_2, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} \leq \frac{\sup_{t \in [0,1]} \|\partial_{x_1}(F(x_1 + th, x_2))\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})}\|h\|_{\mathcal{X}_1} + \|\partial_{x_1} F(x_1, x_2)\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})}\|h\|_{\mathcal{X}_1}}{\|h\|_{\mathcal{X}_1}}.$$

As $\mathcal{U}$ is open, for any $x_1 \in \mathcal{U}$ and h sufficiently small, we have $x_1 + th \in \mathcal{U}$ for all $t \in [0, 1]$. This implies $\sup_{t \in [0,1]} \|\partial_{x_1}(F(x_1 + th, x_2))\|_{\mathcal{L}(\mathcal{X}_1, \mathcal{Y})} \leq g(x_2)$. Thus,

$$\frac{\|R_1(x_1, x_2, h)\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}_1}} \leq 2g(x_2).$$

Applying the dominated convergence theorem thus gives $\|R(x_1, h)\|_{\mathcal{Y}}/\|h\|_{\mathcal{X}_1} \to 0$, and so $DF_1(x_1) = \Psi(x_1) = \mathbb{E}_{x_2 \sim \nu}[D_{x_1} F(x_1, x_2)]$. ■

Using Theorem 51, we can proceed to consider conditional expectations for functions in $\mathcal{FC}^\infty$.

Lemma 52 (Interchanging Fréchet differentiation and conditional expectation)**.** *Suppose $F \in \mathcal{FC}^\infty$ and let $V_r = (v_i)_{i=1}^r$ be an orthonormal set in $\mathcal{E}$ for which $(C_{\mathcal{X}}^{-1} v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, the conditional expectation $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] \in C_b^\infty(\mathcal{X}, \mathcal{Y})$, and its derivatives are given by*

$$D^k \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D^k F(Q_r(\cdot), \dots, Q_r(\cdot))|\sigma(V_r^\dagger)]$$

for any $k \in \mathbb{N}$, where $D^k F(x)(Q_r(\cdot), \dots, Q_r(\cdot)) : (h_1, \dots, h_k) \mapsto D^k F(x)(Q_r h_1, \dots, Q_r h_k)$.

Proof We start with the first derivative. The conditional expectation is given by

$$\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x) = \int F(Q_r x + (I - Q_r)y)\gamma(dy).$$

First, since the derivatives of $F \in \mathcal{FC}^\infty$ are continuous and of finite rank, the mappings $x \mapsto D^k F(x)$ take values in a separable subspace of $\mathcal{L}_k(\mathcal{X}, \mathcal{Y})$ and hence are Bochner measurable. By boundedness of F and its derivatives, the function $(x, y) \mapsto F(Q_r x + (I - Q_r)y)$ satisfies the hypotheses of Theorem 51. We can therefore interchange differentiation and integration to obtain

$$D\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x) = \int DF(Q_r x + (I - Q_r)y)Q_r\gamma(dy) = \mathbb{E}_\gamma[DFQ_r|\sigma(V_r^\dagger)].$$

Let L_k denote the bound on $\|D^k F(x)\|_{\mathcal{L}_k(\mathcal{X},\mathcal{Y})}$. To see boundedness of the derivative, we have

$$\|D\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x)\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})} \le \int \|DF(Q_r x + (I - Q_r)y)Q_r\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}\gamma(dy) \le L_1\|Q_r\|_{\mathcal{L}(\mathcal{X},\mathcal{X})}.$$

On the other hand, to see continuity, we have

$$\begin{aligned}
&\|D\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x_2) - D\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x_1)\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})} \\
&\quad \le \int \|DF(Q_r x_2 + (I - Q_r)y)Q_r - DF(Q_r x_1 + (I - Q_r)y)Q_r\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}\gamma(dy) \\
&\quad \le \int L_2\|Q_r\|^2_{\mathcal{L}(\mathcal{X},\mathcal{X})}\|x_2 - x_1\|_{\mathcal{X}}\ \gamma(dy) \\
&\quad = L_2\|Q_r\|^2_{\mathcal{L}(\mathcal{X},\mathcal{X})}\|x_2 - x_1\|_{\mathcal{X}},
\end{aligned}$$

where the last inequality follows from the mean value theorem and boundedness of the second derivative. Higher-order derivatives follow by induction, starting from the hypothesis that

$$D^k \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D^k F(Q_r(\cdot), \dots, Q_r(\cdot))|\sigma(V_r^\dagger)]$$

for some $k \ge 1$. Then, we can view $x \mapsto D^k F(x)(Q_r(\cdot), \dots, Q_r(\cdot))$ as another separably-valued map with bounded derivatives, for which the same arguments can be used to show

$$D\mathbb{E}_\gamma[D^k F(\underbrace{Q_r(\cdot), \dots, Q_r(\cdot)}_{k \text{ times}})|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D^{k+1} F(\underbrace{Q_r(\cdot), \dots, Q_r(\cdot)}_{k+1 \text{ times}})|\sigma(V_r^\dagger)].$$

Thus, $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ has continuous and bounded kth order derivatives for all $k \in \mathbb{N}$.

■

Finally, the result can be extended to Sobolev differentiation of functions in H_γ^m through a density argument, giving us Lemma 17.

Proof (of Lemma 17) By the density argument, we can find a sequence of $\mathcal{FC}^\infty$ functions, $(F_n)_{n=1}^\infty$ convergent to F in H^m_γ. This also means $\mathbb{E}_\gamma[F_n|\sigma(V_r^\dagger)]$ converges to $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ in L^2_γ, since

$$\begin{aligned}
&\|\mathbb{E}_\gamma[F_n|\sigma(V_r^\dagger)] - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|^2_{L^2_\gamma} \\
&\quad = \int \left\| \int F_n(Q_r x + (I-Q_r)y) - F(Q_r x + (I-Q_r)y)\gamma(dy) \right\|^2 \gamma(dx) \\
&\quad \leq \iint \|F_n(Q_r x + (I-Q_r)y) - F(Q_r x + (I-Q_r)y)\|^2 \gamma(dy)\gamma(dx) \\
&\quad = \|F_n - F\|^2_{L^2_\gamma}.
\end{aligned}$$

By Lemma 52, we can interchange the conditional expectation and Fréchet differentiation for $F_n \in \mathcal{FC}^\infty$. Since this coincides with the Sobolev derivative, we have $D_\mathcal{E}\mathbb{E}_\gamma[F_n|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D_\mathcal{E}F_nQ_r|\sigma(V_r^\dagger)]$. Moreover, we can verify $\mathbb{E}_\gamma[D_\mathcal{E}F_nQ_r|\sigma(V_r^\dagger)]$ converges to $\mathbb{E}_\gamma[D_\mathcal{E}FQ_r|\sigma(V_r^\dagger)]$, since

$$\begin{aligned}
&\|\mathbb{E}_\gamma[D_\mathcal{E}F_nQ_r|\sigma(V_r^\dagger)] - \mathbb{E}_\gamma[D_\mathcal{E}FQ_r|\sigma(V_r^\dagger)]\|^2_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))} \\
&\quad = \int \left\| \int D_\mathcal{E}F_n(Q_r x + (I-Q_r)y)Q_r - D_\mathcal{E}F(Q_r x + (I-Q_r)y)Q_r\gamma(dy) \right\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \gamma(dx) \\
&\quad \leq \iint \|D_\mathcal{E}F_n(Q_r x + (I-Q_r)y) - D_\mathcal{E}F(Q_r x + (I-Q_r)y)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \|Q_r\|^2_{\mathcal{L}(\mathcal{E},\mathcal{E})} \gamma(dy)\gamma(dx) \\
&\quad = \|D_\mathcal{E}F_n - D_\mathcal{E}F\|^2_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}.
\end{aligned}$$

Applying Lemma 47, we obtain $D_\mathcal{E}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] = \mathbb{E}_\gamma[D_\mathcal{E}FQ_r|\sigma(V_r^\dagger)]$. Moreover, repeatedly applying this yields the desired result for higher-order derivatives.

■

A corollary of Lemma 17 is that the conditional expectation is contractive in the H^m_γ norm.

Corollary 53. *Suppose $F \in H^m_\gamma$ for some $m \in \mathbb{N}_0$ and let $V_r = (v_i)_{i=1}^r$ be an orthonormal set in $\mathcal{E}$ for which $(C_\mathcal{X}^{-1}v_i)_{i=1}^r$ are well-defined in $\mathcal{X}$. Then, $\|\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{H^m_\gamma} \leq \|F\|_{H^m_\gamma}$.*

Proof As shown in the proof of Lemma 17, the conditional expectation is contractive in the L^2_γ norm. Thus, for any $k \leq m$, we have

$$\begin{aligned}
\|D^k_\mathcal{E}\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|^2_{L^2(\gamma,\mathrm{HS}_k(\mathcal{E},\mathcal{Y}))} &= \|\mathbb{E}_\gamma[D^k_\mathcal{E}F(Q_r(\cdot),\dots,Q_r(\cdot))|\sigma(V_r^\dagger)]\|^2_{L^2(\gamma,\mathrm{HS}_k(\mathcal{E},\mathcal{Y}))} \\
&\leq \|D^k_\mathcal{E}F(Q_r(\cdot),\dots,Q_r(\cdot))\|^2_{L^2(\gamma,\mathrm{HS}_k(\mathcal{E},\mathcal{Y}))} \\
&\leq \|D^k_\mathcal{E}F\|^2_{L^2(\gamma,\mathrm{HS}_k(\mathcal{E},\mathcal{Y}))}.
\end{aligned}$$

■

A.5 Wiener Chaos Expansions and Polynomial Forms

In this section, we show that nth-order polynomial forms on $\mathcal{X}$ satisfy the derivative inverse inequality (Assumption 10) and the Hessian inverse inequality (Assumption 11) with constants $K_D = n$ and $K_H = n-1$. Let $H_n : \mathbb{R} \to \mathbb{R}$ denote the nth Hermite polynomial, defined for $n = 0, 1, \dots$ by

$$H_n(x) = \frac{(-1)^n}{\sqrt{n!}} \exp\left(\frac{x^2}{2}\right) \frac{d^n}{dx^n} \exp\left(-\frac{x^2}{2}\right).$$

With this choice of normalization, the Hermite polynomials form an orthonormal basis of $L^2(\gamma_1)$ where γ_1 is the standard Gaussian measure on $\mathbb{R}$. Moreover, one has $H_n'(x) = \sqrt{n} H_{n-1}(x)$, which implies that the derivatives of Hermite polynomials are also orthogonal under the $L^2(\gamma_1)$ inner product. To simplify notation, we will also let $H_n(x) := 0$ for $n < 0$.

We can also define multivariate Hermite polynomials as tensor products of univariate Hermite polynomials. Recall that for a multi-index $\alpha = (\alpha_1, \alpha_2, \dots)$, $\alpha_i \in \mathbb{N} \cup \{0\}$, we have $|\alpha| = \sum_{i=1}^{\infty} \alpha_i$ and $\operatorname{supp} \alpha = \{i \in \mathbb{N} : \alpha_i \neq 0\}$. Then, for $\alpha \in \Lambda := \{\alpha : |\alpha| < \infty\}$, we define

$$H_\alpha(x) = \prod_{i \in \operatorname{supp} \alpha} H_{\alpha_i}(x_i).$$

In the case of separable infinite-dimensional Hilbert spaces $\mathcal{X}$ and $\mathcal{Y}$, the family of Hermite polynomials H_α, $\alpha \in \Lambda$, along with an orthonormal basis of the Cameron–Martin space $\mathcal{E}$, can be used to construct an orthonormal basis of $L^2_\gamma = L^2(\gamma, \mathcal{Y})$. For example, given the PCA basis $(v_i^{\mathrm{PCA}})_{i=1}^{\infty}$ (normalized in $\mathcal{E}$), one defines the basis functions $H_{\gamma,\alpha} : \mathcal{X} \to \mathbb{R}$,

$$H_{\gamma,\alpha}(x) := \prod_{i \in \operatorname{supp} \alpha} H_{\alpha_i}\left(\left\langle x, C_{\mathcal{X}}^{-1} v_i^{\mathrm{PCA}} \right\rangle_{\mathcal{X}}\right).$$

Any function $F \in L^2_\gamma$ can then be represented as

$$F(x) = \sum_{\alpha \in \Lambda} c_\alpha H_{\gamma,\alpha}(x)$$

with coefficients $c_\alpha \in \mathcal{Y}$, which is known as the Wiener chaos expansion. Moreover,

$$\|F\|^2_{L^2_\gamma} = \sum_{\alpha \in \Lambda} \|c_\alpha\|^2_{\mathcal{Y}}.$$

Note that the derivatives of $H_{\gamma,\alpha}$ along v_k^{PCA}, which are orthonormal in $\mathcal{E}$, are given by

$$D_{\mathcal{E}} H_{\gamma,\alpha}(x) v_k^{\mathrm{PCA}} = \sqrt{\alpha_k} H_{\gamma,\alpha - e_k}(x), \tag{45}$$

where the multi-index $\alpha - e_k$ is defined as

$$(\alpha - e_k)_j = \begin{cases} \alpha_j - 1 & j = k, \\ \alpha_j & j \neq k. \end{cases}$$

This means that $D_{\mathcal{E}} H_{\gamma,\alpha}$ are also orthogonal in $L^2(\gamma, \mathrm{HS}(\mathcal{E}, \mathbb{R}))$, and

$$\|D_{\mathcal{E}} H_{\gamma,\alpha}(x)\|^2_{L^2(\gamma, \mathrm{HS}(\mathcal{E},\mathbb{R}))} = \sum_{k=1}^{\infty} \alpha_k \int H_{\gamma,\alpha-e_k}(x)^2 \gamma(dx) = |\alpha| \|H_{\gamma,\alpha}\|^2_{L^2_\gamma} = |\alpha| \tag{46}$$

In fact, repeatedly applying the derivative formula (45) gives that $D^k_{\mathcal{E}} H_{\gamma,\alpha}$ are orthogonal in $L^2(\gamma, \mathrm{HS}_k(\mathcal{E}, \mathbb{R}))$ for any $k \in \mathbb{N}$, and

$$\|D^k_{\mathcal{E}} H_{\gamma,\alpha}(x)\|^2_{L^2(\gamma, \mathrm{HS}_k(\mathcal{E},\mathbb{R}))} = |\alpha|(|\alpha| - 1) \cdots (|\alpha| - k + 1). \tag{47}$$

We refer to Nualart (2006, Chapter 1) for a more complete exposition.

Using the above properties of the Hermite polynomials, we can show that nth order polynomial forms on $\mathcal{X}$ satisfy the derivative inverse inequality and the Hessian inverse inequality with constants $K_D = n$ and $K_H = n - 1$.

Proposition 54. *Let $F \in L^2_\gamma$ be represented by Hermite polynomials up to degree $n \in \mathbb{N}$,*

$$F(x) = \sum_{|\alpha| \leq n} c_\alpha H_{\gamma,\alpha}(x).$$

Then, F satisfies the derivative inverse inequality (12) *with constant $K_D = n$ and the Hessian inverse inequality* (13) *with constant $K_H = n - 1$.*

Proof We first note that since the Wiener chaos expansion for F is of finite polynomial order, $F \in H^k_\gamma$ for any $k \geq 0$. We start by considering the derivative inverse inequality. For $F = \sum_{|\alpha| \leq n} c_\alpha H_{\gamma,\alpha}$, its mean-shifted form is given by $F - \bar{F} = \sum_{1 \leq |\alpha| \leq n} c_\alpha H_{\gamma,\alpha}$. Then, given any $u \in \mathcal{Y}$, we consider the function $u^* \circ (F - \bar{F})$, where $u^*(y) = \langle y, u \rangle_{\mathcal{Y}}$. This has the representation $u^* \circ (F - \bar{F}) = \sum_{1 \leq |\alpha| \leq n} \langle u, c_\alpha \rangle_{\mathcal{Y}} H_{\gamma,\alpha}$. Now,

$$\langle u, H_{\mathcal{Y}} u \rangle_{\mathcal{Y}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}(u^* \circ F)\|^2_{\mathrm{HS}(\mathcal{E},\mathbb{R})}] = \mathbb{E}_\gamma[\|D_{\mathcal{E}}(u^* \circ (F - \bar{F}))\|^2_{\mathrm{HS}(\mathcal{E},\mathbb{R})}].$$

Orthogonality of the derivatives of Hermite polynomials along with (46) (see also Nualart (2006, Proposition 1.2.2)) gives

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}(u^* \circ (F - \bar{F}))\|^2_{\mathrm{HS}(\mathcal{E},\mathbb{R})}] \leq n \|u^* \circ (F - \bar{F})\|^2_{L^2(\gamma)} = n \langle u, C_{\mathcal{Y}} u \rangle_{\mathcal{Y}},$$

which shows that $K_D = n$.

The Hessian inverse inequality is shown similarly. We first note that $D_{\mathcal{E}} F$ has a Wiener chaos expansion of degree $n - 1$, $D_{\mathcal{E}} F = \sum_{|\alpha| \leq n-1} A_\alpha H_{\gamma,\alpha}$, where $A_\alpha \in \mathrm{HS}(\mathcal{E}, \mathcal{Y})$. Moreover, let $G := D_{\mathcal{E}} F v = \sum_{|\alpha| \leq n-1} A_\alpha v H_{\gamma,\alpha}$, for which we have

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}(D_{\mathcal{E}} F v)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] = \mathbb{E}_\gamma[\|D_{\mathcal{E}} G\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \leq (n-1)\mathbb{E}_\gamma[\|G\|^2_{\mathcal{Y}}] = (n-1)\mathbb{E}_\gamma[\|D_{\mathcal{E}} F v\|^2_{\mathcal{Y}}].$$

Thus, $K_H = n - 1$. ■

Appendix B. Universal Approximation Theorem in Sobolev Norms

B.1 Original Theorem of Hornik

Hornik (1991) proved a universal approximation result for scalar-valued functions using a single-layer, arbitrary-width neural network in any weighted Sobolev norm on finite-dimensional input spaces with a finite measure ν. For $f \in C^m(\mathbb{R}^d, \mathbb{R})$, this Sobolev norm is defined in Hornik (1991) as

$$\|f\|_{W^{m,p}(\nu,\mathbb{R})} = \left(\int_{\mathbb{R}^d} |f(x)|^p d\nu(x) + \sum_{1 \leq |\alpha| \leq m} \int_{\mathbb{R}^d} |\partial^\alpha f(x)|^p d\nu(x) \right)^{\frac{1}{p}},$$

where we are using the multi-index notation for derivatives, with $\alpha = (\alpha_1, \ldots, \alpha_d)$ and

$$\partial^\alpha f(x) = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \ldots \partial x_d^{\alpha_d}}(x).$$

The original result (Hornik, 1991, Theorem 4) considers activation functions that are non-constant with bounded value and derivatives up to some order $m \geq 0$, and states that single-layer neural networks with such activation functions are universal approximators of the space

$$C^{m,p}_\nu(\mathbb{R}^{d_{\mathrm{in}}}) := \{f \in C^m(\mathbb{R}^{d_{\mathrm{in}}}) : \|f\|_{W^{m,p}(\nu,\mathbb{R})} < \infty\}.$$

Recall that $C_b^m(\mathbb{R})$ denotes the space of functions with bounded derivatives up to order m, i.e., $\psi \in C_b^m(\mathbb{R})$ if $\psi \in C^m(\mathbb{R})$ and for any $k = 0, \dots, m$, there exists $M_k > 0$ such that $|\psi^{(k)}(t)| \leq M_k$ for all $t \in \mathbb{R}$. We can paraphrase Hornik (1991, Theorem 4) as follows.

Theorem 55 (Universal approximation with smooth, bounded activation functions)**.** *Suppose ν is a finite measure on $\mathbb{R}^{d_{\text{in}}}$ and $g \in C_\nu^{m,p}(\mathbb{R}^{d_{\text{in}}})$ for some $m \in \mathbb{N}_0$, $p \in [1, \infty)$. Let $\psi \in C_b^m(\mathbb{R})$ be non-constant. Then, for any $\epsilon > 0$, there exists a single-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ and some width $d_W \in \mathbb{N}$ such that*

$$\|g - \varphi_{\psi,\theta}\|_{W^{m,p}(\nu,\mathbb{R})} \leq \epsilon.$$

B.2 Extension to Smooth Non-saturating Activation Functions

In practice, smooth and non-saturating activation functions, such as GELU, softplus, and SiLU, are often used instead of bounded activation functions. Thus, we will extend the above result to include activation functions belonging to $\mathbb{A}^m$ as defined in Definition 1. The extension is based on the fact that for any $\psi \in \mathbb{A}^m$, we can construct a new activation function $\tilde{\psi}$ that is non-constant and has bounded derivatives up to order m through a linear combination of shifted versions of ψ. This allows us to identify single-layer neural networks using the activation function $\tilde{\psi}$ as single-layer neural networks using the activation function ψ, which, combined with Theorem 55, yields the desired result.

To this end, we start by defining the one-step difference operator $\mathtt{D}_h$ as

$$(\mathtt{D}_h f)(t) = f(t + h) - f(t).$$

We note that for any f that is non-polynomial, there exists some shift h such that $\mathtt{D}_h f$ is non-constant.

Lemma 56. *Suppose $f \in C^0(\mathbb{R})$ is non-polynomial. Then, there exists $h \in \mathbb{R}$ such that $\mathtt{D}_h f$ is non-constant.*

Proof We proceed by contradiction. If $(\mathtt{D}_h f)(t)$ is constant for all $h \in \mathbb{R}$, then we have $f(t + h) - f(t) = c(h)$ for some function $c : \mathbb{R} \to \mathbb{R}$. Taking $t = 0$, we see that $c(h) = f(h) - f(0)$ is continuous. Moreover, using two increments $h_1 \in \mathbb{R}$ and $h_2 \in \mathbb{R}$, we have

$$c(h_1 + h_2) = f(h_1 + h_2) - f(0) = f(0 + h_1 + h_2) - f(h_1) + f(h_1) - f(0) = c(h_2) + c(h_1),$$

and hence c is additive. Since continuous and additive functions are necessarily linear, this implies that for any $t \in \mathbb{R}$, $f(t) = f(0 + t) = c(t) + f(0)$, so that f is an affine polynomial and contradicts the initial assumption. ■

We can then construct $\tilde{\psi} \in C_b^m$ by repeatedly applying the difference operator to $\psi \in \mathbb{A}^m$, with the shifts chosen to ensure that the resulting function is non-polynomial.

Lemma 57. *Let $m \in \mathbb{N}_0$ and suppose $f \in C^{m,1}(\mathbb{R})$ is non-polynomial. Then, there exists $h_0, \dots, h_m$ such that*

$$g(t) := (\mathtt{D}_{h_m} \dots \mathtt{D}_{h_0} f)(t),$$

is non-polynomial and belongs to $C_b^m(\mathbb{R})$.

Proof We proceed by induction. First, consider $k = 0$, such that $f \in C^0(\mathbb{R})$ is non-polynomial and Lipschitz continuous. Let $L > 0$ denote the Lipschitz constant. For any $h \in \mathbb{R}$, we have $g = \mathtt{D}_h f \in C_b^0(\mathbb{R})$ since $|\mathtt{D}_h f(t)| = |f(t+h) - f(t)| \leq L|h|$. Additionally, since f is non-polynomial, by Lemma 56, there exists some $h_0 \in \mathbb{R}$ such that $g = \mathtt{D}_{h_0} f$ is non-constant (and hence non-polynomial as g is bounded). Choosing this h_0 yields the desired result for $k = 0$.

Now, suppose the result holds for some $k-1$ with $k \geq 1$ and consider a non-polynomial function $f \in C^{k,1}(\mathbb{R})$. This implies that $f'(t) \in C^{k-1,1}(\mathbb{R})$ and is also non-polynomial. By the induction hypothesis, there exist $h_0, \ldots, h_{k-1}$ such that $\mathtt{D}_{h_{k-1}} \ldots \mathtt{D}_{h_0} f'(t)$ is non-polynomial and in $C_b^{k-1}(\mathbb{R})$. Let $g_{k-1} := \mathtt{D}_{h_{k-1}} \ldots \mathtt{D}_{h_0} f$, such that $g'_{k-1} = \mathtt{D}_{h_{k-1}} \ldots \mathtt{D}_{h_0} f' \in C_b^{k-1}(\mathbb{R})$. In particular, boundedness of g'_{k-1} implies that g_{k-1} is Lipschitz continuous. Moreover, since g'_{k-1} is non-polynomial, g_{k-1} is also non-polynomial. Therefore, by the same reasoning for the $k = 0$ case, there exists $h_k \in \mathbb{R}$ such that $g = \mathtt{D}_{h_k} g_{k-1}$ is non-polynomial and in $C_b^0(\mathbb{R})$, which combined with the fact that $g' \in C_b^{k-1}(\mathbb{R})$, implies that $g \in C_b^k(\mathbb{R})$, yielding the desired result. Thus, the result holds for all $m \geq 0$. ■

We can then use the bounded activation function $\tilde{\psi}$ based on Lemma 57 with the original universal approximation result in Theorem 55 to obtain a single-layer neural network with activation function $\tilde{\psi}$ that approximates g in the desired Sobolev norm.

Theorem 58 (Shallow universal approximation with $\mathbb{A}^m$ activation functions)**.** *Suppose ν is a finite measure on $\mathbb{R}^{d_{\text{in}}}$ and $g \in C_\nu^{m,p}(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}})$ for some $m \in \mathbb{N}_0$, $p \in [1, \infty)$, and let $\psi \in \mathbb{A}^m$. Then, for any $\epsilon > 0$, there exists a single-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ and some width $d_W \in \mathbb{N}$ such that*

$$\|g - \varphi_{\psi,\theta}\|_{W^{m,p}(\nu, \mathbb{R}^{d_{\text{out}}})} \leq \epsilon.$$

Moreover, $\varphi_{\psi,\theta} \in C_b^m(\mathbb{R}^{d_{\text{in}}}, \mathbb{R}^{d_{\text{out}}})$.

Proof Consider first $d_{\text{out}} = 1$. Since $\psi \in \mathbb{A}^m$, we have that $\psi \in C^{k,1}(\mathbb{R})$ for some $k \geq m$ and ψ is non-polynomial. Thus, we can define a new activation $\tilde{\psi} = \mathtt{D}_{h_k} \ldots \mathtt{D}_{h_0} \psi$ as in Lemma 57, which is non-polynomial and belongs to $C_b^k(\mathbb{R}) \subset C_b^m(\mathbb{R})$. Then, by Theorem 55, there exists a single-layer neural network of width $\tilde{d}_W$ with activation function $\tilde{\psi}$,

$$\varphi_{\tilde{\psi},\tilde{\theta}}(x) = \widetilde{\mathbf{W}}_2 \tilde{\psi}(\widetilde{\mathbf{W}}_1 x + \widetilde{\mathbf{b}}_1) + \widetilde{\mathbf{b}}_2$$

with weights $\widetilde{\mathbf{W}}_1 \in \mathbb{R}^{\tilde{d}_W \times d_{\text{in}}}$, $\widetilde{\mathbf{W}}_2 \in \mathbb{R}^{1 \times \tilde{d}_W}$, and biases $\widetilde{\mathbf{b}}_1 \in \mathbb{R}^{\tilde{d}_W}$, $\widetilde{\mathbf{b}}_2 \in \mathbb{R}$, such that $\|g - \varphi_{\tilde{\psi},\tilde{\theta}}\|_{W^{m,p}(\nu,\mathbb{R})} \leq \epsilon$. Evidently, $\varphi_{\tilde{\psi},\tilde{\theta}} \in C_b^m(\mathbb{R}^{d_{\text{in}}}, \mathbb{R})$ since $\tilde{\psi} \in C_b^m(\mathbb{R})$. Moreover, $\tilde{\psi} = \mathtt{D}_{h_k} \ldots \mathtt{D}_{h_0} \psi$ is a linear combination of shifted versions of ψ with at most 2^{k+1} terms, such that

$$\tilde{\psi}(t) = \sum_{i=1}^{2^{k+1}} a_i \psi(t + b_i),$$

with suitable $a_i, b_i \in \mathbb{R}$ for $i = 1, \ldots, 2^{k+1}$. This implies that $\varphi_{\tilde{\psi},\tilde{\theta}}$ can be written as

$$\varphi_{\tilde{\psi},\tilde{\theta}}(x) = \widetilde{\mathbf{W}}_2 \sum_{i=1}^{2^{k+1}} a_i \psi(\widetilde{\mathbf{W}}_1 x + \widetilde{\mathbf{b}}_1 + b_i \mathbf{1}_{\tilde{d}_W}) + \widetilde{\mathbf{b}}_2 = \mathbf{W}_2 \psi(\mathbf{W}_1 x + \mathbf{b}_1) + \mathbf{b}_2,$$

where $\mathbf{1}_{\tilde{d}_W} = (1, \ldots, 1) \in \mathbb{R}^{\tilde{d}_W}$. That is, $\varphi_{\tilde{\psi},\tilde{\theta}}$ corresponds to a single-layer neural network $\varphi_{\psi,\theta}$ with activation function ψ and width $d_W = 2^{k+1}\tilde{d}_W$, whose weights are

$$\mathbf{W}_1 = \begin{bmatrix} \widetilde{\mathbf{W}}_1 \\ \vdots \\ \widetilde{\mathbf{W}}_1 \end{bmatrix} \in \mathbb{R}^{d_W \times d_{\text{in}}}, \qquad \mathbf{W}_2 = [a_1 \widetilde{\mathbf{W}}_2, a_2 \widetilde{\mathbf{W}}_2, \ldots, a_{2^{k+1}} \widetilde{\mathbf{W}}_2] \in \mathbb{R}^{1 \times d_W},$$

and biases are

$$\mathbf{b}_1 = \begin{bmatrix} \widetilde{\mathbf{b}}_1 + b_1 \mathbf{1}_{\tilde{d}_W} \\ \widetilde{\mathbf{b}}_1 + b_2 \mathbf{1}_{\tilde{d}_W} \\ \vdots \\ \widetilde{\mathbf{b}}_1 + b_{2^{k+1}} \mathbf{1}_{\tilde{d}_W} \end{bmatrix} \in \mathbb{R}^{d_W}, \qquad \mathbf{b}_2 = \widetilde{\mathbf{b}}_2 \in \mathbb{R}.$$

The case $d_{\text{out}} > 1$ follows by applying the same construction to each output component using the appropriate error tolerance and parallelizing the resulting networks. ■

B.3 Extension to Deep Neural Networks

Additionally, the result can be extended to deep neural networks by considering approximations to the identity function $\mathrm{Id}(x) = x$.

Lemma 59. *Suppose $\varphi \in C_b^m(\mathbb{R}^{d_{\text{in}}})$ for some $m \in \mathbb{N}_0$ and $\psi \in \mathbb{A}^m$ is an activation function. Then, for any $\epsilon > 0$, there exists a single-layer neural network Id_θ with activation function ψ such that $\mathrm{Id}_\theta \circ \varphi \in C_b^m(\mathbb{R}^{d_{\text{in}}})$ and*

$$\|\mathrm{Id}_\theta \circ \varphi - \varphi\|_{C_b^m(\mathbb{R}^{d_{\text{in}}})} \leq \epsilon. \tag{48}$$

A fortiori, given a finite measure ν on $\mathbb{R}^{d_{\text{in}}}$ and $p \in [1, \infty)$, for any $\epsilon > 0$, there exists a single-layer neural network Id_θ with activation function ψ such that $\mathrm{Id}_\theta \circ \varphi \in C_b^m(\mathbb{R}^{d_{\text{in}}})$ and

$$\|\mathrm{Id}_\theta \circ \varphi - \varphi\|_{W^{m,p}(\nu,\mathbb{R})} \leq \epsilon. \tag{49}$$

Proof First, let M_k denote the bound on the kth derivative of φ. That is, for any $k = 0, \ldots, m$ and any multi-index α with $|\alpha| = k$, we have $|\partial^\alpha \varphi(x)| \leq M_k$ for all $x \in \mathbb{R}^{d_{\text{in}}}$. We then construct an identity approximation Id_θ on the interval $[-M_0, M_0]$, which contains the range of φ. By conventional universal approximation results on compact sets such as Pinkus (1999, Theorem 4.1), for any $\delta > 0$, there exists a single-layer neural network Id_θ with activation function ψ such that $\|\mathrm{Id}_\theta - \mathrm{Id}\|_{C^m([-M_0,M_0])} \leq \delta$.

Given this construction, we can then look to bound the error between $\mathrm{Id}_\theta \circ \varphi$ and φ in the $C_b^m(\mathbb{R}^{d_{\text{in}}})$ norm. Consider first the $k = 0$ case, where we have

$$|\mathrm{Id}_\theta \circ \varphi(x) - \varphi(x)| \leq \delta, \qquad \forall x \in \mathbb{R}^{d_{\text{in}}}, \tag{50}$$

since $\varphi(x) \in [-M_0, M_0]$ for all $x \in \mathbb{R}^{d_{\text{in}}}$. For the first derivative, we have $\frac{\partial}{\partial x_i}(\mathrm{Id}_\theta \circ \varphi)(x) = \mathrm{Id}'_\theta(\varphi(x)) \frac{\partial \varphi}{\partial x_i}$, such that

$$\left| \frac{\partial}{\partial x_i}(\mathrm{Id}_\theta \circ \varphi)(x) - \frac{\partial \varphi}{\partial x_i}(x) \right| \leq \left| \mathrm{Id}'_\theta(\varphi(x)) - 1 \right| \left| \frac{\partial \varphi}{\partial x_i}(x) \right| \leq \delta M_1, \qquad \forall x \in \mathbb{R}^{d_{\text{in}}}. \tag{51}$$

For the higher-order derivatives, we recall the Faà di Bruno formula for multivariate functions (Hardy, 2006). For $f : \mathbb{R}^d \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, this takes the form

$$\frac{\partial^k (g \circ f)}{\partial x_{i_1} \ldots \partial x_{i_k}} = \sum_{\pi \in \Pi} g^{(|\pi|)}(f) \prod_{J \in \pi} \frac{\partial^{|J|} f}{\prod_{j \in J} \partial x_{i_j}},$$

where Π is the set of partitions of $\{1, \ldots, k\}$ and $|\pi|, |J|$ refer to the number of elements in π and J, respectively. Note that there is only one partition of size one, i.e., $\pi_k = \{\{1, \ldots, k\}\}$, which yields

the term $g'(f)\frac{\partial^k f}{\partial x_{i_1}\dots\partial x_{i_k}}$. We can therefore decompose the sum as

$$\frac{\partial^k (g\circ f)}{\partial x_{i_1}\dots\partial x_{i_k}} = g'(f)\frac{\partial^k f}{\partial x_{i_1}\dots\partial x_{i_k}} + \sum_{\pi\in\Pi\setminus\{\pi_k\}} g^{(|\pi|)}(f)\prod_{J\in\pi}\frac{\partial^{|J|} f}{\prod_{j\in J}\partial x_{i_j}}.$$

Applying this to the composition $\mathrm{Id}_\theta\circ\varphi$, the error in each component of the kth derivative is given by

$$\frac{\partial^k}{\partial x_{i_1}\dots\partial x_{i_k}}(\mathrm{Id}_\theta\circ\varphi) - \frac{\partial^k}{\partial x_{i_1}\dots\partial x_{i_k}}(\varphi) = (\mathrm{Id}'_\theta\circ\varphi - 1)\frac{\partial^k\varphi}{\partial x_{i_1}\dots\partial x_{i_k}} + \sum_{\pi\in\Pi\setminus\{\pi_k\}}\mathrm{Id}_\theta^{(|\pi|)}(\varphi)\prod_{J\in\pi}\frac{\partial^{|J|}\varphi}{\prod_{j\in J}\partial x_{i_j}}.$$

Again, the first term can be bounded by

$$\left|((\mathrm{Id}'_\theta\circ\varphi)(x)-1)\frac{\partial^k\varphi(x)}{\partial x_{i_1}\dots\partial x_{i_k}}\right| \le \delta M_k, \qquad \forall x\in\mathbb{R}^{d_{\text{in}}},$$

while each term in the sum can be bounded by

$$\left|\mathrm{Id}_\theta^{(|\pi|)}(\varphi(x))\prod_{J\in\pi}\frac{\partial^{|J|}\varphi(x)}{\prod_{j\in J}\partial x_{i_j}}\right| \le \delta\prod_{J\in\pi}M_{|J|}, \qquad \forall x\in\mathbb{R}^{d_{\text{in}}},$$

Thus, the overall error in the kth derivative can be bounded by

$$\left|\frac{\partial^k}{\partial x_{i_1}\dots\partial x_{i_k}}(\mathrm{Id}_\theta\circ\varphi)(x) - \frac{\partial^k}{\partial x_{i_1}\dots\partial x_{i_k}}(\varphi)(x)\right| \le \delta M_k + \sum_{\pi\in\Pi\setminus\{\pi_k\}}\delta\prod_{J\in\pi}M_{|J|}, \qquad \forall x\in\mathbb{R}^{d_{\text{in}}}. \quad (52)$$

To obtain the bound in the $C_b^m(\mathbb{R}^{d_{\text{in}}})$ norm, (48), it suffices to choose δ and hence Id_θ such that the upper bounds in (50), (51), and (52) for $k=2,\dots,m$ are simultaneously less than ϵ. The bound in the $W^{m,p}(\nu,\mathbb{R})$, (49), follows by analogous reasoning. ■

To complete the proof of Theorem 2, it remains to extend the single-layer approximation result of Theorem 58 to deep neural networks.

Proof (of Theorem 2) Beginning with the single-layer approximation result of Theorem 58, we can construct a deep neural network by appending the identity approximation Id_{θ_2} to the output of the single-layer network. We focus on the case of $d_{\text{out}}=1$, since the case of $d_{\text{out}}>1$ follows by applying the same construction to each output component and parallelizing the resulting networks.

We first illustrate this for the case of $d_L=2$ layers. Let $\varphi_1=\varphi_{\theta_1}\in C_b^m(\mathbb{R}^{d_{\text{in}}})$ be the single-layer approximation to g as defined in Theorem 58. Then, we define $\varphi_2=\mathrm{Id}_{\theta_2}\circ\varphi_1$ as the deep neural network with $d_L=2$ layers. The approximation error is given by

$$\|g-\varphi_2\|_{W^{m,p}(\nu,\mathbb{R})} \le \|g-\varphi_1\|_{W^{m,p}(\nu,\mathbb{R})} + \|\varphi_1-\mathrm{Id}_{\theta_2}\circ\varphi_1\|_{W^{m,p}(\nu,\mathbb{R})}.$$

By Theorem 58, we can choose φ_1 such that $\|g-\varphi_1\|_{W^{m,p}(\nu,\mathbb{R})}\le\epsilon/2$, and by Lemma 59, we can choose Id_{θ_2} such that $\|\mathrm{Id}_{\theta_2}\circ\varphi_1-\varphi_1\|_{W^{m,p}(\nu,\mathbb{R})}\le\epsilon/2$, such that $\|g-\varphi_2\|_{W^{m,p}(\nu,\mathbb{R})}\le\epsilon$.

For $d_L>2$, since $\mathrm{Id}_{\theta_2}\circ\varphi_1$ remains in $C_b^m(\mathbb{R}^{d_{\text{in}}})$, the same reasoning can be applied to append additional identity approximations to the output of φ_2 to construct a deep neural network with d_L layers, $\varphi_{\psi,\theta}=\mathrm{Id}_{\theta_{d_L}}\circ\cdots\circ\mathrm{Id}_{\theta_2}\circ\varphi_1$, using the appropriate error tolerance for φ_1 and the identity approximations to ensure that the total accumulated error is less than ϵ. ■

B.4 Approximation Rates for ReLU^k Activation in Sobolev Norms

As discussed in Theorem 3, when specifically working with ReLU^k to approximate functions in the Gaussian Sobolev space H^m_γ in finite dimensions, we can obtain bounds for the neural network size. These follow from standard arguments by first considering the Wiener chaos expansion of the target function, and subsequently emulating the truncated expansion with a ReLU^k neural network. While the proof is standard, we provide details for completeness. To this end, we first show when $g \in H^l(\gamma, \mathbb{R}^{d_{\text{out}}})$ for some $l > m$, the Wiener chaos expansion of g converges in $H^m(\gamma, \mathbb{R}^{d_{\text{out}}})$ at the rate $N^{-(l-m)/2}$ when truncated to polynomial degrees up to N.

Proposition 60. *Let $l, m \in \mathbb{N}_0$ with $l > m$, and let $\gamma = \mathcal{N}(0, I_{d_{\text{in}}})$ be the standard Gaussian measure on $\mathbb{R}^{d_{\text{in}}}$. Consider a function $g \in H^l(\gamma, \mathbb{R}^{d_{\text{out}}})$ with Wiener chaos expansion $g = \sum_{n=0}^{\infty} \sum_{|\alpha|=n} c_\alpha H_\alpha$, where H_α is the d_{in}-variate Hermite polynomial of degree α. Let $g_N := \sum_{n=0}^{N} \sum_{|\alpha|=n} c_\alpha H_\alpha$ be the truncation of the Wiener chaos expansion to total polynomial degrees up to N. Then, there exists a constant $\kappa = \kappa(l, m)$ such that for any $N \in \mathbb{N}$, we have*

$$\|g - g_N\|^2_{H^m(\gamma, \mathbb{R}^{d_{\text{out}}})} \leq \kappa N^{-(l-m)} \|g\|^2_{H^l(\gamma, \mathbb{R}^{d_{\text{out}}})}.$$

Proof With $\gamma = \mathcal{N}(0, I_{d_{\text{in}}})$, the Cameron–Martin space coincides with $\mathbb{R}^{d_{\text{in}}}$, and $H^m(\gamma, \mathbb{R}^{d_{\text{out}}})$ functions admit a Wiener chaos expansion in terms of the d_{in}-variate Hermite polynomials H_α. Moreover, by (47), we have the spectral characterization of the $H^m(\gamma, \mathbb{R}^{d_{\text{out}}})$ norm,

$$\kappa_1(m) \sum_{n=0}^{\infty} \sum_{|\alpha|=n} (1+n)^m \|c_\alpha\|_2^2 \leq \|g\|^2_{H^m(\gamma, \mathbb{R}^{d_{\text{out}}})} \leq \kappa_2(m) \sum_{n=0}^{\infty} \sum_{|\alpha|=n} (1+n)^m \|c_\alpha\|_2^2,$$

with equivalence constants $\kappa_1(m), \kappa_2(m) > 0$ that depend on m. The truncation error is then bounded by

$$\|g - g_N\|^2_{H^m(\gamma, \mathbb{R}^{d_{\text{out}}})} \leq \kappa_2(m) \sum_{n=N+1}^{\infty} \sum_{|\alpha|=n} (1+n)^m \|c_\alpha\|_2^2.$$

On the other hand, since $g \in H^l(\gamma, \mathbb{R}^{d_{\text{out}}})$, we have that

$$\sum_{n=0}^{\infty} \sum_{|\alpha|=n} (1+n)^l \|c_\alpha\|_2^2 \leq \kappa_1^{-1}(l) \|g\|^2_{H^l(\gamma, \mathbb{R}^{d_{\text{out}}})} < \infty. \tag{53}$$

Thus, we rewrite the truncation error bound as

$$\|g - g_N\|^2_{H^m(\gamma, \mathbb{R}^{d_{\text{out}}})} \leq \kappa_2(m) \sum_{n=N+1}^{\infty} \sum_{|\alpha|=n} (1+n)^m \|c_\alpha\|_2^2 = \kappa_2(m) \sum_{n=N+1}^{\infty} (1+n)^{m-l} \sum_{|\alpha|=n} (1+n)^l \|c_\alpha\|_2^2.$$

Using the bound in (53) along with the fact that $(n+1)^{m-l} \leq N^{m-l}$ for all $n \geq N+1$, we have

$$\|g - g_N\|^2_{H^m(\gamma, \mathbb{R}^{d_{\text{out}}})} \leq \kappa_1^{-1}(l) \kappa_2(m) N^{-(l-m)} \|g\|^2_{H^l(\gamma, \mathbb{R}^{d_{\text{out}}})},$$

where we can take $\kappa(l, m) = \kappa_1^{-1}(l) \kappa_2(m)$ as the constant in the statement. ■

We can then construct the ReLU^k neural network approximation by emulating the truncated Wiener chaos expansion. To this end, it is known that ReLU^k can exactly represent entire sets of polynomials in an efficient manner. For example, Opschoor et al. (2021) show that given a downward closed multi-index set Λ, for any scalar-valued polynomial within this set, there exists a ReLU^k neural network that can represent this polynomial exactly, where the size of the neural network depends only on the size of the multi-index set, $|\Lambda|$, and not on the input dimension d_{in}.

Proposition 61 (Proposition 2.14 in Opschoor et al., 2021). *Fix $d_{\text{in}} \in \mathbb{N}$ and $k \in \mathbb{N}$, $k \geq 2$. Then, there exists a constant $\kappa_{\text{scalar}} = \kappa_{\text{scalar}}(k)$ such that for any finite downward closed multi-index set $\Lambda \subset \mathbb{N}_0^{d_{\text{in}}}$ and any scalar-valued polynomial $p(x) = \sum_{\alpha \in \Lambda} c_\alpha x^\alpha$, there exists a ReLU^k neural network $\varphi_{\psi,\theta} : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}$ such that*

$$\varphi_{\psi,\theta}(x) = p(x), \quad \forall x \in \mathbb{R}^{d_{\text{in}}},$$

with $\text{depth}(\varphi_{\psi,\theta}) \leq \kappa_{\text{scalar}} \log_2(|\Lambda|)$ and $\text{size}(\varphi_{\psi,\theta}) \leq \kappa_{\text{scalar}}|\Lambda|$.

For vector-valued polynomials, we can simply emulate each output component separately by a ReLU^k neural network and parallelize the resulting networks.

Corollary 62 (Vector-valued polynomial emulation). *Let $N \in \mathbb{N}$, $d_{\text{in}}, d_{\text{out}} \in \mathbb{N}$, and $k \in \mathbb{N}, k \geq 2$ be given. Then, there exist constants $\kappa_{\text{depth}} = \kappa_{\text{depth}}(d_{\text{in}}, k)$ and $\kappa_{\text{size}} = \kappa_{\text{size}}(d_{\text{in}}, d_{\text{out}}, k)$ such that for any polynomial function $p : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ of total degree at most $N \in \mathbb{N}$, there exists a ReLU^k neural network $\varphi_{\psi,\theta}$ such that*

$$\varphi_{\psi,\theta}(x) = p(x), \quad \forall x \in \mathbb{R}^{d_{\text{in}}},$$

with $\text{depth}(\varphi_{\psi,\theta}) \leq \kappa_{\text{depth}} (\log_2(N) + 1)$ and $\text{size}(\varphi_{\psi,\theta}) \leq \kappa_{\text{size}} N^{d_{\text{in}}}$.

Proof We consider the downward-closed multi-index set $\Lambda_N = \{\alpha \in \mathbb{N}_0^{d_{\text{in}}} : |\alpha| \leq N\}$. Then, $p(x) = (p_1(x), \ldots, p_{d_{\text{out}}}(x))$, where each p_i is a scalar-valued polynomial of the form $p_i(x) = \sum_{\alpha \in \Lambda_N} c_\alpha^{(i)} x^\alpha$ for some coefficients $c_\alpha^{(i)} \in \mathbb{R}$. By Proposition 61, for each $i = 1, \ldots, d_{\text{out}}$, there exists a ReLU^k neural network $\tilde{\varphi}_i$ such that $\tilde{\varphi}_i(x) = p_i(x)$ for all $x \in \mathbb{R}^{d_{\text{in}}}$, with $\text{size}(\tilde{\varphi}_i) \leq \kappa_{\text{scalar}}(k)|\Lambda_N|$ and $\text{depth}(\tilde{\varphi}_i) \leq \kappa_{\text{scalar}}(k) \log_2(|\Lambda_N|)$.

Since ReLU^k can be used to exactly represent the identity function $\text{Id}(x) = x$ as a single-layer neural network of fixed size $\kappa_{\text{Id}} = \kappa_{\text{Id}}(k)$ (Opschoor et al., 2021, Proposition 2.3), we can append additional identity layers to $\tilde{\varphi}_i$, i.e., $\varphi_i = \text{Id} \circ \cdots \circ \text{Id} \circ \tilde{\varphi}_i$, such that each φ_i is a ReLU^k neural network with the same depth, $\max_{i=1,\ldots,d_{\text{out}}} \text{depth}(\tilde{\varphi}_i)$. We can then define the vector-valued network $\varphi_{\psi,\theta} = (\varphi_1, \ldots, \varphi_{d_{\text{out}}})$ such that $\varphi_{\psi,\theta}(x) = p(x)$ for all $x \in \mathbb{R}^{d_{\text{in}}}$. Moreover, $\varphi_{\psi,\theta}$ is a feedforward neural network with

$$\begin{aligned}
&\text{depth}(\varphi_{\psi,\theta}) \leq \kappa_{\text{scalar}}(k) \log_2(|\Lambda_N|), \\
&\quad \text{size}(\varphi_{\psi,\theta}) \leq d_{\text{out}}(1 + \kappa_{\text{Id}}(k)) \left(\kappa_{\text{scalar}}(k)|\Lambda_N| + \kappa_{\text{Id}}(k)\kappa_{\text{scalar}}(k) \log_2(|\Lambda_N|)\right) \leq \kappa_1(d_{\text{out}}, k)|\Lambda_N|.
\end{aligned}$$

Here, we have accounted for the additional identity layers in the size bound, of which there are at most $\kappa_{\text{scalar}}(k) \log_2(|\Lambda_N|)$, using the fact that the direct composition of n neural networks $f_1, \ldots, f_n$ has size at most $\sum_{i=1}^n \text{size}(f_i) + \sum_{i=1}^{n-1} \text{size}(f_{i+1}) \cdot \text{size}(f_i)$ (see, for example, Opschoor et al., 2021, Definition 2.2). Finally, since $|\Lambda_N| = \binom{N+d_{\text{in}}}{d_{\text{in}}} \leq \kappa_\Lambda(d_{\text{in}}) N^{d_{\text{in}}}$, with a constant $\kappa_\Lambda(d_{\text{in}}) > 0$ that depends on d_{in}, we have that

$$\text{depth}(\varphi_{\psi,\theta}) \leq \kappa_{\text{depth}}(d_{\text{in}}, k) \left(\log_2(N) + 1\right), \qquad \text{size}(\varphi_{\psi,\theta}) \leq \kappa_{\text{size}}(d_{\text{in}}, d_{\text{out}}, k) N^{d_{\text{in}}}.$$

■

We now combine the above two results to complete the proof of Theorem 3.

Proof (of Theorem 3) It suffices to consider $g \neq 0$, since the result for $g = 0$ is trivial. For $g \neq 0$, we first consider the truncated Wiener chaos expansion g_N of g to polynomial degrees up to N. By Proposition 60, we can choose

$$N = \left\lceil \kappa^{\frac{1}{l-m}} \left(\epsilon^{-1} \|g\|_{H^l_\gamma}\right)^{\frac{2}{l-m}} \right\rceil$$

where $\lceil\cdot\rceil$ denotes the ceiling function and $\kappa = \kappa(l,m)$ is the constant from Proposition 60. This choice of N ensures that the truncation error satisfies $\|g - g_N\|^2_{H^m_\gamma(\mathbb{R}^{d_{\text{in}}},\mathbb{R}^{d_{\text{out}}})} \le \epsilon^2$.

Furthermore, by Corollary 62, there exists a ReLU^k neural network $\varphi_{\psi,\theta}$ such that $\varphi_{\psi,\theta}(x) = g_N(x)$ for all $x \in \mathbb{R}^{d_{\text{in}}}$. To this end, we note that

$$N \le \kappa^{\frac{1}{l-m}} \left(\epsilon^{-1}\|g\|_{H^l_\gamma}\right)^{\frac{2}{l-m}} + 1 \le \tilde{\kappa}(l,m)\left(\epsilon^{-1}\|g\|_{H^l_\gamma} + 1\right)^{\frac{2}{l-m}},$$

with $\tilde{\kappa}(l,m) = \kappa(l,m)^{\frac{1}{l-m}} + 1$, where the second inequality follows from the fact that $at^p + 1 \le (a+1)(t+1)^p$ for any $a, t, p \ge 0$. We can estimate the size and depth of this neural network using the bounds from Corollary 62. For the depth, we have

$$\begin{aligned}
\text{depth}(\varphi_{\psi,\theta}) &\le \kappa_{\text{depth}}(d_{\text{in}},k)\left(\log_2(N)+1\right)\\
&\le \kappa_{\text{depth}}(d_{\text{in}},k)\left(\log_2\left(\tilde{\kappa}(l,m)(\epsilon^{-1}\|g\|_{H^l_\gamma}+1)^{\frac{2}{l-m}}\right)+1\right)\\
&\le \kappa_{\text{depth}}(d_{\text{in}},k)\left(\log_2(\tilde{\kappa}(l,m)) + \frac{2}{l-m}\log_2\left(\epsilon^{-1}\|g\|_{H^l_\gamma}+1\right)+1\right)\\
&\le \kappa_1(d_{\text{in}},k,l,m)\left(\log_2\left(\epsilon^{-1}\|g\|_{H^l_\gamma}+1\right)+1\right).
\end{aligned}$$

For the size, we make use of the fact that $(t+1)^p \le 2^p(t^p+1)$ for any $t, p \ge 0$ to obtain

$$\begin{aligned}
\text{size}(\varphi_{\psi,\theta}) &\le \kappa_{\text{size}}(d_{\text{in}},d_{\text{out}},k)\tilde{\kappa}(l,m)^{d_{\text{in}}}\left(\epsilon^{-1}\|g\|_{H^l_\gamma}+1\right)^{\frac{2d_{\text{in}}}{l-m}}\\
&\le \kappa_2(d_{\text{in}},d_{\text{out}},k,l,m)\left(\left(\epsilon^{-1}\|g\|_{H^l_\gamma}\right)^{\frac{2d_{\text{in}}}{l-m}}+1\right).
\end{aligned}$$

Thus, the desired result holds with constants $\kappa_1(d_{\text{in}},k,l,m)$ and $\kappa_2(d_{\text{in}},d_{\text{out}},k,l,m)$. ■

Appendix C. Proofs for the Error Analysis

C.1 Proof of Lemma 18

Proof (of Lemma 18) We leverage the universal approximation result of Theorem 2 with a density argument. Let $\epsilon > 0$ be arbitrary and consider any $\eta > 0$. Since $\mathcal{FC}^\infty$ is dense in H^m_γ, we can find $F_\eta \in \mathcal{FC}^\infty$ such that $\|F - F_\eta\|_{H^m_\gamma} \le \eta$. This also means that $\|P_r\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] - P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)]\|_{H^m_\gamma} \le \eta$. We then focus on approximating $P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)]$.

By Lemma 52, since $F_\eta \in \mathcal{FC}^\infty$, we have $\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)] \in C^\infty_b$. We therefore look for an approximation of the function $g_\eta(s) := U_r^\dagger\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](V_r s)$ by a neural network, where $g_\eta \in C^\infty_b(\mathbb{R}^r,\mathbb{R}^r)$. In particular, Theorem 2 implies that there exists a neural network $\varphi = \varphi_{\psi,\theta}$ approximating g_η such that $\|g_\eta - \varphi\|_{H^m(\gamma_r,\mathbb{R}^r)} \le \eta$, where $\gamma_r = \mathcal{N}(0,I_r)$ refers to the standard Gaussian on $\mathbb{R}^r$. Moreover, the L^2_γ error between $P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)]$ and $U_r\circ\varphi\circ V_r^\dagger$ can be expressed as

$$\int \|P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](x) - U_r\varphi(V_r^\dagger x)\|^2_{\mathcal{Y}}\gamma(dx) = \int \|U_r^\dagger\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](V_rV_r^\dagger x) - \varphi(V_r^\dagger x)\|_2^2\gamma(dx),$$

since $\|P_r u\|^2_{\mathcal{Y}} = \|U_r^\dagger u\|_2^2$. We can now perform a change of variable to $s = V_r^\dagger x$. Since V_r is orthonormal in $\mathcal{E}$, s is distributed according to γ_r, such that

$$\int \|P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](x) - U_r\varphi(V_r^\dagger x)\|^2_{\mathcal{Y}}\gamma(dx) = \int \|U_r^\dagger\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](V_r s) - \varphi(s)\|_2^2\gamma_r(ds)$$

Furthermore, since we have defined $g_\eta = U_r^\dagger \mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](V_r s)$, we have

$$\int \|U_r^\dagger \mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](V_r s) - \varphi(s)\|_2^2 \gamma_r(ds) = \int \|g_\eta(s) - \varphi(s)\|_2^2 \gamma_r(ds).$$

Similarly, for kth derivative,

$$\begin{aligned}
&\int \|D^k P_r(\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](x)) - D^k(U_r\varphi(V_r^\dagger x))\|^2_{\mathrm{HS}_k(\mathcal{E},\mathcal{Y})}\gamma(dx) \\
&\quad = \int \|P_r\mathbb{E}_\gamma[D^k F_\eta|\sigma(V_r^\dagger)](Q_r x)(Q_r\cdot,\ldots,Q_r\cdot) - U_r D^k\varphi(V_r^\dagger x)(V_r^\dagger\cdot,\ldots,V_r^\dagger\cdot)\|^2_{\mathrm{HS}_k(\mathcal{E},\mathcal{Y})}\gamma(dx)
\end{aligned}$$

where we have interchanged differentiation and conditional expectation using Lemma 17 in Appendix A.4. We then follow the same approach, projecting the norm into the reduced subspace and then performing the change of variables $s = V_r^\dagger x$ to obtain a similar bound for the derivatives,

$$\begin{aligned}
&\int \|D^k P_r(\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)](x)) - D^k(U_r\varphi(V_r^\dagger x))\|^2_{\mathrm{HS}_k(\mathcal{E},\mathcal{Y})}\gamma(dx) \\
&\quad = \int \|U_r^\dagger\mathbb{E}_\gamma[D^k F_\eta|\sigma(V_r^\dagger)](V_r V_r^\dagger x)(V_r\cdot,\ldots,V_r\cdot) - D^k\varphi(V_r^\dagger x)\|^2_{F_k}\gamma(dx) \\
&\quad = \int \|U_r^\dagger\mathbb{E}_\gamma[D^k F_\eta|\sigma(V_r^\dagger)](V_r s)(V_r\cdot,\ldots,V_r\cdot) - D^k\varphi(s)\|^2_{F_k}\gamma_r(ds) \\
&\quad = \int \|D^k g_\eta(s) - D^k\varphi(s)\|^2_{F_k}\gamma_r(ds).
\end{aligned}$$

where we have used $\|\cdot\|_{F_k} = \|\cdot\|_{\mathrm{HS}_k(\mathbb{R}^r,\mathbb{R}^r)}$ to denote the tensor Frobenius norms on $\mathbb{R}^r$, noting that $\|\cdot\|_{F_1} = \|\cdot\|_F$ is the standard matrix Frobenius norm.

Finally, since the Fréchet derivatives coincide with the weak derivatives, we have

$$\|P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)] - U_r\circ\varphi\circ V_r^\dagger\|_{H^m_\gamma} = \|g_\eta - \varphi\|_{H^m(\gamma_r,\mathbb{R}^r)} \le \eta.$$

Combined with the fact that $\|P_r\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] - P_r\mathbb{E}_\gamma[F_\eta|\sigma(V_r^\dagger)]\|_{H^m_\gamma} \le \eta$, it suffices to choose $\eta = \epsilon/2$. ■

C.2 Proof of Lemma 19

Proof We begin with the unshifted case, $b = 0$. For $F \in H^l_\gamma$, we can directly define $g(s) := U_r^\dagger\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](V_r s)$, which is well-defined for γ_r-almost every s. Then, $g \in H^l(\gamma_r,\mathbb{R}^r)$ and $\|g\|_{H^l(\gamma_r,\mathbb{R}^r)} \le \|F\|_{H^l_\gamma}$. Indeed, the density and change-of-variable arguments used in the proof of Lemma 18 show that g is the $H^l(\gamma_r,\mathbb{R}^r)$ limit of the sequence of functions $g_\eta(s) \in C_b^\infty(\mathbb{R}^r,\mathbb{R}^r)$, and $\|g\|_{H^l(\gamma_r,\mathbb{R}^r)} = \|P_r\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{H^l_\gamma} \le \|F\|_{H^l_\gamma}$. By Theorem 3, we can find a ReLU^k network φ such that $\|g-\varphi\|_{H^m(\gamma_r,\mathbb{R}^r)} \le \epsilon$ with

$$\mathrm{depth}(\varphi) \le \kappa_1\left(\log_2\left(\epsilon^{-1}\|F\|_{H^l_\gamma}+1\right)+1\right), \qquad \mathrm{size}(\varphi) \le \kappa_2\left(\left(\epsilon^{-1}\|F\|_{H^l_\gamma}\right)^{\frac{2r}{l-m}}+1\right),$$

where we have directly expressed the bounds in terms of $\|F\|_{H^l_\gamma}$, and $\kappa_1 = \kappa_1(r,k,l,m)$ and $\kappa_2 = \kappa_2(r,k,l,m)$ are based on the constants from Theorem 3 (note that $d_{\mathrm{in}} = d_{\mathrm{out}} = r$). By the same change-of-variable argument as in the proof of Lemma 18, we have that $\|P_r\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] - U_r\circ\varphi\circ V_r^\dagger\|_{H^m_\gamma} = \|g-\varphi\|_{H^m(\gamma_r,\mathbb{R}^r)} \le \epsilon$.

For the shifted case, $b \neq 0$, we take $\varphi_b := \varphi - U_r^\dagger b$ as the approximating neural network, which corresponds to shifting the final layer bias of φ by $U_r^\dagger b$. We then have

$$\|P_r \mathbb{E}_\gamma[F - b|\sigma(V_r^\dagger)] - U_r \circ \varphi_b \circ V_r^\dagger\|_{H^m_\gamma} = \|P_r \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)] - U_r \circ \varphi \circ V_r^\dagger\|_{H^m_\gamma} \leq \epsilon,$$

where φ_b has the same depth as φ, while its size is at most $\text{size}(\varphi)+r$, accounting for the possibility of introducing up to r additional nonzero parameters in the bias term of the final layer. Thus, the depth and size estimates also hold for the shifted case by absorbing r into the size constant κ_2. ■

C.3 Proof of Theorem 12

The universal approximation of projected conditional expectations in H^m_γ can be used to obtain the generic universal approximation result of Theorem 12 as follows.

Proof (Theorem 12) Let ϵ be arbitrary. Given $F \in H^m_\gamma$ and the output basis $(u_i)_{i=1}^\infty$, we have

$$\|F\|^2_{L^2_\gamma} = \mathbb{E}_\gamma\left[\sum_{i=1}^\infty |\langle u_i, F\rangle_\mathcal{Y}|^2\right] = \sum_{i=1}^\infty \mathbb{E}_\gamma\left[|\langle u_i, F\rangle_\mathcal{Y}|^2\right] < \infty$$

where the interchange of sum and expectation is possible since all terms are non-negative. Moreover, for any $r > 0$, we have

$$\|F - P_r F\|^2_{L^2_\gamma} = \mathbb{E}_\gamma\left[\sum_{i=r+1}^\infty |\langle u_i, F\rangle_\mathcal{Y}|^2\right] = \sum_{i=r+1}^\infty \mathbb{E}_\gamma\left[|\langle u_i, F\rangle_\mathcal{Y}|^2\right].$$

Thus, $\|F - P_r F\|_{L^2_\gamma}$ converges monotonically to zero as $r \to \infty$. Analogously, for any $k \leq m$,

$$\|D^k_\mathcal{E} F\|^2_{L^2(\gamma, \text{HS}_k(\mathcal{E},\mathcal{Y}))} = \sum_{j=1}^\infty \sum_{i_1,\ldots,i_k=1}^\infty \mathbb{E}_\gamma\left[\left|\left\langle u_j, D^k_\mathcal{E} F(w_{i_1}, \ldots, w_{i_k})\right\rangle_\mathcal{Y}\right|^2\right] < \infty, \tag{54}$$

using any orthonormal basis $(w_i)_{i=1}^\infty$ of $\mathcal{E}$, such that

$$\|D^k_\mathcal{E} F - P_r D^k_\mathcal{E} F\|^2_{L^2(\gamma, \text{HS}_k(\mathcal{E},\mathcal{Y}))} = \sum_{j=r+1}^\infty \sum_{i_1,\ldots,i_k=1}^\infty \mathbb{E}_\gamma\left[\left|\left\langle u_j, D^k_\mathcal{E} F(w_{i_1}, \ldots, w_{i_k})\right\rangle_\mathcal{Y}\right|^2\right] < \infty, \tag{55}$$

also converges monotonically to zero. This allows us to find some $r_\mathcal{Y}$ such that $\|F - P_{r_\mathcal{Y}} F\|_{H^m_\gamma} \leq \epsilon/3$. For the input basis $(v_i)_{i=1}^\infty$, we then consider a rank $r_\mathcal{X} \geq r_\mathcal{Y}$, such that the conditional expectation $\mathbb{E}_\gamma[F|\sigma(V_{r_\mathcal{X}}{}^\dagger)]$ also satisfies $\|F - \mathbb{E}_\gamma[F|\sigma(V_{r_\mathcal{X}}{}^\dagger)]\|_{H^m_\gamma} \leq \epsilon/3$. This is possible since the conditional expectations converge to F as $r \to \infty$ by Bogachev (1998, Theorem 5.4.5). We can now increase the output rank to $r_\mathcal{X}$ in order to make the input and output ranks equal, i.e., $r = r_\mathcal{X} \geq r_\mathcal{Y}$. Note that

$$\|F - P_r F\|_{H^m_\gamma} \leq \|F - P_{r_\mathcal{Y}} F\|_{H^m_\gamma} \leq \epsilon/3$$

due to (54) and (55) and

$$\|P_r F - \mathbb{E}_\gamma[P_r F|\sigma(V_r^\dagger)]\|_{H^m_\gamma} \leq \|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{H^m_\gamma} \leq \epsilon/3,$$

since $\|P_r\|_{\mathcal{L}(\mathcal{Y},\mathcal{Y})} = 1$.

Lemma 18 then implies the existence of a neural network $\varphi_{\psi,\theta}$ such that the neural operator $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger$ approximates the projected conditional expectation to error

$$\|\mathbb{E}_\gamma[P_r F|\sigma(V_r^\dagger)] - \widetilde{F}_\theta\|_{H^m_\gamma} \leq \epsilon/3.$$

Thus, combining the error terms in a triangle inequality yields the desired error bound,

$$\|F - \widetilde{F}_\theta\|_{H^m_\gamma} \leq \|F - P_r F\|_{H^m_\gamma} + \|P_r F - \mathbb{E}_\gamma[P_r F|\sigma(V_r^\dagger)]\|_{H^m_\gamma} + \|\mathbb{E}_\gamma[P_r F|\sigma(V_r^\dagger)] - \widetilde{F}_\theta\|_{H^m_\gamma} \leq \epsilon.$$

■

C.4 Proofs for the Error Decompositions

C.4.1 Proof of Proposition 20

Proof (of Proposition 20) The results follow from decomposing each term into its component within the projection subspace and its orthogonal complement. We begin with the L^2_γ error. For $\widetilde{F} = U_r \circ \varphi \circ V_r^\dagger + \widehat{F}$, we can write the misfit $\|F(x) - \widetilde{F}(x)\|^2_{\mathcal{Y}}$ as

$$\|F(x) - \widetilde{F}(x)\|^2_{\mathcal{Y}} = \|P_r(F(x) - \widetilde{F}(x))\|^2_{\mathcal{Y}} + \|(I - P_r)(F(x) - \widehat{F})\|^2_{\mathcal{Y}},$$

since $(I - P_r)U_r = 0$. Moreover, using the decomposition $F - \widehat{F} = F - \bar{F} + \bar{F} - \widehat{F}$, we have

$$\|(I - P_r)(F - \widehat{F})\|^2_{\mathcal{Y}} = \|(I - P_r)(F - \bar{F})\|^2_{\mathcal{Y}} + 2\left\langle (I - P_r)(F - \bar{F}), (\bar{F} - \widehat{F})\right\rangle_{\mathcal{Y}} + \|(I - P_r)(\bar{F} - \widehat{F})\|^2_{\mathcal{Y}}.$$

We recognize that $\|I - P_r\|_{\mathcal{L}(\mathcal{Y},\mathcal{Y})} = 1$ implies $\|(I - P_r)(\bar{F} - \widehat{F})\|^2_{\mathcal{Y}} \leq \|\bar{F} - \widehat{F}\|^2_{\mathcal{Y}}$. Then, taking the expectation over γ removes the cross term and yields the bound (18).

For the derivative error, we have $D\widetilde{F}(x) = U_r D\varphi(V_r^\dagger x)V_r^\dagger$. Thus, using $(v_i)_{i=1}^\infty$ as the basis for evaluating the Hilbert–Schmidt norm, we have

$$\begin{aligned}
\|D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} &= \sum_{i=1}^\infty \|(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))v_i\|^2_{\mathcal{Y}} \\
&= \sum_{i=1}^r \|(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))v_i\|^2_{\mathcal{Y}} + \sum_{i=r+1}^\infty \|(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))v_i\|^2_{\mathcal{Y}} \\
&= \|(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} + \|D_{\mathcal{E}}F(x)(I - Q_r)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})},
\end{aligned}$$

since $D_{\mathcal{E}}\widetilde{F}(x)(I - Q_r) = 0$. By analogous reasoning, we can further decompose the first term into

$$\begin{aligned}
\|(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} &= \|P_r(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F}(x))Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \\
&\quad + \|(I - P_r)D_{\mathcal{E}}F(x)Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}.
\end{aligned}$$

Finally, since $\|Q_r\|_{\mathcal{L}(\mathcal{E},\mathcal{E})} = 1$, we have $\|(I - P_r)D_{\mathcal{E}}F(x)Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \leq \|(I - P_r)D_{\mathcal{E}}F(x)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}$. Combining the individual components and taking the expectation yields the bound (19).

To obtain (20) and (21) for the terms involving $\widetilde{F}$, we make use of Parseval's identity with the basis $(u_i)_{i=1}^\infty$. For the function value, we have

$$\|P_r(F - \widetilde{F})\|^2_{\mathcal{Y}} = \|U_r U_r^\dagger(F - U_r\varphi(V_r^\dagger x) - \widehat{F})\|^2_{\mathcal{Y}} = \|U_r^\dagger(F - \widehat{F}) - \varphi(V_r^\dagger x)\|^2_2.$$

For the derivative, we additionally use $(v_i)_{i=1}^\infty$ as the basis for the $\mathrm{HS}(\mathcal{E},\mathcal{Y})$ norm, giving

$$\|P_r(D_{\mathcal{E}}F(x) - D_{\mathcal{E}}\widetilde{F})Q_r\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} = \|U_r^\dagger D_{\mathcal{E}}F(x)V_r - D\varphi(V_r^\dagger x)\|^2_F.$$

■

C.4.2 Proof of Corollary 21

Corollary 21 then follows as a result of the error decomposition in Proposition 20 along with the universal approximation of conditional expectations in Lemmas 18 and 19.

Proof (of Corollary 21) By Proposition 20, we can decompose the approximation error into

$$\|F - \widetilde{F}_\theta\|_{L^2_\gamma}^2 \leq \|P_r(F - \widetilde{F}_\theta)\|_{L^2_\gamma}^2 + \|(I - P_r)(F - \bar{F})\|_{L^2_\gamma}^2 + \|\bar{F} - \widehat{F}\|_{\mathcal{Y}}^2,$$

and

$$\begin{aligned}|F - \widetilde{F}_\theta|_{H^1_\gamma}^2 \leq &\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_\theta)Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 \\ &+ \|(I - P_r)D_{\mathcal{E}}F\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 + \|D_{\mathcal{E}}F(I - Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2.\end{aligned}$$

We first make use of the conditional expectation $\widetilde{F}_r = \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]$ as an intermediate to the approximation F, such that $P_r\widetilde{F}_r = \mathbb{E}_\gamma[P_rF|\sigma(V_r^\dagger)]$ is the optimal L^2_γ approximation to P_rF within the closed subspace of $\sigma(V_r^\dagger)$-measurable functions. Since $\widetilde{F}_\theta$ is also $\sigma(V_r^\dagger)$-measurable, the variational characterization of the conditional expectation gives orthogonality in the sense that $\langle P_r(F - \widetilde{F}_r), P_r(\widetilde{F}_\theta - \widetilde{F}_r)\rangle_{L^2_\gamma} = 0$. Thus, the error $\|P_r(F - \widetilde{F}_\theta)\|_{L^2_\gamma}^2$ can be decomposed as

$$\|P_r(F - \widetilde{F}_\theta)\|_{L^2_\gamma}^2 \leq \|P_r(F - \widetilde{F}_r)\|_{L^2_\gamma}^2 + \|P_r(\widetilde{F}_r - \widetilde{F}_\theta)\|_{L^2_\gamma}^2.$$

For the derivative error, we note that $P_rD_{\mathcal{E}}\widetilde{F}_rQ_r = \mathbb{E}_\gamma[P_rD_{\mathcal{E}}FQ_r|\sigma(V_r^\dagger)]$ by Lemma 17. Therefore, by analogous reasoning, we have

$$\begin{aligned}\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_\theta)Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 \leq &\|P_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_r)Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 \\ &+ \|P_r(D_{\mathcal{E}}\widetilde{F}_r - D_{\mathcal{E}}\widetilde{F}_\theta)Q_r\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2.\end{aligned}$$

By Lemma 18, there exists a neural network $\varphi_{\psi,\theta}$ such that $U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger$ approximates $P_r(\widetilde{F}_r - \widehat{F}) = \mathbb{E}_\gamma[P_r(F - \widehat{F})|\sigma(V_r^\dagger)]$ in $\|\cdot\|_{H^1_\gamma}^2$ to error ϵ. This means that $\widetilde{F}_\theta = U_r \circ \varphi_{\psi,\theta} \circ V_r^\dagger + \widehat{F}$ satisfies the desired error bounds. If $\psi = \mathrm{ReLU}^k$ with $k \geq 2$ is used, we can instead make use of Lemma 19 with the error tolerance $\epsilon^{1/2}$ to obtain $\varphi_{\psi,\theta}$, such that the desired error bounds hold with

$$\mathrm{depth}(\varphi_{\psi,\theta}) \leq \kappa_1\left(\log_2\left(\epsilon^{-\frac{1}{2}}\|F\|_{H^m_\gamma} + 1\right) + 1\right), \qquad \mathrm{size}(\varphi_{\psi,\theta}) \leq \kappa_2\left(\left(\epsilon^{-1}\|F\|_{H^m_\gamma}^2\right)^{\frac{r}{m-1}} + 1\right).$$

Here, $\kappa_1 = \kappa_1(r,k,m)$ and $\kappa_2 = \kappa_2(r,k,m)$ are based on the constants from Lemma 19 with l and m in Lemma 19 corresponding to m and 1 in this result. ■

C.5 Proof of the Subspace Poincaré Inequality, Theorem 28

We begin by stating the Poincaré inequality, which holds for Gaussian measures on separable Hilbert spaces (Bogachev, 1998).

Theorem 63 (Poincaré inequality for Gaussian measures)**.** *Suppose $F \in H^1_\gamma$. Then,*

$$\mathbb{E}_\gamma[\|F - \bar{F}\|_{\mathcal{Y}}^2] \leq \mathbb{E}_\gamma[\|D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2],$$

where $\bar{F} = \mathbb{E}_\gamma[F]$ is the mean of F.

The subspace Poincaré inequality then follows from the Poincaré inequality itself.

Proof (Of Theorem 28) The procedure is almost identical to that of Zahm et al. (2020), with justifications to ensure the operations are valid in the separable Hilbert space setting. We first consider the case where $F \in \mathcal{FC}^\infty$, for which we can define $G(x,y) := F(Q_r x + (I - Q_r)y)$, such that $\mathbb{E}_{y\sim\gamma}[G(x,y)] = \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x)$. Since Q_r and $I - Q_r$ are continuous linear operators, we have

$$\partial_y G(x,y) = DF(Q_r x + (I - Q_r)y)(I - Q_r).$$

Moreover, since $Q_r x + (I - Q_r)y \sim \gamma$ whenever $(x,y) \sim \gamma \otimes \gamma$, we know

$$\iint \|G(x,y)\|_{\mathcal{Y}}^2 \gamma(dy)\gamma(dx) = \iint \|F(Q_r x + (I - Q_r)y)\|_{\mathcal{Y}}^2 \gamma(dy)\gamma(dx) = \|F\|_{L^2_\gamma}^2 < \infty.$$

An analogous calculation can be made for $\partial_y G$. Therefore, we have $\int |G(x,y)|^2 \gamma(dy) < \infty$ and $\int \|\partial_y G(x,y)\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2 \gamma(dy) < \infty$ for almost every x. This suggests that $G(x,\cdot) \in H^1_\gamma$ for almost every x, with a Sobolev derivative equal to $\partial_y G(x,\cdot)$. Applying the Poincaré inequality on separable Hilbert spaces gives

$$\int \|G(x,y) - \mathbb{E}_{y\sim\gamma}[G(x,y)]\|_{\mathcal{Y}}^2 \gamma(dy) \le \int \|\partial_y G(x,y)\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2 \gamma(dy)$$

for almost every x. Integrating again over $x \sim \gamma$ gives

$$\begin{aligned}\iint \|F(Q_r x + (I - Q_r)y) - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x)\|_{\mathcal{Y}}^2 \gamma(dy)\gamma(dx) \\ \le \iint \|DF(Q_r x + (I - Q_r)y)(I - Q_r)\|_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}^2 \gamma(dy)\gamma(dx).\end{aligned}$$

Since $\mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](x) = \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)](Q_r x + (I - Q_r)y)$ and $Q_r x + (I - Q_r)y \sim \gamma$, we obtain the inequality (30) for $F \in \mathcal{FC}^\infty$.

We use a density argument to extend the result to arbitrary $F \in H^1_\gamma$. We first note that (30) is equivalently written as

$$\|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{L^2_\gamma} \le \|D_{\mathcal{E}} F(I - Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}.$$

From here, consider an approximating sequence $F_n \in \mathcal{FC}^\infty$ converging to F in H^1_γ. An application of the triangle inequality gives

$$\|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{L^2_\gamma} \le \|F_n - \mathbb{E}_\gamma[F_n|\sigma(V_r^\dagger)]\|_{L^2_\gamma} + \|F - F_n\|_{L^2_\gamma} + \|\mathbb{E}_\gamma[F - F_n|\sigma(V_r^\dagger)]\|_{L^2_\gamma}.$$

Applying the subspace Poincaré inequality for F_n yields

$$\|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{L^2_\gamma} \le \|D_{\mathcal{E}} F_n(I - Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))} + \|F - F_n\|_{L^2_\gamma} + \|\mathbb{E}_\gamma[F - F_n|\sigma(V_r^\dagger)]\|_{L^2_\gamma}.$$

Another application of the triangle inequality yields

$$\begin{aligned}\|F - \mathbb{E}_\gamma[F|\sigma(V_r^\dagger)]\|_{L^2_\gamma} \le &\|D_{\mathcal{E}} F(I - Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))} + \|(D_{\mathcal{E}} F_n - D_{\mathcal{E}} F)(I - Q_r)\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))} \\ &+ \|F - F_n\|_{L^2_\gamma} + \|\mathbb{E}_\gamma[F - F_n|\sigma(V_r^\dagger)]\|_{L^2_\gamma}.\end{aligned}$$

Since $F_n \to F$ in H^1_γ, the latter three terms involving the difference of F_n and F can be made arbitrarily small (see also Corollary 53). This leads to the subspace Poincaré inequality for general $F \in H^1_\gamma$. ■

C.6 Proofs for the Sampling Error Bounds

C.6.1 Proofs for the Monte Carlo Errors, Lemmas 33 and 35

Here, we present the proofs in the analyses of the sampling errors in Section 3.7. We begin with the proofs for the Monte Carlo error bounds, Lemma 33 and Lemma 35. The proof of Lemma 33 is reproduced below from Bhattacharya et al. (2021).

Proof (of Lemma 33) Given the assumption that the second moment is bounded, we have

$$\begin{aligned}
\mathbb{E}_N\left[\|\hat{z}-\bar{z}\|_{\mathcal{H}}^2\right] &= \mathbb{E}_N\left[\left\langle \frac{1}{N}\sum_{k=1}^N z_k - \bar{z}, \frac{1}{N}\sum_{k=1}^N z_k - \bar{z}\right\rangle_{\mathcal{H}}\right] \\
&= \frac{1}{N}\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^2] + \frac{N^2-N}{N^2}\langle \bar{z},\bar{z}\rangle_{\mathcal{H}} - 2\langle \bar{z},\bar{z}\rangle_{\mathcal{H}} + \langle \bar{z},\bar{z}\rangle_{\mathcal{H}} \\
&= \frac{1}{N}\left(\mathbb{E}_{z\sim\nu}\left[\|z\|_{\mathcal{H}}^2 - \|\bar{z}\|_{\mathcal{H}}^2\right]\right) \\
&= \frac{1}{N}\left(\mathbb{E}_{z\sim\nu}\left[\|z-\bar{z}\|_{\mathcal{H}}^2\right]\right),
\end{aligned}$$

where we have made use of the independence of z_i, z_j for $i \neq j$. ■

We then proceed with the proof for the error in the fourth moments.

Proof (of Lemma 35) Assume first that z is a centered random variable, i.e., $\bar{z} = 0$. Then,

$$\mathbb{E}_N[\|\hat{z}-\bar{z}\|_{\mathcal{H}}^4] = \frac{1}{N^4}\mathbb{E}_N\left[\left\langle\sum_{i=1}^N z_i, \sum_{j=1}^N z_j\right\rangle_{\mathcal{H}}\left\langle\sum_{k=1}^N z_k, \sum_{l=1}^N z_l\right\rangle_{\mathcal{H}}\right] = \frac{1}{N^4}\sum_{i,j,k,l=1}^N \mathbb{E}_N\left[\langle z_i, z_j\rangle_{\mathcal{H}}\langle z_k, z_l\rangle_{\mathcal{H}}\right].$$

For $(z_i)_{i=1}^N$ that are i.i.d., the expectation is zero whenever a single index differs from the remaining three, e.g., $i \neq j,k,l$, since

$$\mathbb{E}_N\left[\langle z_i, z_j\rangle_{\mathcal{H}}\langle z_k, z_l\rangle_{\mathcal{H}}\right] = \langle \mathbb{E}_N[z_i], \mathbb{E}_N[z_j\langle z_k, z_l\rangle_{\mathcal{H}}]\rangle_{\mathcal{H}} = \langle \mathbb{E}_{z\sim\nu}[z], \mathbb{E}_N[z_j\langle z_k, z_l\rangle_{\mathcal{H}}]\rangle_{\mathcal{H}} = 0.$$

Thus, the only non-zero terms are those where all indices are equal, i.e., $i=j=k=l$, and those where two pairs of indices are equal, e.g., $i=j\neq k=l$. The former contributes N terms of the form $\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^4]$. The latter case corresponds to $\binom{N}{2}\cdot\binom{4}{2} = 3N(N-1)$ terms, where for every term, the Cauchy–Schwarz inequality implies that

$$\mathbb{E}_N\left[\langle z_i, z_j\rangle_{\mathcal{H}}\langle z_k, z_l\rangle_{\mathcal{H}}\right] \leq \mathbb{E}_N\left[\|z_i\|_{\mathcal{H}}\|z_j\|_{\mathcal{H}}\|z_k\|_{\mathcal{H}}\|z_l\|_{\mathcal{H}}\right] = \mathbb{E}_N\left[\|z_i\|_{\mathcal{H}}^2\right]\mathbb{E}_N\left[\|z_k\|_{\mathcal{H}}^2\right] = \mathbb{E}_{z\sim\nu}\left[\|z\|_{\mathcal{H}}^2\right]^2,$$

where we have again made use of independence. Combining the two contributions, we arrive at

$$\mathbb{E}_N\left[\|\hat{z}-\bar{z}\|_{\mathcal{H}}^4\right] \leq \frac{N\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^4] + 3N(N-1)\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^2]^2}{N^4} \leq \frac{K_z}{N^2},$$

where it suffices to take

$$K_z = \mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^4] + 3\mathbb{E}_{z\sim\nu}[\|z\|_{\mathcal{H}}^2]^2.$$

When z is uncentered, we note that

$$\hat{z}-\bar{z} = \frac{1}{N}\sum_{k=1}^N(z_k-\bar{z}) = \frac{1}{N}\sum_{k=1}^N \tilde{z}_k$$

where $\tilde{z}_k = z_k - \bar{z}$ are centered i.i.d. random variables. Thus, applying the result for the centered case above to the random variable $\tilde{z}$ yields the desired result. ■

C.6.2 Proofs for PCA and DIS Sampling Errors, Propositions 36 and 37

The proofs for Propositions 36 and 37 then follow from the previous results, Lemmas 33 and 35.

Proof (of Proposition 36) Bhattacharya et al. (2021, Lemma B.2) considers the uncentered case. To derive the result for the centered case, we decompose the covariance estimator $\widehat{C}_{\mathcal{Y}}$ as

$$\begin{aligned}\widehat{C}_{\mathcal{Y}} &= \frac{1}{N}\sum_{k=1}^{N}(F(x_k)-\widehat{F})\otimes(F(x_k)-\widehat{F}) \\ &= \frac{1}{N}\sum_{k=1}^{N}(F(x_k)-\bar{F}+\bar{F}-\widehat{F})\otimes(F(x_k)-\bar{F}+\bar{F}-\widehat{F}) \\ &= \frac{1}{N}\sum_{k=1}^{N}(F(x_k)-\bar{F})\otimes(F(x_k)-\bar{F})-(\bar{F}-\widehat{F})\otimes(\bar{F}-\widehat{F}).\end{aligned}$$

Let $\bar{C}_{\mathcal{Y}} := \frac{1}{N}\sum_{k=1}^{N}(F(x_k)-\bar{F})\otimes(F(x_k)-\bar{F})$. Then,

$$\|C_{\mathcal{Y}}-\widehat{C}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})} \leq 2\|C_{\mathcal{Y}}-\bar{C}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})} + 2\|(\widehat{F}-\bar{F})\otimes(\widehat{F}-\bar{F})\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}. \tag{56}$$

In particular, the first term reverts to the case of uncentered covariances $\|C_{\mathcal{Y}}-\bar{C}_{\mathcal{Y}}\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}$, which we present here for completeness. Since $\|y\otimes y\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})} = \|y\|^2_{\mathcal{Y}}$ for $y\in\mathcal{Y}$, we have

$$\mathbb{E}_\gamma[\|(F-\bar{F})\otimes(F-\bar{F})\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] = \mathbb{E}_\gamma[\|F-\bar{F}\|^4_{\mathcal{Y}}] < \infty.$$

This allows us to apply Lemma 33 using the random variable $z=(F-\bar{F})\otimes(F-\bar{F})$, taking values in the Hilbert space $\mathrm{HS}(\mathcal{Y},\mathcal{Y})$, whose mean is $C_{\mathcal{Y}} = \mathbb{E}_\gamma[(F-\bar{F})\otimes(F-\bar{F})]$. This gives the bound

$$\mathbb{E}_N[\|C_{\mathcal{Y}}-\bar{C}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] \leq \frac{\mathbb{E}_\gamma[\|(F-\bar{F})\otimes(F-\bar{F})-C_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}]}{N}.$$

For the second term in (56), we have

$$\|(\widehat{F}-\bar{F})\otimes(\widehat{F}-\bar{F})\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})} = \|\widehat{F}-\bar{F}\|^2_{\mathcal{Y}}$$

such that

$$\mathbb{E}_N[\|(\widehat{F}-\bar{F})\otimes(\widehat{F}-\bar{F})\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] = \mathbb{E}_N[\|\widehat{F}-\bar{F}\|^4_{\mathcal{Y}}] \leq \frac{\|F-\bar{F}\|^4_{L^4_\gamma}+3\|F-\bar{F}\|^4_{L^2_\gamma}}{N^2}.$$

Combining the bounds for the two terms from (56) yields

$$\mathbb{E}_N[\|C_{\mathcal{Y}}-\widehat{C}_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] \leq \frac{2\mathbb{E}_\gamma[\|(F-\bar{F})\otimes(F-\bar{F})-C_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}]}{N} + \frac{2\|F-\bar{F}\|^4_{L^4_\gamma}+6\|F-\bar{F}\|^4_{L^2_\gamma}}{N^2}.$$

Since N^{-2} decays faster than N^{-1}, we can simply take the constant as

$$M_{C_{\mathcal{Y}}} = 2\mathbb{E}_\gamma[\|(F-\bar{F})\otimes(F-\bar{F})-C_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] + 2\|F-\bar{F}\|^4_{L^4_\gamma}+6\|F-\bar{F}\|^4_{L^2_\gamma}.$$

■

The strategy for the sampling error estimates in the DIS case is similar (and in fact, simpler, since we do not need to account for the centering by the empirical mean).

Proof (of Proposition 37) Recall that for $H_{\mathcal{X}} = \mathbb{E}_\gamma[D_{\mathcal{E}}F^* D_{\mathcal{E}}F]$ and $H_{\mathcal{Y}} = \mathbb{E}_\gamma[D_{\mathcal{E}}F D_{\mathcal{E}}F^*]$, their empirical estimators

$$\widehat{H}_{\mathcal{X}} = \frac{1}{N}\sum_{k=1}^{N} D_{\mathcal{E}}F(x_k)^* D_{\mathcal{E}}F(x_k), \qquad \widehat{H}_{\mathcal{Y}} = \frac{1}{N}\sum_{k=1}^{N} D_{\mathcal{E}}F(x_k) D_{\mathcal{E}}F(x_k)^*,$$

are sample averages of the random variables $D_{\mathcal{E}}F^* D_{\mathcal{E}}F$ and $D_{\mathcal{E}}F D_{\mathcal{E}}F^*$ that take values in the Hilbert spaces $\mathrm{HS}(\mathcal{E},\mathcal{E})$ and $\mathrm{HS}(\mathcal{Y},\mathcal{Y})$, respectively. Note that for Hilbert–Schmidt operators $A \in \mathrm{HS}(\mathcal{H}_1,\mathcal{H}_2)$ and $B \in \mathrm{HS}(\mathcal{H}_2,\mathcal{H}_3)$, which map between separable Hilbert spaces $\mathcal{H}_1,\mathcal{H}_2,\mathcal{H}_3$, the norm of their composition satisfies $\|BA\|_{\mathrm{HS}(\mathcal{H}_1,\mathcal{H}_3)} \le \|A\|_{\mathrm{HS}(\mathcal{H}_1,\mathcal{H}_2)}\|B\|_{\mathrm{HS}(\mathcal{H}_2,\mathcal{H}_3)}$. Thus, we have

$$\|D_{\mathcal{E}}F^* D_{\mathcal{E}}F\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})} \le \|D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}, \qquad \|D_{\mathcal{E}}F D_{\mathcal{E}}F^*\|_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})} \le \|D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}.$$

Squaring and taking expectations gives us

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F^* D_{\mathcal{E}}F\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}] \le \mathbb{E}_\gamma[\|D_{\mathcal{E}}F\|^4_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] < \infty$$

and

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}}F D_{\mathcal{E}}F^*\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}] \le \mathbb{E}_\gamma[\|D_{\mathcal{E}}F\|^4_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] < \infty$$

by our moment assumption. This allows us to apply Lemma 33, which yields the desired result with constants

$$M_{H_{\mathcal{X}}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}F^* D_{\mathcal{E}}F - H_{\mathcal{X}}\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{E})}], \qquad M_{H_{\mathcal{Y}}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}F D_{\mathcal{E}}F^* - H_{\mathcal{Y}}\|^2_{\mathrm{HS}(\mathcal{Y},\mathcal{Y})}].$$

■

C.7 Proof of Theorem 13

We now conclude with the proof of the main theorem, Theorem 13, which combines all the results from the previous sections.

Proof First note that under the assumption that $F \in H^m_\gamma \cap W^{1,4}_\gamma$, for almost every realization of the samples $\mathcal{S}_N = (x_k)_{k=1}^N \overset{iid}{\sim} \gamma$, the empirical estimator $\widehat{F} \in \mathcal{Y}$, the empirical covariance $\widehat{C}_{\mathcal{Y}}$ and sensitivity operator $\widehat{H}_{\mathcal{Y}}$ lead to orthonormal bases $(\widehat{u}_i)_{i=1}^\infty$ in $\mathcal{Y}$, and the empirical sensitivity operator $\widehat{H}_{\mathcal{X}}$ leads to an orthonormal basis $(\widehat{v}_i^{\mathrm{DIS}})_{i=1}^\infty$ in $\mathcal{E}$. Additionally, Assumption 8 ensures that $(C_{\mathcal{X}}^{-1}\widehat{v}_i^{\mathrm{DIS}})_{i=1}^r$ are elements of $\mathcal{X}$. We can therefore apply Corollary 21 to obtain the existence of a neural network $\varphi_{\psi,\theta}$ such that (22) and (23) hold for any $\epsilon > 0$, with corresponding estimates for depth and size when using $\psi = \mathrm{ReLU}^k$ with $k \ge 2$.

The remaining terms in (22) and (23) are the mean estimator error, $\|\bar{F} - \widehat{F}\|^2_{\mathcal{Y}}$, which satisfies the Monte Carlo error bound of Corollary 34,

$$\mathbb{E}_N[\|\bar{F} - \widehat{F}\|^2_{\mathcal{Y}}] \le \frac{\|F - \bar{F}\|^2_{L^2_\gamma}}{N},$$

ridge function and reconstruction errors due to the input dimension reduction $\widehat{V}_r$,

$$\|F - \widetilde{F}_r\|^2_{L^2_\gamma}, \quad \|\widehat{P}_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_r)\widehat{Q}_r\|^2_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}, \quad \|D_{\mathcal{E}}F(I - \widehat{Q}_r)\|^2_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))},$$

and reconstruction errors due to the output dimension reduction $\widehat{U}_r$,

$$\|(I - \widehat{P}_r)(F - \bar{F})\|^2_{L^2_\gamma}, \quad \|(I - \widehat{P}_r)D_{\mathcal{E}}F\|^2_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}.$$

We can then consider each dimension reduction strategy separately, substituting in the results from the reconstruction, ridge function, and sampling errors developed in Section 3 to obtain the final approximation errors.

Empirical output PCA $U_r = \widehat{U}_r^{\text{PCA}}$: For the output reconstruction error, Proposition 24 gives

$$\|(I - \widehat{P}_r^{\text{PCA}})(F - \bar{F})\|_{L^2_\gamma}^2 \leq \sum_{i=r+1}^{\infty} \lambda_i^{\text{PCA}} + \min\left\{\sqrt{2r}\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{PCA}} - \lambda_{r+1}^{\text{PCA}}}\right\}.$$

To bound the derivative reconstruction error, we additionally require Assumption 10, which combined with Proposition 25 gives

$$\begin{aligned}&\|(I - \widehat{P}_r^{\text{PCA}})D_{\mathcal{E}}F\|_{L^2(\gamma,\text{HS}(\mathcal{E},\mathcal{Y}))}^2 \\ &\quad\leq K_D\left(\sum_{i=r+1}^{\infty} \lambda_i^{\text{PCA}} + \min\left\{\sqrt{2r}\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{PCA}} - \lambda_{r+1}^{\text{PCA}}}\right\}\right).\end{aligned}$$

Taking expectations with respect to the sampling error with the help of Proposition 36 gives

$$\mathbb{E}_N\left(\min\left\{\sqrt{2r}\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|C_{\mathcal{Y}} - \widehat{C}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{PCA}} - \lambda_{r+1}^{\text{PCA}}}\right\}\right) \leq \min\left\{\sqrt{\frac{2rM_{C_{\mathcal{Y}}}}{N}}, \frac{2M_{C_{\mathcal{Y}}}}{(\lambda_r^{\text{PCA}} - \lambda_{r+1}^{\text{PCA}})N}\right\}$$

with $M_{C_{\mathcal{Y}}} = 2\mathbb{E}_\gamma[\|(F - \bar{F}) \otimes (F - \bar{F}) - C_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2] + 2\|F - \bar{F}\|_{L^4_\gamma}^4 + 6\|F - \bar{F}\|_{L^2_\gamma}^4$. Thus, the overall constants are $K_{\mathcal{Y}}^{(1)} = 1$, $K_{\mathcal{Y}}^{(2)} = K_D$, and $M_{\mathcal{Y}} = 2\mathbb{E}_\gamma[\|(F - \bar{F}) \otimes (F - \bar{F}) - C_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2] + 2\|F - \bar{F}\|_{L^4_\gamma}^4 + 6\|F - \bar{F}\|_{L^2_\gamma}^4$.

Empirical output DIS $U_r = \widehat{U}_r^{\text{DIS}}$: From Propositions 26 and 27, we have

$$\|(I - \widehat{P}_r^{\text{DIS}})(F - \bar{F})\|_{L^2_\gamma}^2 \leq \sum_{i=r+1}^{\infty} \lambda_i^{\text{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{DIS}} - \lambda_{r+1}^{\text{DIS}}}\right\},$$

$$\|(I - \widehat{P}_r^{\text{DIS}})D_{\mathcal{E}}F\|_{L^2(\gamma,\text{HS}(\mathcal{E},\mathcal{Y}))}^2 \leq \sum_{i=r+1}^{\infty} \lambda_i^{\text{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{DIS}} - \lambda_{r+1}^{\text{DIS}}}\right\}.$$

Taking expectations with respect to the sampling error using Proposition 37 gives

$$\mathbb{E}_N\left[\min\left\{\sqrt{2r}\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}, \frac{2\|H_{\mathcal{Y}} - \widehat{H}_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2}{\lambda_r^{\text{DIS}} - \lambda_{r+1}^{\text{DIS}}}\right\}\right] \leq \min\left\{\sqrt{\frac{2rM_{H_{\mathcal{Y}}}}{N}}, \frac{2M_{H_{\mathcal{Y}}}}{(\lambda_r^{\text{DIS}} - \lambda_{r+1}^{\text{DIS}})N}\right\},$$

where $M_{H_{\mathcal{Y}}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}FD_{\mathcal{E}}F^* - H_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2]$. Thus, the overall constants are $K_{\mathcal{Y}}^{(1)} = K_{\mathcal{Y}}^{(2)} = 1$ and $M_{\mathcal{Y}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}FD_{\mathcal{E}}F^* - H_{\mathcal{Y}}\|_{\text{HS}(\mathcal{Y},\mathcal{Y})}^2]$.

Exact input PCA $V_r = V_r^{\text{PCA}}$: Under Assumption 8, the results in Proposition 23, Corollary 29, and Proposition 31 give

$$\begin{aligned}\|D_{\mathcal{E}}F(I - Q_r^{\text{PCA}})\|_{L^2(\gamma,\text{HS}(\mathcal{E},\mathcal{Y}))}^2 &\leq \mathbb{E}_\gamma\left[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2\right]\sum_{i=r+1}^{\infty}\mu_i^{\text{PCA}}, \\ \|F - \widetilde{F}_r\|_{L^2_\gamma}^2 &\leq \mathbb{E}_\gamma\left[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2\right]\sum_{i=r+1}^{\infty}\mu_i^{\text{PCA}}, \\ \|\widehat{P}_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_r)Q_r^{\text{PCA}}\|_{L^2(\gamma,\text{HS}(\mathcal{E},\mathcal{Y}))}^2 &\leq \mathbb{E}_\gamma\left[\|D^2F\|_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}^2\right]\mathbb{E}_\gamma[\|x\|_{\mathcal{X}}^2]\sum_{i=r+1}^{\infty}\mu_i^{\text{PCA}}.\end{aligned}$$

Moreover, since there is no sampling error in the exact PCA, these bounds are deterministic, such that $M_{\mathcal{X}} = 0$, $K_{\mathcal{X}}^{(1)} = \mathbb{E}_\gamma[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2]$, and $K_{\mathcal{X}}^{(2)} = \mathbb{E}_\gamma[\|D^2F\|_{\mathcal{L}_2(\mathcal{X},\mathcal{Y})}^2]\mathbb{E}_\gamma[\|x\|_{\mathcal{X}}^2] + \mathbb{E}_\gamma[\|DF\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})}^2]$.

Empirical input DIS $V_r = \widehat{V}_r^{\mathrm{DIS}}$: Proposition 26 and Corollary 29 give

$$\|F - \widetilde{F}_r\|_{L^2_\gamma}^2 \le \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\},$$

$$\|D_{\mathcal{E}}F(I - \widehat{Q}_r^{\mathrm{DIS}})\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 \le \sum_{i=r+1}^{\infty} \mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\},$$

To bound the derivative ridge function error, we additionally need Assumption 11, which combined with Proposition 32 gives

$$\begin{aligned}&\|\widehat{P}_r(D_{\mathcal{E}}F - D_{\mathcal{E}}\widetilde{F}_r)\widehat{Q}_r^{\mathrm{DIS}}\|_{L^2(\gamma,\mathrm{HS}(\mathcal{E},\mathcal{Y}))}^2 \\ &\quad\le K_H\left(\sum_{i=r+1}^{\infty} \mu_i^{\mathrm{DIS}} + \min\left\{\sqrt{2r}\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\}\right).\end{aligned}$$

Taking expectations with respect to the sampling distribution and applying Proposition 37 then yields

$$\mathbb{E}_N\left[\min\left\{\sqrt{2r}\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}, \frac{2\|H_{\mathcal{X}} - \widehat{H}_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2}{\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}}}\right\}\right] \le \min\left\{\sqrt{\frac{2rM_{H_{\mathcal{X}}}}{N}}, \frac{2M_{H_{\mathcal{X}}}}{(\mu_r^{\mathrm{DIS}} - \mu_{r+1}^{\mathrm{DIS}})N}\right\},$$

where $M_{H_{\mathcal{X}}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}F^* D_{\mathcal{E}}F - H_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2]$. Thus, the overall constants are $K_{\mathcal{X}}^{(1)} = 1$, $K_{\mathcal{X}}^{(2)} = K_H + 1$, and $M_{\mathcal{X}} = \mathbb{E}_\gamma[\|D_{\mathcal{E}}F^* D_{\mathcal{E}}F - H_{\mathcal{X}}\|_{\mathrm{HS}(\mathcal{E},\mathcal{E})}^2]$.

As suggested in Remark 14, we have used Assumption 8 to ensure the DIS yields $C_{\mathcal{X}}^{-1}\widehat{v}_i^{\mathrm{DIS}}$ that are elements of $\mathcal{X}$. We do not need the full Assumption 8 in the input DIS result. Instead, we can relax the assumption by directly assuming that F is such that the eigenvalue problem involving $\widehat{H}_{\mathcal{X}}$ yields $C_{\mathcal{X}}^{-1}\widehat{v}_i^{\mathrm{DIS}}$ that are indeed elements of $\mathcal{X}$. ■

Appendix D. Proofs for the Analytic Examples

D.1 Proofs for the Coefficient Map with Nonlinear Transformations

D.1.1 Verification of the Moment Bounds, Proposition 40

Proof (of Proposition 40) We first note that Fréchet differentiability of F can be shown from first principles by verifying the explicit expressions for the derivatives given in (37). From (37), we then see that D^nF is continuous. To this end, consider any $x, z \in \ell^2$ and $v_1, \ldots, v_n \in \ell^2$, for which

$$\|\left(D^nF(z) - D^nF(x)\right)(v_1, \ldots, v_n)\|_{\ell^2}^2 = \sum_{k=1}^{\infty} \frac{(g^{(n)}(z^{(k)}) - g^{(n)}(x^{(k)}))^2}{k^4}(v_1^{(k)})^2 \ldots (v_n^{(k)})^2.$$

Repeatedly applying Hölder's inequality and using the fact that $\|v\|_{\ell^\infty} \le \|v\|_{\ell^2}$ gives the bound

$$\|\left(D^nF(z) - D^nF(x)\right)(v_1, \ldots, v_n)\|_{\ell^2}^2 \le \sup_{k\in\mathbb{N}} |g^{(n)}(z^{(k)}) - g^{(n)}(x^{(k)})|^2 \prod_{i=1}^{n} \|v_i\|_{\ell^2}^2.$$

That is, $\|D^nF(z) - D^nF(x)\|_{\mathcal{L}_n(\ell^2,\ell^2)} \le \sup_{k\in\mathbb{N}} |g^{(n)}(z^{(k)}) - g^{(n)}(x^{(k)})|$. Consider now a convergent sequence $x_j \to x$ in ℓ^2. Then, $x_j \to x$ in ℓ^∞, which implies that $x_j^{(k)} \to x^{(k)}$ uniformly in $k \in \mathbb{N}$ and

all components of the converging sequence are contained in a compact interval, say, $[-R, R] \subset \mathbb{R}$ for some $R > 0$. The continuity of $g^{(n)}$ then implies uniform continuity over $[-R, R]$, which gives $\sup_{k\in\mathbb{N}} \left|g^{(n)}(x_j^{(k)}) - g^{(n)}(x^{(k)})\right|^2 \to 0$. Thus, $F \in C^m(\mathcal{X}, \mathcal{Y})$. Note that continuity in turn implies that $F : \mathcal{X} \to \mathcal{Y}$ and its derivatives $D^n F : \mathcal{X} \to \mathcal{L}_n(\mathcal{X}, \mathcal{Y})$ are strongly measurable with respect to γ. Here, it suffices to observe that F and $D^n F$ are Borel measurable and separably valued, the latter of which follows from the fact that their ranges are separable, being continuous images of the separable space $\mathcal{X}$.

Moreover, by the boundedness of $g^{(n)}(t)$ for $n = 1, \ldots, m$, we can show by analogous reasoning that $\|D^n F(x)\|_{\mathcal{L}_n(\mathcal{X},\mathcal{Y})} \leq M$ for all $1 \leq n \leq m$ and all $x \in \mathcal{X}$. Here, the $n = 1$ case additionally implies that F is Lipschitz continuous, which in turn implies that F has finite moments of any order under γ since Fernique's theorem asserts that $\mathbb{E}_\gamma[\|x\|_{\mathcal{X}}^p] < \infty$ for any $p \in [1, \infty)$. Therefore, we have $F \in L^p(\gamma, \mathcal{Y})$ and $D^n F \in L^p(\gamma, \mathcal{L}_n(\mathcal{X}, \mathcal{Y}))$ for any $n = 1, \ldots, m$ and any $p \in [1, \infty)$. Finally, since the embedding from $\mathcal{E}$ to $\mathcal{X}$ is Hilbert–Schmidt, we have $F \in W_\gamma^{m,p}$ for any $p \in [1, \infty)$.

■

D.1.2 Verification of the Inverse Inequalities, Proposition 41

Proof (of Proposition 41) We make use of the explicit expressions for the derivatives of F and the fact that $x^{(k)}$ are independent Gaussian random variables to compute the expectations in the inverse inequalities.

Derivative inverse inequality. To see that the derivative inverse inequality holds, we test the operators $C_{\mathcal{Y}}$ and $H_{\mathcal{Y}}$ with arbitrary $u \in \mathcal{Y}$ and compare the resulting expressions. To begin, we note that the expectation of the output is zero, i.e.,

$$\mathbb{E}_\gamma[F(x)] = \left(\frac{\mathbb{E}_\gamma[g(x^{(k)})]}{k^2}\right)_{k=1}^{\infty} = 0$$

as g is odd. For the covariance operator, we have

$$\mathbb{E}_\gamma[\langle u, F - \bar{F}\rangle_{\mathcal{Y}}^2] = \mathbb{E}_{x\sim\gamma}[\langle u, F(x)\rangle_{\mathcal{Y}}^2] = \mathbb{E}_{x\sim\gamma}\left[\left(\sum_{k=1}^{\infty} \frac{g(x^{(k)})}{k^2} u^{(k)}\right)^2\right].$$

Since $x^{(k)}$ are independent Gaussian random variables and $g(x^{(k)})$ are centered, the cross terms in the square of the sum vanish, leaving

$$\mathbb{E}_\gamma[\langle u, F - \bar{F}\rangle_{\mathcal{Y}}^2] = \sum_{k=1}^{\infty} \frac{(u^{(k)})^2}{k^4} \mathbb{E}_{x\sim\gamma}\left[g(x^{(k)})^2\right]. \tag{57}$$

Similar calculations can be made for the derivative covariance $H_{\mathcal{Y}}$. We have that

$$\mathbb{E}_{x\sim\gamma}[\|D_{\mathcal{E}} F(x)^* u\|_{\mathcal{E}}^2] = \mathbb{E}_{x\sim\gamma}[\|C_{\mathcal{X}}^{1/2} DF(x)^* u\|_{\mathcal{X}}^2] = \sum_{k=1}^{\infty} \frac{(u^{(k)})^2}{k^4} \sigma_k^2 \mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2]. \tag{58}$$

Comparing (57) to (58), we see that it suffices to find a constant $K_D > 0$ such that

$$\sigma_k^2 \mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \leq K_D \mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2] \qquad \forall k \in \mathbb{N}, \tag{59}$$

which would then imply that the derivative inverse inequality holds with the constant K_D.

To this end, boundedness of the derivatives of g implies that

$$|g(x) - x| \leq M|x|^2 \quad \forall x \in \mathbb{R}, \tag{60a}$$
$$|g'(x) - 1| \leq M|x| \quad \forall x \in \mathbb{R}. \tag{60b}$$

Using (60a) inside the second moment gives us a lower bound

$$\begin{aligned}\mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2] &= \mathbb{E}_{x\sim\gamma}[|x^{(k)}|^2] + 2\mathbb{E}_{x\sim\gamma}[(g(x^{(k)}) - x^{(k)})x^{(k)}] + \mathbb{E}_{x\sim\gamma}[(g(x^{(k)}) - x^{(k)})^2] \\ &\geq \mathbb{E}_{x\sim\gamma}\left[|x^{(k)}|^2\right] - 2M\mathbb{E}_{x\sim\gamma}\left[|x^{(k)}|^3\right] - M^2\mathbb{E}_{x\sim\gamma}\left[|x^{(k)}|^4\right] \\ &\geq \sigma_k^2 - 4\sqrt{\frac{2}{\pi}}M\sigma_k^3 - 3M^2\sigma_k^4. \end{aligned} \tag{61}$$

Analogously, using (60b) inside the second moment gives us an upper bound

$$\begin{aligned}\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] &= 1 + 2\mathbb{E}_{x\sim\gamma}[(g'(x^{(k)}) - 1)\cdot 1] + \mathbb{E}_{x\sim\gamma}[(g'(x^{(k)}) - 1)^2] \\ &\leq 1 + 2M\mathbb{E}_{x\sim\gamma}[|x^{(k)}|] + M^2\mathbb{E}_{x\sim\gamma}[|x^{(k)}|^2] \\ &\leq 1 + 2\sqrt{\frac{2}{\pi}}M\sigma_k + M^2\sigma_k^2, \end{aligned} \tag{62}$$

or equivalently

$$\sigma_k^2\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \leq \sigma_k^2 + 2\sqrt{\frac{2}{\pi}}M\sigma_k^3 + M^2\sigma_k^4. \tag{63}$$

Here, we have made use of expressions for moments of the Gaussian random variable $x^{(k)} \sim \mathcal{N}(0, \sigma_k^2)$ in the final steps. Since $\sigma_k \to 0$, there exists $K \in \mathbb{N}$ such that for all $k \geq K$, σ_k is sufficiently small and

$$\mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2] \geq \frac{\sigma_k^2}{2}, \qquad \sigma_k^2\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \leq 2\sigma_k^2,$$

which implies that $\sigma_k^2\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \leq 4\mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2]$ for all $k \geq K$. For the finitely many $k < K$, we can choose a sufficiently large $\widetilde{K}_D$ such that $\sigma_k^2\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \leq \widetilde{K}_D\mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2]$ for all $k < K$, noting that $\mathbb{E}_{x\sim\gamma}[g(x^{(k)})^2] > 0$ since g' is continuous and $g'(0) = 1$. Combining the two cases, we have that (59) holds with $K_D := \max\{4, \widetilde{K}_D\}$, which in turn implies that the derivative inverse inequality holds with the same constant K_D.

Hessian inverse inequality. We can verify the Hessian inverse inequality by a similar approach. Given any $v \in \mathcal{E}$, we have

$$\mathbb{E}_{x\sim\gamma}[\|D_{\mathcal{E}}F(x)v\|_{\mathcal{Y}}^2] = \sum_{k=1}^{\infty}\frac{(v^{(k)})^2}{k^4}\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2]. \tag{64}$$

For the Hilbert–Schmidt norm of Hessian action, we consider the PCA basis $(w_k)_{k=1}^{\infty}$ of $\mathcal{E}$ defined by $w_k := \sigma_k e_k$, where e_k is the standard Euclidean basis in ℓ^2 with a one in the kth position and zeros elsewhere. Then, we have that

$$\mathbb{E}_{x\sim\gamma}\left[\sum_{k=1}^{\infty}\|D_{\mathcal{E}}^2F(x)(v, w_k)\|_{\mathcal{Y}}^2\right] = \mathbb{E}_{x\sim\gamma}\left[\sum_{k=1}^{\infty}\sum_{j=1}^{\infty}\frac{(v^{(j)})^2(w_k^{(j)})^2}{j^4}g''(x^{(j)})^2\right].$$

Using the fact that $w_k^{(j)} = \sigma_k$ if $j = k$ and zero otherwise, we have, after interchanging the sum and expectation,

$$\mathbb{E}_{x\sim\gamma}\left[\sum_{k=1}^{\infty}\|D_{\mathcal{E}}^2F(x)(v, w_k)\|_{\mathcal{Y}}^2\right] = \sum_{k=1}^{\infty}\left[\frac{(v^{(k)})^2}{k^4}\sigma_k^2\mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2]\right]. \tag{65}$$

Comparing (64) to (65), it suffices to find a constant $K_H > 0$ such that

$$\sigma_k^2 \mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2] \leq K_H \mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \qquad \forall k \in \mathbb{N}, \tag{66}$$

which then implies that the Hessian inverse inequality holds with the constant K_H.

This can be shown by analogous calculations to the derivative inverse inequality. For the derivative, we have

$$\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \geq 1 - 2\sqrt{\frac{2}{\pi}} M\sigma_k - M^2\sigma_k^2, \tag{67}$$

and for the Hessian, we simply have

$$\sigma_k^2 \mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2] \leq M^2\sigma_k^2. \tag{68}$$

Thus, there exists some $K \in \mathbb{N}$ such that whenever $k \geq K$ (σ_k is sufficiently small), we have that

$$\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] \geq 1/2, \qquad \sigma_k^2 \mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2] \leq 1/2$$

such that $\sigma_k^2 \mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2] \leq \mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2]$ for all $k \geq K$. For the finitely many $k < K$, we can choose $\widetilde{K}_H$ sufficiently large such that $\sigma_k^2 \mathbb{E}_{x\sim\gamma}[g''(x^{(k)})^2] \leq \widetilde{K}_H \mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2]$ for all $k < K$, again noting that $\mathbb{E}_{x\sim\gamma}[g'(x^{(k)})^2] > 0$ since g' is continuous and $g'(0) = 1$. Therefore, (66) holds with $K_H := \max\{1, \widetilde{K}_H\}$, which implies that the Hessian inverse inequality holds with the same constant K_H.

■

D.2 Proofs for the Semilinear Elliptic PDE Examples

D.2.1 Well-posedness and Differentiability of the Solution Operator

We begin by establishing the well-posedness of the solution operator S defined for the semilinear elliptic PDE (38) under Assumption 42, which is a consequence of the standard theory of monotone operators.

Proposition 64 (Well-posedness of the solution operator)**.** *Consider the semilinear elliptic PDE* (38) *with the nonlinearity g satisfying Assumption 42. Then, for any $f \in H^{-1}(\Omega)$, there exists a unique solution $u \in H_0^1(\Omega)$ that solves the PDE* (38)*. That is, $S : H^{-1}(\Omega) \to H_0^1(\Omega)$ is a well-defined solution operator.*

Proof Let $\mathcal{B}(u) := -\Delta u + g(u)$ be the nonlinear operator defined on $H_0^1(\Omega)$ such that the PDE can be written as $\mathcal{B}(u) = f$. We can verify that $\mathcal{B}$ is a strongly monotone, coercive, and hemicontinuous operator, which implies that for any $f \in H^{-1}(\Omega)$, there exists a unique solution $u \in H_0^1(\Omega)$ such that $\mathcal{B}(u) = f$ (Tröltzsch, 2010, Theorem 4.1).

To see strong monotonicity, note that for any $u, v \in H_0^1(\Omega)$, we have

$$\begin{aligned}
\langle \mathcal{B}(u) - \mathcal{B}(v), u - v \rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)} &= \int_\Omega |\nabla(u-v)|^2 + (g(u) - g(v))(u-v) d\Omega \\
&\geq \int_\Omega |\nabla(u-v)|^2 d\Omega \\
&= \|u - v\|^2_{H_0^1(\Omega)},
\end{aligned} \tag{69}$$

where $\langle \cdot, \cdot \rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)}$ denotes the duality pairing between $H^{-1}(\Omega)$ and $H_0^1(\Omega)$. Here, we have used the fact that g is monotonically increasing, which implies that $(g(u(s))-g(v(s)))(u(s)-v(s)) \geq 0$ for all $s \in \Omega$. Coercivity follows from monotonicity and the fact that $g(0) = 0$ by taking $v = 0$ in (69), which gives us

$$\langle \mathcal{B}(u), u \rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)} \geq \|u\|^2_{H_0^1(\Omega)}.$$

Finally, $\mathcal{B}$ is hemicontinuous, i.e., $[0,1] \ni t \mapsto \langle \mathcal{B}(u+tv), w \rangle_{H^{-1}(\Omega),H_0^1(\Omega)}$ is continuous for all $u, v, w \in H_0^1(\Omega)$. Evidently, the linear part of $\mathcal{B}$, i.e., $-\Delta$, is continuous. On the other hand, the nonlinear part $g(u) : H_0^1(\Omega) \to H^{-1}(\Omega)$ is continuous since g is in $C^m(\mathbb{R})$ with the appropriate bounds on its growth and $H_0^1(\Omega) \hookrightarrow L^\infty(\Omega) \hookrightarrow H^{-1}(\Omega)$. As such, we can treat the nonlinear part as a continuous Nemytskii operator from $H_0^1(\Omega) \hookrightarrow L^\infty$ to $L^\infty \to H^{-1}(\Omega)$. The overall operator $\mathcal{B}$ is then hemicontinuous as the sum of two continuous operators. ■

In fact, we also have that S is Lipschitz continuous since we have strong monotonicity. Alternatively, Lipschitz continuity can be obtained by the global bound on the first derivative operator, which we show in the following proposition on the differentiability of the solution operator. The proof is based on standard arguments using the implicit function theorem on Banach spaces, but we include it here for completeness.

Proposition 65 (Differentiability of the solution operator)**.** *Let $S : H^{-1}(\Omega) \to H_0^1(\Omega)$ be the solution operator considered in Proposition 64 with the nonlinearity g satisfying Assumption 42 for some $m \in \mathbb{N}, m \geq 2$. Then, $S \in C^m(H^{-1}(\Omega), H_0^1(\Omega))$ and its first two Fréchet derivatives are given by the following:*

$$w = DS(f)\phi \text{ solves } -\Delta w + g'(S(f))w = \phi \tag{70}$$

and

$$z = D^2S(f)(\phi_1, \phi_2) \text{ solves } -\Delta z + g'(S(f))z = -g''(S(f))w_1 w_2, \tag{71}$$

where $w_i = DS(f)\phi_i$ for $i = 1, 2$. Moreover, we have that

$$\begin{aligned} \|DS(f)\|_{\mathcal{L}(H^{-1}(\Omega),H_0^1(\Omega))} &\leq 1, \\ \|D^2S(f)\|_{\mathcal{L}_2(H^{-1}(\Omega),H_0^1(\Omega))} &\lesssim \|g''(S(f))\|_{L^\infty(\Omega)}, \end{aligned}$$

where $\lesssim$ denotes an inequality up to a constant independent of $f \in H^{-1}(\Omega)$.

Proof We define the residual operator $R : H_0^1(\Omega)\times H^{-1}(\Omega) \to H^{-1}(\Omega)$ as $R(y, f) := -\Delta y + g(y) - f$, such that the solution operator S satisfies $R(S(f), f) = 0$ for all $f \in H^{-1}(\Omega)$. We note that R is m-times continuously differentiable since the linear parts are smooth and the nonlinear part $g(y)$ is m-times continuously differentiable as a Nemytskii operator from $L^\infty(\Omega)$ to $L^\infty(\Omega)$, which continuously embeds into $H^{-1}(\Omega)$. Moreover, we can compute the following derivatives of R:

$$\begin{aligned} D_y R(y, f)w &= -\Delta w + g'(y)w, \\ D_f R(y, f)\phi &= -\phi, \end{aligned}$$

for any $w \in H_0^1(\Omega)$ and $\phi \in H^{-1}(\Omega)$.

We let $\mathcal{A}_y w := -\Delta w + g'(y)w$. Since g is monotonic, we have $g'(y) \geq 0$, such that the operator $\mathcal{A}_y$ is both coercive and continuous. Here, coercivity follows from

$$\langle \mathcal{A}_y w, w \rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)} = \int_\Omega |\nabla w|^2 + g'(y)w^2 d\Omega \geq \|w\|^2_{H_0^1(\Omega)}, \tag{72}$$

while continuity follows from

$$\left|\langle \mathcal{A}_y w, z\rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)}\right| = \left|\int_\Omega \nabla w\cdot\nabla z + g'(y)wz\, d\Omega\right| \leq (1+\kappa_p^2\|g'(y)\|_{L^\infty(\Omega)})\|w\|_{H_0^1(\Omega)}\|z\|_{H_0^1(\Omega)} \tag{73}$$

where κ_p is the Poincaré constant, i.e., $\|v\|_{L^2(\Omega)} \leq \kappa_p\|v\|_{H_0^1(\Omega)}$ for all $v \in H_0^1(\Omega)$. Given any $y \in H_0^1(\Omega)$, the embedding of $H_0^1(\Omega)$ into $L^\infty(\Omega)$ implies that $\|g'(y)\|_{L^\infty(\Omega)}$ is finite and hence $\mathcal{A}_y$ is continuous. The Lax–Milgram theorem implies that $D_y R$ is invertible with a bounded inverse. By the implicit function theorem, it follows that $S \in C^m(H^{-1}(\Omega), H_0^1(\Omega))$ and its derivatives are given by implicit differentiation as in (70) and (71).

Boundedness of the derivative operators. Let us first consider the derivative operator $DS(f) : H^{-1}(\Omega) \to H_0^1(\Omega)$. We have that $w = DS(f)\phi$ solves the linearized PDE, $\mathcal{A}_y w = \phi$, where $y = S(f)$ is the solution of the original PDE. The Lax–Milgram theorem also gives a bound on the solution of the linearized PDE in terms of the coercivity constant (which is one in this case), i.e.,

$$\|w\|_{H_0^1(\Omega)} \leq \|\phi\|_{H^{-1}(\Omega)}. \tag{74}$$

Thus, we have that $\|DS(f)\|_{\mathcal{L}(H^{-1}(\Omega),H_0^1(\Omega))} \leq 1$ for all $f \in H^{-1}(\Omega)$. This additionally implies that S is Lipschitz continuous with Lipschitz constant at most one.

Similarly, for the second derivatives, we have that for $z = D^2S(f)(\phi_1,\phi_2)$,

$$\mathcal{A}_y z = -g''(y)w_1w_2, \qquad \mathcal{A}_y w_i = \phi_i, \quad i = 1,2.$$

Here, $g''(y)w_1w_2$ is an element of $H^{-1}(\Omega)$ in the form

$$\langle g''(y)w_1w_2, v\rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)} = \int_\Omega g''(y)w_1w_2v d\Omega$$

for any $v \in H_0^1(\Omega)$. Moreover, by Hölder's inequality and the embedding of $H_0^1(\Omega)$ into $L^4(\Omega)$, we have

$$\begin{aligned}|\langle g''(y)w_1w_2, v\rangle_{H^{-1}(\Omega)\times H_0^1(\Omega)}| &= \left|\int_\Omega g''(y)w_1w_2v d\Omega\right| \leq \|g''(y)\|_{L^\infty(\Omega)}\|w_1\|_{L^4(\Omega)}\|w_2\|_{L^4(\Omega)}\|v\|_{L^2(\Omega)}\\ &\lesssim \|g''(y)\|_{L^\infty(\Omega)}\|w_1\|_{H_0^1(\Omega)}\|w_2\|_{H_0^1(\Omega)}\|v\|_{H_0^1(\Omega)},\end{aligned}$$

such that $\|g''(y)w_1w_2\|_{H^{-1}(\Omega)} \lesssim \|g''(y)\|_{L^\infty(\Omega)}\|w_1\|_{H_0^1(\Omega)}\|w_2\|_{H_0^1(\Omega)}$, where the constant is independent of y and hence f. Together with (74), this yields

$$\|z\|_{H_0^1(\Omega)} \lesssim \|g''(y)\|_{L^\infty(\Omega)}\|\phi_1\|_{H^{-1}(\Omega)}\|\phi_2\|_{H^{-1}(\Omega)},$$

which is the desired bound for $D^2S(f)$. ■

We can also obtain bounds on the higher-order derivatives of S in terms of the derivatives of g evaluated at the solution $S(f)$ by analogous arguments to the second derivative case.

Proposition 66 (Higher-order derivatives of the solution operator)**.** *Let $S : H^{-1}(\Omega) \to H_0^1(\Omega)$ be the solution operator considered in Proposition 64 with the nonlinearity g satisfying Assumption 42 for some $m \in \mathbb{N}, m \geq 2$. Then,*

$$\|D^nS(f)\|_{\mathcal{L}_n(H^{-1}(\Omega),H_0^1(\Omega))} \lesssim \left(1 + \sum_{k=2}^n \|g^{(k)}(S(f))\|_{L^\infty(\Omega)}\right)^{p_n}, \tag{75}$$

for $n = 2,3,\ldots,m$, where p_n is some positive integer that depends on n.

Proof The proof follows by induction. The base case of $n = 2$ is given by Proposition 65. Suppose now the result holds for all orders of derivatives up to $n-1$ for some $2 < n \leq m$. We can obtain an expression for the action of the nth derivative, $D^n S(f)(v_1, \ldots, v_n)$, by implicit differentiation, i.e., by considering the equation $D_f^n(R(S(f), f))(v_1, \ldots, v_n) = 0$. For the semilinear elliptic PDE of (38), we have by Faà di Bruno's formula

$$D_f^n(R(S(f),f))(v_1,\ldots,v_n) = -\Delta D^n S(f)(v_1,\ldots,v_n) + g'(S(f))D^n S(f)(v_1,\ldots,v_n) + \sum_{\pi\in\Pi\setminus\{\pi_n\}} g^{(|\pi|)}(S(f)) \prod_{J\in\pi} D^{|J|}S(f)((v_j)_{j\in J}), \tag{76}$$

where Π is the set of all partitions of $\{1, 2, \ldots, n\}$ and π_n is the partition consisting of only one set $\{1, 2, \ldots, n\}$. Note that we are using $D^k S(f)((v_j)_{j\in J})$ to denote the action of the kth derivative of S at f on the tuple of vectors $(v_j)_{j\in J}$.

From (76), we see that analogous to the second derivative case, we can obtain the nth derivative action $w = D^n S(f)(v_1, \ldots, v_n)$ as the solution of the linearized PDE, linearized about $y = S(f)$, with the right-hand side given by the sum of products of lower-order derivatives of S and derivatives of g. That is,

$$\mathcal{A}_y w = -\sum_{\pi\in\Pi\setminus\{\pi_n\}} g^{(|\pi|)}(S(f)) \prod_{J\in\pi} D^{|J|}S(f)((v_j)_{j\in J}).$$

We can bound the H^{-1} norm of the right-hand side using the bounds on the lower-order derivatives of S from the induction hypothesis along with the embedding $L^\infty(\Omega) \hookrightarrow H^{-1}(\Omega)$. First, we have

$$\begin{aligned}
&\left\| \sum_{\pi\in\Pi\setminus\{\pi_n\}} g^{(|\pi|)}(S(f)) \prod_{J\in\pi} D^{|J|}S(f)((v_j)_{j\in J}) \right\|_{H^{-1}(\Omega)} \lesssim \left\| \sum_{\pi\in\Pi\setminus\{\pi_n\}} g^{(|\pi|)}(S(f)) \prod_{J\in\pi} D^{|J|}S(f)((v_j)_{j\in J}) \right\|_{L^\infty(\Omega)} \\
&\lesssim \sum_{\pi\in\Pi\setminus\{\pi_n\}} \|g^{(|\pi|)}(S(f))\|_{L^\infty(\Omega)} \prod_{J\in\pi} \left(\|D^{|J|}S(f)\|_{\mathcal{L}_{|J|}(H^{-1}(\Omega),H_0^1(\Omega))} \prod_{j\in J} \|v_j\|_{H^{-1}(\Omega)} \right) \\
&= \left(\sum_{\pi\in\Pi\setminus\{\pi_n\}} \|g^{(|\pi|)}(S(f))\|_{L^\infty(\Omega)} \prod_{J\in\pi} \|D^{|J|}S(f)\|_{\mathcal{L}_{|J|}(H^{-1}(\Omega),H_0^1(\Omega))} \right) \prod_{j=1}^n \|v_j\|_{H^{-1}(\Omega)}.
\end{aligned}$$

For the partitions $\pi \in \Pi \setminus \{\pi_n\}$ we have $1 < |\pi| \leq n$ and $1 \leq |J| < n$ for all $J \in \pi$. For $|J| = 1$, we have $\|DS(f)\|_{\mathcal{L}(H^{-1}(\Omega),H_0^1(\Omega))} \leq 1$ by Proposition 65, while for $2 \leq |J| < n$, we can apply the induction hypothesis, i.e.,

$$\|D^{|J|}S(f)\|_{\mathcal{L}_{|J|}(H^{-1}(\Omega),H_0^1(\Omega))} \lesssim \left(1 + \sum_{k=2}^{|J|} \|g^{(k)}(S(f))\|_{L^\infty(\Omega)} \right)^{p_{|J|}}.$$

Thus, the overall bound on the right-hand side is of the form

$$\left\| \sum_{\pi\in\Pi\setminus\{\pi_n\}} g^{(|\pi|)}(S(f)) \prod_{J\in\pi} D^{|J|}S(f)((v_j)_{j\in J}) \right\|_{H^{-1}(\Omega)} \lesssim \left(1 + \sum_{k=2}^{n} \|g^{(k)}(S(f))\|_{L^\infty(\Omega)} \right)^{p_n} \prod_{j=1}^n \|v_j\|_{H^{-1}(\Omega)},$$

for some positive integer p_n that depends on n. Combining this with the bound (74) gives

$$\|D^n S(f)(v_1,\ldots,v_n)\|_{H_0^1(\Omega)} \lesssim \left(1 + \sum_{k=2}^{n} \|g^{(k)}(S(f))\|_{L^\infty(\Omega)} \right)^{p_n} \prod_{j=1}^n \|v_j\|_{H^{-1}(\Omega)},$$

which yields the operator-norm bound for $D^n S$ in (75).

■

D.2.2 Verification of the Moment Assumptions, Proposition 43

Proof (of Proposition 43) Since $S \in C^m(H^{-1}(\Omega), H_0^1(\Omega))$ by Propositions 65 and 66, we also have that $F = S \circ \iota_{\mathcal{X}} \in C^m(\mathcal{X}, \mathcal{Y})$. The moment bounds for F can be obtained as follows. First, boundedness of $\iota_{\mathcal{X}}$ along with the bounds from Propositions 65 and 66 imply that for any $x \in \mathcal{X}$,

$$\|F(x)\|_{\mathcal{Y}} \lesssim \|x\|_{\mathcal{X}}, \tag{77}$$

$$\|DF(x)\|_{\mathcal{L}(\mathcal{X},\mathcal{Y})} \lesssim 1, \tag{78}$$

$$\|D^n F(x)\|_{\mathcal{L}_n(\mathcal{X},\mathcal{Y})} \lesssim \left(1 + \sum_{k=2}^{n} \|g^{(k)}(S(\iota_{\mathcal{X}} x))\|_{L^\infty(\Omega)}\right)^{p_n}, \quad n = 2, 3, \ldots, m. \tag{79}$$

From Assumption 42, we have that $|g^{(k)}(s)| \leq M(1 + |s|^q)$ for some $M > 0$ and $q \geq 0$, such that

$$\|g^{(k)}(S(\iota_{\mathcal{X}} x))\|_{L^\infty(\Omega)} \lesssim 1 + \|S(\iota_{\mathcal{X}} x)\|^q_{L^\infty(\Omega)}. \tag{80}$$

The embedding of $H_0^1(\Omega)$ into $L^\infty(\Omega)$ gives us $\|S(\iota_{\mathcal{X}} x)\|_{L^\infty(\Omega)} \lesssim \|S(\iota_{\mathcal{X}} x)\|_{H_0^1(\Omega)}$, and the Lipschitz continuity of S gives us $\|S(\iota_{\mathcal{X}} x)\|_{H_0^1(\Omega)} \lesssim \|\iota_{\mathcal{X}} x\|_{H^{-1}(\Omega)}$, which is further bounded by $\|x\|_{L^2(\Omega)}$ due to the boundedness of the embedding. Substituting this into (80) gives us

$$\|D^n F(x)\|_{\mathcal{L}_n(\mathcal{X},\mathcal{Y})} \lesssim (1 + \|x\|^q_{\mathcal{X}})^{p_n}, \qquad n = 2, 3, \ldots, m, \tag{81}$$

for any $x \in \mathcal{X}$.

To obtain the moment bounds for F and its derivatives, we first note that F and $D^n F$, $1 \leq n \leq m$, are strongly measurable. Indeed, it again suffices to observe that F and $D^n F$ are continuous mappings from the separable space $\mathcal{X}$ to $\mathcal{Y}$ and $\mathcal{L}_n(\mathcal{X}, \mathcal{Y})$, respectively, such that they are Borel measurable and separably valued, and hence strongly measurable. Moreover, since Fernique's theorem implies that all moments of $\|x\|_{\mathcal{X}}$, $x \sim \mathcal{N}(0, C_{\mathcal{X}})$ are finite, the bounds in (77), (78), and (81) imply that $F \in L^p(\gamma, \mathcal{Y})$ and $D^n F \in L^p(\gamma, \mathcal{L}_n(\mathcal{X}, \mathcal{Y}))$ for any $p \geq 1$. Finally, together with the Hilbert–Schmidt embedding of $\mathcal{E}$ into $\mathcal{X}$, these bounds also imply that $F \in W^{m,p}(\gamma, \mathcal{Y})$ for any $p \geq 1$.

■

D.2.3 Verification of the Hessian Inverse Inequality, Proposition 44

Proof (of Proposition 44) Let $v \in \mathcal{E}$ be arbitrary. We first derive a lower bound for $\|D_{\mathcal{E}} F(x) v\|^2_{\mathcal{Y}}$ that is analogous to a coercivity result. Recall that $w = D_{\mathcal{E}} F(x) v$ solves the linearized PDE $\mathcal{A}_y w = \iota_{\mathcal{X}} v$, where $\mathcal{A}_y w = -\Delta w + g'(y) w$ and $y = F(x)$. Moreover, by the continuity of $\mathcal{A}_y$ as in (73), we have

$$\|w\|_{H_0^1(\Omega)} \geq \frac{1}{1 + \kappa_p^2 \|g'(y)\|_{L^\infty(\Omega)}} \|\mathcal{A}_y w\|_{H^{-1}(\Omega)}.$$

In particular, for $y = F(x)$, using arguments analogous to those used to derive (81), we have that $\|g'(y)\|_{L^\infty(\Omega)} \lesssim (1 + \|x\|^q_{\mathcal{X}})$. Therefore, there exists a constant $\kappa_1 > 0$ such that

$$\|D_{\mathcal{E}} F(x) v\|^2_{\mathcal{Y}} = \|w\|^2_{\mathcal{Y}} \geq \frac{\kappa_1}{(1 + \|x\|^q_{\mathcal{X}})^2} \|\iota_{\mathcal{X}} v\|^2_{H^{-1}(\Omega)}.$$

Taking the expectation with respect to γ gives us

$$\mathbb{E}_\gamma[\|D_{\mathcal{E}} F(x) v\|^2_{\mathcal{Y}}] \geq K_1 \|\iota_{\mathcal{X}} v\|^2_{H^{-1}(\Omega)}, \tag{82}$$

where $K_1 = \mathbb{E}_{x\sim\gamma}\left[\kappa_1(1+\|x\|_{\mathcal{X}}^q)^{-2}\right] > 0$ since the integrand is positive and finite for all $x \in \mathcal{X}$.

For the second derivative, we can derive a matching upper bound for $\|D^2_{\mathcal{E}}F(x)(v,\cdot)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}$. To this end, we take $(w_i)_{i=1}^\infty$ to be the eigenfunctions of $C_{\mathcal{X}}$ normalized in the $\mathcal{E}$ norm (i.e., the PCA basis), such that $\|w_i\|^2_{\mathcal{X}} = \mu_i^{\mathrm{PCA}}$. Then, we have that

$$\begin{aligned}\|D^2_{\mathcal{E}}F(x)(v,w_i)\|^2_{\mathcal{Y}} &= \|D^2 S(\iota_{\mathcal{X}}x)(\iota_{\mathcal{X}}v,\iota_{\mathcal{X}}w_i)\|^2_{H^1_0(\Omega)}\\ &\lesssim \|g''(S(\iota_{\mathcal{X}}x))\|^2_{L^\infty(\Omega)}\|\iota_{\mathcal{X}}v\|^2_{H^{-1}(\Omega)}\|\iota_{\mathcal{X}}w_i\|^2_{H^{-1}(\Omega)}\\ &\lesssim (1+\|x\|^{2q}_{\mathcal{X}})\|\iota_{\mathcal{X}}v\|^2_{H^{-1}(\Omega)}\|\iota_{\mathcal{X}}w_i\|^2_{H^{-1}(\Omega)}.\end{aligned}$$

Moreover, we have that $\|\iota_{\mathcal{X}}w_i\|^2_{H^{-1}(\Omega)} \le \kappa_p^2\|w_i\|^2_{\mathcal{X}} \lesssim \mu_i^{\mathrm{PCA}}$. Summing over the basis $(w_i)_{i=1}^\infty$ then yields

$$\|D^2_{\mathcal{E}}F(x)(v,\cdot)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} = \sum_{i=1}^\infty \|D^2_{\mathcal{E}}F(x)(v,w_i)\|^2_{\mathcal{Y}} \lesssim (1+\|x\|^{2q}_{\mathcal{X}})\|\iota_{\mathcal{X}}v\|^2_{H^{-1}(\Omega)}\sum_{i=1}^\infty \mu_i^{\mathrm{PCA}} \tag{83}$$

Since $\sum_{i=1}^\infty \mu_i^{\mathrm{PCA}} = \mathrm{Tr}(C_{\mathcal{X}}) < \infty$, there exists a constant $\kappa_2 > 0$ such that

$$\|D^2_{\mathcal{E}}F(x)(v,\cdot)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})} \le \kappa_2(1+\|x\|^q_{\mathcal{X}})^2\|\iota_{\mathcal{X}}v\|^2_{H^{-1}(\Omega)}.$$

Taking expectations yields

$$\mathbb{E}_\gamma[\|D^2_{\mathcal{E}}F(x)(v,\cdot)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \le K_2\|\iota_{\mathcal{X}}v\|^2_{H^{-1}(\Omega)}, \tag{84}$$

where $K_2 = \kappa_2\mathbb{E}_{x\sim\gamma}\left[(1+\|x\|^q_{\mathcal{X}})^2\right] < \infty$.

Finally, comparing (82) and (84) gives us

$$\mathbb{E}_\gamma[\|D^2_{\mathcal{E}}F(x)(v,\cdot)\|^2_{\mathrm{HS}(\mathcal{E},\mathcal{Y})}] \le \frac{K_2}{K_1}\mathbb{E}_\gamma[\|D_{\mathcal{E}}F(x)v\|^2_{\mathcal{Y}}].$$

Thus, F satisfies the Hessian inverse inequality with $K_H = K_2/K_1$. ■

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